國立臺灣大學電機資訊學院資訊網路與多媒體研究所

碩士論文

Graduate Institute of Networking and Multimedia

College of Electrical Engineering and Computer Science

National Taiwan University

Master's Thesis

以平均速度流匹配（MeanFlow）進行公里尺度容許對流模式之生成式模擬：應用於臺灣對流尺度天氣預報的一至二步殘差生成

Kilometer-Scale Convection-Allowing Model Emulation using Average-Velocity Flow Matching: One- to Two-Step MeanFlow Residual Generation for Taiwan Convective-Scale Weather Prediction

鍾鎮嶸

Desmond Cheong Zheng Yong

指導教授: 周承復博士 博士

Advisor: Cheng-Fu Chou, Ph.D.

中華民國 115 年 6 月

June 2026

doi:10.6342/NTU202601285

國立臺灣大學碩士學位論文

口試委員會審定書

以平均速度流匹配（MeanFlow）進行公里尺度容許對流模式之生成式模擬：應用於臺灣對流尺度天氣預報的一至二步殘差生成

Kilometer-Scale Convection-Allowing Model Emulation using Average-Velocity Flow Matching: One- to Two-Step MeanFlow Residual Generation for Taiwan Convective-Scale Weather Prediction

本論文係鍾鎮嶸君（R13944064）在國立臺灣大學資訊網路與多媒體研究所完成之碩士學位論文，於民國 115 年 6 月 23 日承下列考試委員審查通過及口試及格，特此證明

doi:10.6342/NTU202601285

口試委員：_____

（指導教授）

_____ _____

_____ _____

_____ _____

_____ _____

所　　長：_____

doi:10.6342/NTU202601285

Acknowledgements

I would like to thank my advisor for guidance throughout this thesis, and the members of the lab for the many discussions that shaped this work. I am grateful to the developers of StormCast and PhysicsNeMo for releasing the pretrained checkpoints and training infrastructure that made this study possible, and to the Taiwan Central Weather Administration / NCDR for the RWRF reanalysis data used as training targets. Finally, I thank my family for their support throughout the program.

doi:10.6342/NTU202601285

doi:10.6342/NTU202601285

摘要

近年來，基於擴散模型的生成式天氣預報（例如 GenCast 與 StormCast）已展現出在機率性、銳利化、且可集成的短時預報任務上超越傳統確定性深度學習模型的能力。然而，擴散模型在推論階段需要沿著機率流常微分方程（PF-ODE）執行數十次神經網路前向傳遞（Number of Function Evaluations, NFE），使其在進行長時間自迴歸滾動預測或大規模集成預報時的計算成本過高；同時，擴散殘差在稀疏、重尾的降水通道上仍偏向平滑，難以重現對流尺度的極端降水結構。

本論文以一個以臺灣區域（RWRF）資料集訓練的兩階段 StormCast 模型（凍結的確定性回歸均值加上生成式殘差頭）為基礎，提出並評估 **MeanFlow**（平均速度流匹配）作為其生成式殘差頭：網路直接學習一段完整時間區間上的平均速度 u_θ，使得單次前向傳遞即可橫越整段噪聲到資料的軌跡，推論僅需 1–2 次網路評估、完全不需數值積分。MeanFlow 與一般擴散蒸餾不同，無需教師、僅需單一訓練階段：它透過 MeanFlow 恆等式（以單次 Jacobian–向量積自舉、並停止梯度的目標）直接從資料學習平均速度場，當區間長度趨於零時其目標恰好退化為標準流匹配。我們將此目標移植到 StormCast 的標準化殘差管線上，並保留兩個對照基準：原生的 EDM 擴散殘差頭（需 35 NFE，即我們所欲加速的對象），以及以條件流匹配（Conditional Flow Matching, I-CFM）學習瞬時直線速度場 v_θ、需以少數幾步 Euler 積分取樣的 **FlowCast** 殘差頭——MeanFlow 在 75% 的批次上恰好退化為 FlowCast 的目標，故 MeanFlow 對 FlowCast 構成僅針對

doi:10.6342/NTU202601285

訓練目標本身的對照。針對臺灣 RWRF 目標僅有 4 個高解析度輸出通道（t2m, u10, v10, qpepre）的特性，我們設計逐通道加權的 L2 損失，並在稀疏重尾的降水通道 qpepre 上加入徑向對數功率譜密度（log-PSD）正則化項；我們另外比較了 log1p 降水編碼與原始 mm/h 編碼，以及降水通道權重 w_{pr} 的取捨。

我們在 2022 年整年逐小時驗證資料上，以確定性誤差（RMSE/MAE）、機率分數（CRPS）、降水類別分數（CSI、FSS、HSS、誤報率）、徑向功率譜，以及最長 12 小時的自迴歸滾動誤差等指標，在同訓練樣本預算下將 MeanFlow 與 EDM 擴散基準及 FlowCast 進行正面比較。實驗結果顯示，MeanFlow 僅以 1–2 次網路評估（約為擴散取樣器十分之一、FlowCast 約四分之一的成本）即可達到與擴散基準同級的確定性與類別技能，並取得全研究最佳的溫度精度；其單步降水 CRPS 優於 FlowCast 在掃描範圍內任何步數的數值。其主要代價為集成離散度偏窄，在自迴歸滾動下表現為小幅的 CRPS 落差（於僅 1 次評估時最為明顯，第 2 次評估可補回大部分離散度，故以 $K = 2$ 為建議運行點）。本研究為區域對流尺度生成式預報提供了一套可重現的單階段平均速度流匹配流程與降水導向的評估基準。

關鍵字：平均速度流匹配、MeanFlow、流匹配、擴散模型、單步生成、生成式天氣預報、對流尺度預報、降水臨近預報、StormCast

doi:10.6342/NTU202601285

Abstract

Diffusion-based generative weather forecasters such as GenCast and StormCast have recently surpassed deterministic deep-learning baselines on probabilistic, sharp, and ensemble-capable short-range prediction tasks. However, sampling a diffusion model requires integrating its probability-flow ODE with tens of neural-network function evaluations (NFE) per forecast step, which makes autoregressive rollouts and large ensembles computationally prohibitive for operational regional, convective-scale deployment; moreover, the diffusion residual remains comparatively smooth on the sparse, heavy-tailed precipitation channel, where convective-scale extremes matter most.

This thesis builds on a two-stage StormCast model trained on a Taiwan-domain RWRF dataset — a frozen deterministic regression mean plus a generative residual head — and proposes and evaluates **MeanFlow**, an average-velocity flow-matching objective, as the StormCast generative residual head. The network learns the mean velocity over an entire time interval, so a single forward pass transports the state across the whole noise-to-data

doi:10.6342/NTU202601285

trajectory and sampling needs only one to two network evaluations, with no numerical integration at all. Unlike diffusion distillation, MeanFlow is teacher-free and trained in a single stage: it learns the average-velocity field directly from data through the MeanFlow identity (a single-Jacobian–vector-product, stop-gradient bootstrapped target), and as the interval length shrinks to zero its objective collapses exactly to standard flow matching. We transcribe the objective into StormCast's standardised-residual pipeline and retain two reference heads for comparison: the native EDM diffusion residual (35 NFE — the model we set out to accelerate), and **FlowCast**, an Independent Conditional Flow Matching (I-CFM) head that learns the instantaneous straight-line velocity field v_θ and samples it with a handful of explicit Euler steps. Because MeanFlow's objective reduces exactly to FlowCast's on 75% of each batch, the MeanFlow-versus-FlowCast comparison is a controlled A/B test on the training objective alone. Because the Taiwan RWRF target contains only four high-resolution output channels (`t2m`, `u10`, `v10`, `qpepre`), we can afford channel-specific loss design: a channel-weighted L2 objective plus a radial log power-spectral-density (log-PSD) regularizer on the sparse, heavy-tailed precipitation channel `qpepre`. We additionally study the `log1p` versus raw-mm/h precipitation encoding and the precipitation channel-weight w_{pr}.

We compare MeanFlow head-to-head against the EDM diffusion baseline and the

doi:10.6342/NTU202601285

FlowCast head at a matched training-sample budget on the full 2022 hourly validation year, using a precipitation-aware metric suite: per-channel RMSE/MAE, probabilistic CRPS, categorical precipitation scores (CSI, FSS, HSS, false-alarm ratio) at operational thresholds, radial log-PSD, and autoregressive rollout error out to 12 hours. Our experiments show that MeanFlow reaches the EDM diffusion baseline's deterministic and categorical skill class at one to two network evaluations — roughly a tenth of the diffusion sampler's cost and a quarter of FlowCast's — with the best temperature accuracy in the study and a single-step precipitation CRPS that undercuts FlowCast at any step count. Its principal cost is narrowed ensemble spread, which surfaces as a modest rollout-CRPS gap (largest when exactly one evaluation is used; a second evaluation restores most of the spread, so $K = 2$ is the recommended operating point). The thesis contributes a reproducible single-run average-velocity flow-matching pipeline for the StormCast residual and a precipitation-aware evaluation protocol for regional convective-scale generative forecasting.

Keywords: MeanFlow, Average-Velocity Flow Matching, Flow Matching, Diffusion Models, One-Step Generation, Generative Weather Forecasting, Convective-Scale Forecasting, Precipitation Nowcasting, StormCast

doi:10.6342/NTU202601285

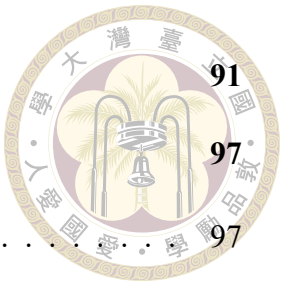
doi:10.6342/NTU202601285

Contents

doi:10.6342/NTU202601285

doi:10.6342/NTU202601285

doi:10.6342/NTU202601285



doi:10.6342/NTU202601285

List of Figures

doi:10.6342/NTU202601285

doi:10.6342/NTU202601285

List of Tables

doi:10.6342/NTU202601285

doi:10.6342/NTU202601285

Denotation

NWP	Numerical Weather Prediction
RWRF	Regional Weather Research and Forecasting (Taiwan domain reanalysis)
EDM	Elucidated Diffusion Model (Karras et al., 2022); the reference diffusion baseline
CFM	Conditional Flow Matching
I-CFM	Independent Conditional Flow Matching (the FlowCast baseline objective)
MeanFlow	Average-velocity flow matching (Geng et al., 2025); the method of this thesis
JVP	Jacobian–vector product (forward-mode, used for the MeanFlow target)
ODE	Ordinary Differential Equation
PF-ODE	Probability-Flow Ordinary Differential Equation

doi:10.6342/NTU202601285

SDE	Stochastic Differential Equation
NFE	Number of Function Evaluations (neural-net forward passes per sample)
EMA	Exponential Moving Average
PSD	Power Spectral Density
CSI	Critical Success Index
FSS	Fractions Skill Score
HSS	Heidke Skill Score
FAR	False-Alarm Ratio
CRPS	Continuous Ranked Probability Score
X_t	High-resolution atmospheric state at time t (target). Channels: `t2m`, `u10`, `v10`, `qpepre`.
S_t	Low-resolution conditioning state at time t (24 channels).
I	Time-invariant fields (land–sea mask, orography).
F_ξ	Deterministic regression network producing mean $M_t = F_\xi(X_{t-1}, S_t, I)$.
M_t	Regression mean prediction.
R_t	Residual $R_t = X_t - M_t$ learned by the generative head.
c	Conditioning bundle for the generative head. All three heads (EDM diffusion, FlowCast, MeanFlow) use concat(X_{t-1}, M_t, I); the orig-

doi:10.6342/NTU202601285

	inal StormCast additionally concatenates the low-resolution background S_t.
D_ψ	EDM diffusion baseline denoiser (EDMPrecond SongUNet).
v_θ	FlowCast (I-CFM baseline) instantaneous velocity field (FlowCastPrecond SongUNet).
u_θ	MeanFlow average-velocity field $u_\theta(z, r, t, c)$ (MeanFlowPrecond SongUNet); the network of this thesis.
σ	EDM noise level.
$\sigma_{\text{min}}, \sigma_{\text{max}}$	Minimum and maximum EDM noise levels (0.002 and 80).
σ_{data}	Data standard-deviation hyperparameter shared by all three heads (0.5).
ρ	Karras schedule exponent (7).
t (or r, t)	Flow time, in $[0, 1]$ (noise at 0, data at 1). MeanFlow uses an interval $[r, t]$ with state time $r \leq t$.
x_0, x_1	Flow endpoints: prior sample $x_0 \sim \mathcal{N}(0, I)$ and standardised residual $x_1 = R_t/\sigma_{\text{data}}$.
v	Instantaneous (I-CFM) velocity field; conditional target $v = x_1 - x_0$.
$u(z_r, r, t)$	Average velocity of the flow over $[r, t]$; the field MeanFlow learns. $\lim_{t\to r} u = v$.
ρ_{MF}	MeanFlow ratio: fraction of each batch with $r < t$ (the bootstrapped branch); 0.25.

doi:10.6342/NTU202601285

σ_{path}	I-CFM conditional-path standard deviation (0.01; MeanFlow uses the pure interpolant, $\sigma_{\text{path}} = 0$).
K	Number of sampling network evaluations (NFE): Euler steps for FlowCast, average-velocity segments for MeanFlow.
β_k	Per-channel loss weight; $w_{\text{pr}} = \beta_{\texttt{qpepre}}$ is the precipitation weight.
λ_{spec}	Weight of the radial log-PSD spectral regulariser.
$P(\kappa)$	Radial power spectral density at wavenumber κ.

doi:10.6342/NTU202601285

Chapter 1 Introduction

1.1 Weather Forecasting and Its Societal Stakes

Accurate short-range weather forecasting is a public-safety service. In Taiwan, a small, densely populated island lying along the western Pacific typhoon track, the stakes are especially concentrated: a single landfalling typhoon can dump more than 1000 mm of rainfall on steep orography in twelve hours, trigger flash floods and landslides, and disrupt transportation, agriculture, and power. Between typhoons, Mei-yu frontal systems, afternoon convective storms, and orographically forced rainfall drive the remainder of the hydrological cycle. For decision-makers who must issue evacuation orders, schedule reservoir releases, or re-route air traffic, the relevant question is almost always probabilistic and local: how much rain, where, within the next six to twelve hours?

The operationally critical quantity — precipitation at convective scales — is also the hardest for weather models to predict. Precipitation fields are sparse (most pixels are dry), heavy-tailed (the few wet pixels span orders of magnitude), and strongly coupled to terrain and mesoscale dynamics that only emerge at kilometre-scale resolution. This combination means that any forecasting system which produces visually smooth outputs, or which regresses toward the mean, will systematically underpredict the very events that matter most.

doi:10.6342/NTU202601285

1.2 A Brief History of Weather Prediction Methods

Modern weather forecasting has passed through four broad eras, each driven by a different scientific or computational advance.

Numerical Weather Prediction (NWP). Since Richardson's early twentieth-century vision and its post-war realisation by Charney, Phillips, and others, operational forecasting has been dominated by numerical integration of the primitive equations of atmospheric motion. State-of-the-art systems such as the ECMWF Integrated Forecasting System (IFS), the NCEP Global Forecast System (GFS), and regional convection-permitting models like WRF and its Taiwan variant RWRF discretise the atmosphere on a three-dimensional grid, solve the governing fluid-dynamics and thermodynamics equations, and close the unresolved sub-grid processes (convection, microphysics, boundary-layer turbulence, radiation) through physical parameterisations. Uncertainty is quantified by ensemble prediction systems that perturb initial conditions and model physics. NWP is physically consistent, well-understood, and remarkably skilful, but convection-permitting runs are computationally expensive, parameterisation error is non-trivial, and operational latency is a fundamental constraint.

Statistical and classical machine-learning post-processing. Long before deep learning, statistical techniques — model output statistics (MOS), analog methods, kriging, and more recently random forests and gradient-boosted trees — have been used to bias-correct and downscale NWP outputs. These approaches treat the NWP forecast as a feature vector and learn a local correction to each observation site. They set an important precedent:

doi:10.6342/NTU202601285

that learned components can complement physical cores, and that the ultimate criterion is predictive skill at points of interest rather than global physical fidelity.

The deep-learning era. Starting in the early 2020s, a new generation of purely data-driven global weather models began to rival and in some cases surpass operational NWP at synoptic scales. FourCastNet [14] used an Adaptive Fourier Neural Operator to produce six-hourly global forecasts at $0.25°$ resolution. Pangu-Weather [2] introduced a three-dimensional hierarchical transformer and demonstrated medium-range skill competitive with IFS. GraphCast [10] adopted an icosahedral-mesh graph neural network and achieved operational-quality ten-day forecasts at a fraction of the inference cost of IFS. At convective scales, MetNet [1, 4, 23] and the Deep Generative Model of Rainfall (DGMR) [17] pushed radar-based nowcasting from deterministic CNNs toward generative models capable of producing sharp, ensemble-like outputs. Most recently, StormCast [15] demonstrated that a convection-permitting diffusion model can learn a km-scale regional forecaster directly from reanalysis data.

The generative turn. A persistent problem of deterministic deep-learning forecasters is that, because they are trained with pixelwise regression losses, they produce fields that are systematically blurry and that underestimate extremes. Generative models resolve this by learning the distribution of plausible forecasts rather than its conditional mean. DGMR showed this qualitatively for radar nowcasting; GenCast [16] made the argument at global scale, demonstrating that a conditional diffusion model produces both sharper and better-calibrated ensemble forecasts than GraphCast. The lesson transfers directly to regional convective forecasting, and it motivates the generative forecaster at the centre of this thesis.

doi:10.6342/NTU202601285

1.3 Diffusion Models for Weather: Intuition

A diffusion model is a generative model trained to reverse a gradual noising process. Given a clean sample x_0 drawn from some data distribution — say, a high-resolution precipitation field over Taiwan — we progressively add Gaussian noise until after enough steps the result is indistinguishable from pure noise. A neural network is then trained to undo one step of this process: given a noised sample and the current noise level, predict the underlying clean signal (equivalently, the score of the noised distribution). At sampling time we start from pure Gaussian noise and apply the learned denoiser repeatedly until we recover a clean sample.

Three properties make this framework attractive for weather forecasting. First, it is naturally probabilistic: different noise seeds produce different but equally plausible samples, giving an ensemble for free. Second, the samples are sharp: because the model is never asked to average over outcomes, it does not regress toward the mean. Third, the denoiser can be conditioned on auxiliary inputs — the previous atmospheric state, coarse boundary fields, invariant land-surface information — which is exactly the interface a forecaster needs. GenCast at global scale and StormCast at regional scale both exploit these properties.

1.4 The Inference-Cost Problem

The cost of this flexibility is paid at inference time. High-quality diffusion sampling requires integrating the model's probability-flow ordinary differential equation (PF-ODE) with a few tens of neural-network forward passes per generated sample — the so-called

doi:10.6342/NTU202601285

number of function evaluations (NFE); the 18-step Heun schedule used by the baseline in this thesis costs 35 NFE per forecast step. For a single forecast step this is tolerable, but operational use cases compound it. An autoregressive 24-hour rollout multiplies the cost by 24. A 50-member ensemble for probabilistic forecasting multiplies it again. A convection-permitting Taiwan-domain forecast with a 50-member ensemble over a 24-hour horizon therefore requires on the order of $50 \times 24 \times 35 = 42{,}000$ forward passes of a large U-Net — which places the entire class of generative forecasters near the edge of, or outside, the operational envelope of a regional forecasting centre that must refresh its nowcast every few minutes between radar volumes.

The cost is not the only issue. The reason diffusion needs so many steps is that its PF-ODE trajectory — the path that carries a pure-noise sample to a clean forecast — is curved. A curved trajectory cannot be integrated accurately in one or two steps; coarse step counts incur visible discretisation error, and that error is worst precisely on the sparse, heavy-tailed precipitation channel, where the score of the noised distribution is ill-conditioned near small noise levels. The practical symptom is that an aggressively sub-sampled diffusion residual reverts toward a smooth, "slightly wet everywhere" field that scores acceptably on RMSE but destroys the convective structure operational users care about.

1.5 From Diffusion to Flow Matching

Both problems — the step count and the curvature — point at the same lever: the shape of the generative trajectory. If the path from noise to data were a straight line rather than a curved diffusion ODE, then a handful of explicit Euler steps would integrate it

doi:10.6342/NTU202601285

almost exactly, and a single step would already be a strong approximation.

Conditional Flow Matching (CFM) [11, 28] is a recently-proposed generative framework that learns exactly such a model. Instead of learning a score and integrating a curved PF-ODE, CFM regresses a deterministic velocity field that transports a Gaussian prior to the data along a straight-line probability path, and it does so with a simulation-free mean-squared-error objective that needs no teacher model at all. Ribeiro and Pucer [18] applied this idea to radar precipitation nowcasting under the name **FlowCast** and reported that a 1-to-10-step flow-matching sampler matched or beat a 100-step diffusion sampler on probabilistic precipitation skill at a fraction of the inference cost. A straight-line flow is integrated accurately in a handful of explicit Euler steps — but it is still integrated, and that residual handful of network evaluations is the last cost a flow-matching forecaster leaves on the table.

This thesis pushes the same lever one decisive notch further with **MeanFlow** [5]. Rather than learning the instantaneous velocity of the flow and then integrating it, MeanFlow learns the flow's average velocity over a whole time interval, so that one network evaluation transports a noise sample across the entire trajectory — there is no ODE solver and no discretisation error to amortise, and sampling collapses to one or two network evaluations. We adapt MeanFlow to the regional convection-permitting StormCast setting on the Taiwan RWRF domain as the generative residual head, keeping StormCast's two-stage decomposition (a frozen deterministic regression mean plus a generative residual head) intact and replacing only the residual head's curved EDM diffusion process. MeanFlow is the method of this thesis; FlowCast's straight-line I-CFM velocity field is retained as the natural flow-matching baseline that MeanFlow generalises — in fact MeanFlow's objective reduces exactly to FlowCast's whenever the time interval has zero length, so the two

doi:10.6342/NTU202601285

differ only by the average-velocity bootstrap and a comparison between them is a controlled A/B on the training objective. The EDM diffusion residual is retained throughout as the reference baseline that we set out to accelerate, against which all heads are measured at a matched training-sample budget, so that every quality and cost difference can be attributed to the choice of generative trajectory rather than to data, architecture, or the regression mean, all of which are held fixed. Crucially, MeanFlow is not a distillation of the diffusion teacher: it is trained from scratch in a single run, which makes it a clean test of whether a better-shaped generative target alone — without any teacher signal — already delivers the one-to-two-evaluation speed and the sharpness that operational regional forecasting needs.

1.6 Contributions

This thesis makes the following contributions:

1. We bring the average-velocity **MeanFlow** objective [5] into weather forecasting — to our knowledge for the first time — and adapt it to the StormCast two-stage regression-plus-residual architecture as the generative residual head: the network learns the displacement of the flow over whole time intervals via the MeanFlow identity (a single-JVP, stop-gradient bootstrapped target), so the residual is sampled in 1–2 network evaluations with no ODE solver, teacher-free and in a single training run, on a four-channel Taiwan regional target.

2. We retain two reference heads on the same backbone and budget so that the MeanFlow comparison isolates the generative trajectory: the native EDM diffusion residual (the 35-NFE model we set out to accelerate), and **FlowCast**, the straight-line

doi:10.6342/NTU202601285

Independent-CFM velocity baseline [11, 18, 28] that MeanFlow generalises. Because MeanFlow's objective collapses exactly to FlowCast's at zero interval length, MeanFlow-versus-FlowCast is a controlled A/B test on the training objective alone.

3. We carry out the engineering needed to make all three residual heads share exactly the frozen regression mean and per-channel normalisation: same data, same F_ξ, same σ_{data}, same backbone, only the EDM process versus the I-CFM velocity field versus the MeanFlow average-velocity field.

4. We exploit the four-channel target with a precipitation-aware loss: a channel-weighted L2 objective plus an auxiliary radial power-spectral-density (log-PSD) regulariser on the sparse, heavy-tailed `qpepre` channel, and we ablate the precipitation channel weight w_{pr} and the `log1p`-versus-raw-mm/h precipitation encoding.

5. We assemble a precipitation-aware evaluation suite — per-channel RMSE/MAE, probabilistic CRPS, categorical skill scores (CSI, FSS, HSS, false-alarm ratio) at operational thresholds, radial log-PSD, and 1–12-hour rollout errors — and run it on the full 2022 hourly validation year of the Taiwan RWRF dataset.

6. We present a head-to-head comparison of MeanFlow against the EDM diffusion residual and the FlowCast velocity baseline at a matched sample budget, including an NFE/wall-clock-versus-quality trade-off and a comparison against a legacy old-StormCast baseline, on a regional convection-permitting forecaster — to our knowledge the first study of average-velocity flow matching on the Taiwan convective-scale domain.

7. Appendix B.3 provides line-by-line derivations of the theoretical machinery MeanFlow rests on — the score–denoiser (Tweedie) identity, the VE-SDE / PF-ODE,

doi:10.6342/NTU202601285

EDM preconditioning, Conditional Flow Matching, and the MeanFlow average-velocity identity — in a form that should let a reader with a working knowledge of Gaussian algebra and the Fokker–Planck equation reconstruct every identity from scratch.

1.7 Thesis Outline

Chapter 2 surveys weather-prediction methods, a condensed account of the diffusion theory underlying the EDM baseline, and conditional flow matching building up to the average-velocity MeanFlow objective at the centre of this thesis, in enough depth to motivate and define every component used later. Chapter 3 describes the dataset, the frozen regression mean and the EDM diffusion and FlowCast baselines, and — as its core — the MeanFlow average-velocity residual pipeline as implemented in the accompanying codebase. Chapter 4 presents the experimental protocol and the head-to-head results across NFE budgets, precipitation thresholds, and rollout horizons. Chapter 5 summarises findings, limitations, and future work.

doi:10.6342/NTU202601285

doi:10.6342/NTU202601285

Chapter 2 Background and Related Work

This chapter assembles the theoretical background needed to understand the remainder of the thesis. It is organised in four parts: a survey of weather-prediction methods from NWP to data-driven models (Sections 2.1–2.4); a condensed account of score-based diffusion — DDPM and DDIM in summary, the continuous variance-exploding SDE / probability-flow ODE, and the EDM formulation used by our diffusion baseline (Sections 2.5–2.7), given only to the depth needed to understand the baseline we accelerate and the shared data scale σ_{data}; the theory of Conditional Flow Matching building up to the average-velocity **MeanFlow** objective at the centre of this thesis — with FlowCast's straight-line velocity field as the flow-matching baseline MeanFlow generalises — together with a direct comparison against the curved diffusion path and a survey of prior generative precipitation models (Sections 2.8–2.10); and an overview of precipitation-aware evaluation metrics (Section 2.11). Throughout the chapter we state the key identities in the main text and defer the derivations the method rests on to Appendix B.3.

doi:10.6342/NTU202601285

2.1 Numerical Weather Prediction

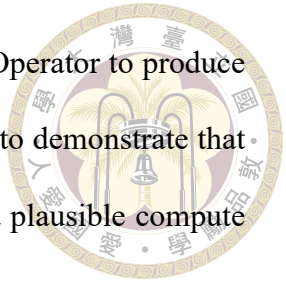
Numerical weather prediction discretises the primitive equations of atmospheric motion — continuity, momentum (Navier–Stokes on a rotating sphere), thermodynamic energy, and water-vapour conservation — onto a three-dimensional grid and advances the state forward in time via explicit or semi-implicit integration. Sub-grid processes not captured by the dynamical core (convection, microphysics, boundary-layer turbulence, radiation, land-surface exchange) are closed through physical parameterisations. Operational systems include ECMWF's IFS (global spectral, $0.1°$ horizontal resolution), NCEP's GFS, and regional limited-area models such as WRF and its Taiwan-tuned variant RWRF, which provide the high-resolution targets used in this thesis.

Uncertainty is quantified by ensemble prediction systems that perturb initial conditions (e.g. via singular-vector or ensemble Kalman methods) and model physics. A convection-permitting (sub-4 km) ensemble with a few tens of members and a 24-hour forecast horizon requires hours of dedicated supercomputer time. The practical consequence is that NWP ensemble size and spatial resolution are traded off against operational latency.

2.2 Data-Driven Global Weather Models

Starting in 2022, a series of deep-learning models demonstrated that a single neural network, trained on the ERA5 reanalysis, can produce global medium-range forecasts that rival operational NWP on many headline metrics at a fraction of the inference cost.

doi:10.6342/NTU202601285



FourCastNet. Pathak et al. [14] used an Adaptive Fourier Neural Operator to produce six-hourly global forecasts at $0.25°$ resolution. It was the first model to demonstrate that a data-driven system could match IFS on synoptic variables within a plausible compute budget.

Pangu-Weather. Bi et al. [2] introduced a hierarchical three-dimensional transformer with a hierarchical time-aggregation scheme and demonstrated medium-range skill competitive with or exceeding IFS on several variables and lead times.

GraphCast. Lam et al. [10] adopted a multi-mesh graph neural network defined on an icosahedral discretisation of the sphere. It produces ten-day $0.25°$ forecasts that are competitive with IFS on the majority of evaluated fields and does so in roughly one minute on a single TPU, compared with hours on a supercomputer for NWP.

These models share an important limitation: because they are trained with deterministic pixelwise losses (typically latitude-weighted MSE), their forecasts become progressively blurrier with lead time and systematically underestimate extreme events. For convection and precipitation, which this thesis cares about most, this is disqualifying.

2.3 Regional and Convective-Scale Forecasting

Global models operate at resolutions too coarse to resolve convection and orographic rainfall over Taiwan. Three lines of work fill the gap.

Nowcasting models. MetNet-1/2/3 [1, 4, 23] produce high-resolution short-range precipitation forecasts by ingesting ground-based radar and satellite observations through

doi:10.6342/NTU202601285

large convolutional/attention backbones. DGMR [17] uses a conditional GAN to produce sharp, ensemble-like radar nowcasts and was the first work to make a strong empirical case that generative modelling is essential for precipitation skill.

StormCast. Pathak et al. [15] introduced a convection-permitting regional diffusion forecaster trained on the High-Resolution Rapid Refresh (HRRR) analysis over CONUS. StormCast decomposes forecasting into two stages:

1. A deterministic regression network F_ξ (a `StormCastUNet`) takes the previous high-resolution state X_{t-1}, a low-resolution conditioning state S_t, and time-invariant fields I, and predicts the conditional mean $M_t = F_\xi(X_{t-1}, S_t, I)$.

2. A residual diffusion network D_ψ (an EDMPrecond SongUNet) models the residual $R_t = X_t - M_t$ conditionally on $c = \text{concat}(X_{t-1}, S_t, M_t, I)$.

At inference time the model is rolled out autoregressively, with the previous diffusion output feeding the next step's regression input. This decomposition has two payoffs: the deterministic network captures what is easy to predict with a single network call, leaving the diffusion model free to focus on the residual uncertainty; and the residual is closer to Gaussian than the raw fields, which stabilises diffusion training.

This thesis uses a Taiwan-adapted StormCast model trained on the RWRF dataset. The adaptation reduces the output state from the 99 channels of the CONUS original to just four: two-metre temperature (`t2m`), ten-metre zonal and meridional winds (`u10`, `v10`), and hourly precipitation (`qpepre`). This channel reduction has a useful side-effect: with only four output channels, per-channel loss engineering becomes cheap, and we exploit this in the precipitation-aware losses of Chapter 3.

doi:10.6342/NTU202601285

2.4 GenCast and the Generative Turn

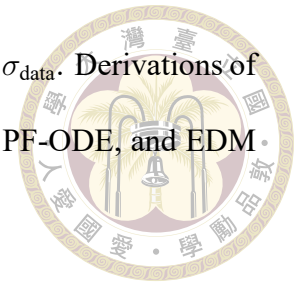
Price et al. [16] introduced GenCast, a global $0.25°$ conditional diffusion forecaster that samples ensemble members by varying the initial noise seed. GenCast was the first diffusion forecaster to clearly surpass the leading deterministic model (GraphCast) across the majority of its evaluation suite, on both deterministic scores (RMSE) and probabilistic scores (CRPS). Its success is the strongest empirical argument for the claim that motivates this thesis: generative modelling — and diffusion in particular — is the right inductive bias for short-to-medium range weather forecasting, but its operational cost is the obstacle to practical deployment.

2.5 Score-Based Diffusion: A Primer

A diffusion model defines a forward stochastic process that progressively corrupts a data sample $x_0 \sim p_{\text{data}}$ with Gaussian noise until the terminal state is indistinguishable from pure noise, then learns a neural network that reverses the corruption. The apparent difficulty of this programme — fitting the full generative reverse process of a high-dimensional distribution — hides a sequence of algebraic identities that together reduce training to a mean-squared-error regression. Because diffusion enters this thesis only as the reference baseline we set out to accelerate (and as the lineage of the curved generative path that flow matching straightens), we give it a condensed treatment: the discrete-time DDPM construction of Ho et al. [8] and its deterministic DDIM sampler [24] are summarised below, then the continuous variance-exploding SDE and probability-flow ODE that unify them [26], and finally the elucidated EDM parameterisation [9] that the base-

doi:10.6342/NTU202601285

line actually uses and from which all three heads inherit the data scale σ_{data}. Derivations of the continuous-time identities the baseline relies on — Tweedie, the PF-ODE, and EDM preconditioning — are collected in Appendix B.3.



2.5.1 DDPM and the Discrete-Time Construction

The denoising diffusion probabilistic model (DDPM) of Ho et al. [8] defines a T-step Markov chain that adds Gaussian noise to a clean sample x_0 according to a variance schedule $\{\beta_t\}$. Writing $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s \le t} \alpha_s$, the chain has a closed-form marginal that lets one jump to any noise level in a single line — so the forward process never has to be simulated:

$$q(x_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\ (1 - \bar{\alpha}_t)\, I\big), \qquad x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon,\ \ \epsilon \sim \mathcal{N}(0, I). \tag{2.1}$$

The reverse process is a learned Gaussian $p_\theta(x_{t-1} \mid x_t)$ matched to the (tractable, x_0-conditioned) forward posterior. After the standard variational-bound algebra and an ϵ-reparameterisation, the per-step KL terms collapse — with no loss-bearing approximation — into a single mean-squared-error objective that simply regresses the added noise:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon}\Big[\big\|\epsilon - \epsilon_\theta\big(\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon,\ t\big)\big\|^2\Big]. \tag{2.2}$$

The punchline is that an intractable maximum-likelihood problem reduces to a supervised denoising regression, with one network call per training step. The cost lands at inference: DDPM's ancestral sampler draws $T \approx 1000$ stochastic reverse steps, far too slow for deployment.

doi:10.6342/NTU202601285

2.5.2 DDIM and the Deterministic ODE View

Song et al. [24] observed that the DDPM loss (2.2) depends only on the marginals $q(x_t \mid x_0)$, not on the particular Markov chain producing them, so the same trained network admits a family of non-Markovian samplers — including a fully deterministic one:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0(x_t, t) + \sqrt{1-\bar{\alpha}_{t-1}}\,\frac{x_t - \sqrt{\bar{\alpha}_t}\,\hat{x}_0(x_t, t)}{\sqrt{1-\bar{\alpha}_t}}, \tag{2.3}$$

with $\hat{x}_0(x_t, t) = (x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(x_t, t))/\sqrt{\bar{\alpha}_t}$ the implied clean-data prediction. Equation (2.3) is exactly one explicit Euler step of an underlying ordinary differential equation, so the same model can be sampled in far fewer steps (typically 20–50 NFE, and toward ≈ 10 with higher-order solvers such as DPM-Solver [13]). DDIM is therefore the simplest first-order integrator of the probability-flow ODE made precise in the next subsection; the persistent curvature of that ODE — which keeps the step count in the 20–50 band for sharp precipitation — is exactly what the straight-line flow-matching path of Section 2.8 removes, and what the average-velocity MeanFlow objective at the centre of this thesis then jumps over entirely.

2.5.3 Continuous-Time View: VE-SDE and PF-ODE

Song et al. [26] unified DDPM, noise-conditional score matching, and their continuous-time limits into a single SDE framework. In the variance-exploding (VE) formulation adopted by EDM and inherited by StormCast, the forward SDE is

$$dx = \sqrt{\tfrac{d\sigma^2(t)}{dt}}\,dW_t, \tag{2.4}$$

doi:10.6342/NTU202601285

so that the closed-form marginal is the same one-line Gaussian corruption

$$x_\sigma = x_0 + \sigma\,\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{2.5}$$

now indexed by a continuous noise level $\sigma \in [0, \sigma_{\max}]$. The remarkable structural fact is that every diffusion SDE admits a deterministic probability-flow ordinary differential equation (PF-ODE) whose marginals coincide with those of the SDE at every time:

$$\frac{dx}{d\sigma} = -\sigma\,\nabla_x \log p_\sigma(x), \tag{2.6}$$

where $\nabla_x \log p_\sigma(x)$ is the (Stein) score of the noised marginal. Equation (2.6) is derived in Appendix B.3, §C.2 by matching the Fokker–Planck equation of the SDE against the continuity equation of the deterministic flow. The PF-ODE plays a central role here: it defines a deterministic trajectory $\sigma \mapsto x(\sigma)$ that links a pure-noise sample $x(\sigma_{\max})$ to its corresponding clean sample $x(\sigma_{\min})$, and any method that shortens the number of sampling steps — a few-step diffusion sampler, or the straight-line flow of Section 2.8 — is, one way or another, trying to traverse this kind of trajectory with fewer function evaluations.

Score–denoiser equivalence (Tweedie). Training a diffusion model amounts to learning the score, typically indirectly through a denoiser $D(x_\sigma; \sigma)$ whose output predicts $\mathbb{E}[x_0 \mid x_\sigma]$. The score and the denoiser are related by Tweedie's formula,

$$\nabla_x \log p_\sigma(x) = \frac{D(x;\sigma) - x}{\sigma^2}, \tag{2.7}$$

which we derive from the MMSE-denoiser identity in Appendix B.3, §C.1. The practical consequence is that denoising-score matching reduces to an ℓ_2 regression of the clean data

doi:10.6342/NTU202601285

from its noised version:

$$\mathcal{L}_{\text{DSM}}(\sigma) = \mathbb{E}_{x_0,\epsilon}\big[\|D_\theta(x_0 + \sigma\epsilon; \sigma) - x_0\|^2\big]. \tag{2.8}$$

Equation (2.8) is the continuous-time counterpart of Equation (2.2): the tractable training objective is a weighted mean-squared error over the noise scale, and the network can equivalently predict the clean data x_0, the noise ϵ, or the $v = \alpha_t\epsilon - \sigma_t x_0$ target of Salimans and Ho [20], the three being linearly interconvertible. The EDM baseline of this thesis uses the x_0-prediction form through the preconditioning of Section 2.6; flow matching, by contrast, regresses a velocity field directly and uses none of these denoiser parameterisations.

2.6 EDM: The Elucidated Formulation

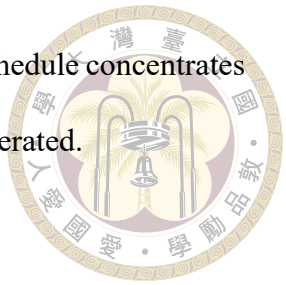
Karras et al. [9] proposed a unified parameterisation of diffusion training and sampling that cleans up a number of ad-hoc choices in earlier DDPM-style models. The EDM diffusion baseline of this thesis uses this formulation in full, and FlowCast borrows its data scale σ_{data} from it (Chapter 3). The EDM preconditioning scalings are not heuristics: they are the unique one-parameter family that makes the denoiser's input statistics, target statistics, and per-σ loss scale simultaneously uniform across σ. A line-by-line derivation of the $c_{\text{skip}}, c_{\text{out}}, c_{\text{in}}, c_{\text{noise}}, \lambda(\sigma)$ formulas below is given in Appendix B.3, §C.3.

Noise schedule. EDM uses a ρ-spaced discretisation of noise levels between σ_{min} and σ_{max}:

$$\sigma_i = \left(\sigma_{\text{max}}^{1/\rho} + \frac{i}{N-1}\left(\sigma_{\text{min}}^{1/\rho} - \sigma_{\text{max}}^{1/\rho}\right)\right)^{\rho}, \tag{2.9}$$

doi:10.6342/NTU202601285

for $i = 0, \ldots, N-1$, with $\sigma_{\min} = 0.002$, $\sigma_{\max} = 80$, and $\rho = 7$. The schedule concentrates noise levels near $\sigma_{\min}$, which is where most of the useful signal is generated.



Preconditioning. A naive denoiser $F_\theta(x_\sigma, \sigma)$ must cope with inputs whose magnitudes vary by orders of magnitude. EDM addresses this by wrapping the inner network in skip, input, output, and noise scalings:

$$D_\theta(x_\sigma; \sigma) = c_{\text{skip}}(\sigma)\, x_\sigma + c_{\text{out}}(\sigma)\, F_\theta\big(c_{\text{in}}(\sigma)\, x_\sigma,\ c_{\text{noise}}(\sigma)\big), \tag{2.10}$$

where

$$c_{\text{skip}}(\sigma) = \sigma_{\text{data}}^2/(\sigma^2 + \sigma_{\text{data}}^2), \qquad c_{\text{out}}(\sigma) = \sigma \cdot \sigma_{\text{data}}/\sqrt{\sigma^2 + \sigma_{\text{data}}^2},$$
$$c_{\text{in}}(\sigma) = 1/\sqrt{\sigma^2 + \sigma_{\text{data}}^2}, \qquad c_{\text{noise}}(\sigma) = \tfrac{1}{4}\log\sigma.$$

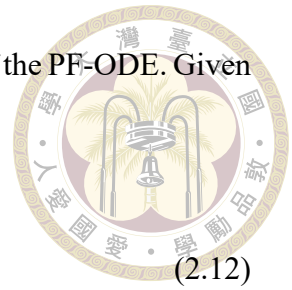
These scalings ensure that F_θ always sees inputs of unit variance and predicts an unnormalised residual, which is numerically stable across the full range of σ.

Training loss. EDM trains with a weighted denoising-score-matching loss:

$$\mathcal{L}_{\text{EDM}} = \mathbb{E}_{x_0, \sigma, \epsilon}\Big[\lambda(\sigma)\, \|D_\theta(x_0 + \sigma\epsilon; \sigma) - x_0\|^2\Big], \tag{2.11}$$

with a log-normal distribution over σ and a Karras-specified weighting $\lambda(\sigma)$.

doi:10.6342/NTU202601285



Heun sampler. EDM samples via a second-order Heun integrator of the PF-ODE. Given x_i at noise level σ_i, the next sample is

$$d_i = (x_i - D_\theta(x_i; \sigma_i))/\sigma_i, \tag{2.12}$$

$$x_{i+1}^{(\text{euler})} = x_i + (\sigma_{i+1} - \sigma_i)\, d_i, \tag{2.13}$$

$$d'_i = (x_{i+1}^{(\text{euler})} - D_\theta(x_{i+1}^{(\text{euler})}; \sigma_{i+1}))/\sigma_{i+1}, \tag{2.14}$$

$$x_{i+1} = x_i + (\sigma_{i+1} - \sigma_i)\, \tfrac{1}{2}(d_i + d'_i). \tag{2.15}$$

Each step uses two denoiser evaluations except the final step where the correction is skipped. With $N = 18$ steps this yields $2N - 1 = 35$ NFE; the original StormCast paper reports a shorter $N \approx 13$ schedule (≈ 25 NFE). The baseline in this thesis keeps the native $N = 18$ schedule, and its 35-NFE Heun sampler is the cost figure that the straight-line FlowCast sampler of Chapter 3 is benchmarked against: every NFE that FlowCast saves at matched forecast quality is a direct operational speedup.

2.7 Conditioning in Diffusion Models

Conditional diffusion models can be implemented either by concatenating the condition c along the channel axis of the denoiser's input, or by classifier-free guidance (CFG) [7], which trains the network jointly on conditional and unconditional tasks and linearly combines their outputs at inference time. StormCast uses the simpler concatenation approach: the conditioning bundle is stacked with the noised target along the channel dimension and fed to the SongUNet. No classifier-free guidance is used. The original StormCast concatenates $c = \text{concat}(X_{t-1}, S_t, M_t, I)$; all three generative heads in this thesis — the EDM diffusion baseline, FlowCast, and MeanFlow — instead condition on the trimmed

doi:10.6342/NTU202601285

bundle concat(X_{t-1}, M_t, I), dropping the low-resolution background S_t identically (Section 3.5). Because the trim is applied to every head alike, the fundamental change between the diffusion baseline and the flow-matching heads remains the generative process itself, not the way conditioning enters the network.

2.8 Flow Matching, FlowCast, and MeanFlow

Conditional Flow Matching (CFM) is a recently-proposed alternative to score-based diffusion that learns the same kind of noise-to-data generative model — a deterministic flow integrated from a simple prior to the data distribution — but with a training objective that bypasses score matching entirely. CFM is the foundation on which the method of this thesis is built: the average-velocity **MeanFlow** objective (Section 3.6) generalises the straight-line CFM velocity field of the **FlowCast** baseline (Section 3.5). We therefore develop CFM in full — the simulation-free training principle, the straight-line I-CFM path of FlowCast, and finally the average-velocity identity of MeanFlow — giving it more depth than the condensed diffusion primer above.

Continuous normalising flows, simulation-free training. A continuous normalising flow (CNF; 3) parameterises a deterministic velocity field $v_\theta(x, t)$, $t \in [0, 1]$, and pushes a prior sample $x_0 \sim \mathcal{N}(0, I)$ along the ODE $\dot{x}_t = v_\theta(x_t, t)$ to obtain x_1, which we would like to distribute as p_{data}. Classical maximum-likelihood CNF training requires evaluating $\log p_\theta(x_1)$ via the Jacobian trace over the whole trajectory — a prohibitive cost at image or video scale. Lipman et al. [11] and Tong et al. [28] instead train v_θ by direct regression against a target vector field $u_t(x \mid z)$ associated with a tractable conditional probability

doi:10.6342/NTU202601285

path $p_t(x \mid z)$ indexed by a latent z:

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\ z\sim q(z),\ x\sim p_t(x|z)}\Big[\|v_\theta(x,t) - u_t(x \mid z)\|^2\Big]. \tag{2.16}$$

The key theoretical result [11, Thm. 2] is that the gradient of (2.16) with respect to θ equals the gradient of the marginal flow-matching loss with the intractable marginal target field — so training the network against a conditional target yields an unbiased estimator of the true field. A derivation of this identity, along with the constants that are dropped because they do not depend on θ, is given in Appendix B.3, §C.5.

Independent CFM (I-CFM) and the straight-line path. The simplest instantiation, Independent CFM [28], takes $z = (x_0, x_1)$ with $x_0 \sim \mathcal{N}(0, I)$ independent of $x_1 \sim p_{\text{data}}$, and defines the straight-line Gaussian probability path

$$p_t(x \mid x_0, x_1) = \mathcal{N}\big((1-t)\,x_0 + t\,x_1,\ \sigma_{\text{path}}^2 I\big), \qquad x_t = (1-t)\,x_0 + t\,x_1 + \sigma_{\text{path}}\,\eta,\ \eta \sim \mathcal{N}(0, I). \tag{2.17}$$

Differentiating the conditional mean with respect to t gives the corresponding target vector field

$$u_t(x \mid x_0, x_1) = x_1 - x_0, \tag{2.18}$$

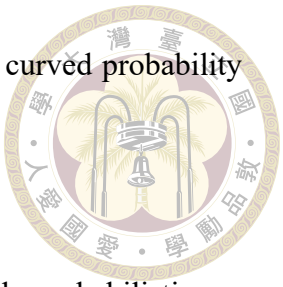
which is independent of x — exactly the straight-line constant-velocity interpolation between the noise sample and the data sample. The training objective becomes

$$\mathcal{L}_{\text{I-CFM}} = \mathbb{E}_{t,x_0,x_1,\eta}\Big[\big\|v_\theta\big((1-t)x_0 + t x_1 + \sigma_{\text{path}}\eta,\ t\big) - (x_1 - x_0)\big\|^2\Big]. \tag{2.19}$$

The path noise $\sigma_{\text{path}} > 0$ "thickens" the otherwise-singular conditional paths and regularises the vector-field fit, following Tong et al. [28]; FlowCast uses $\sigma_{\text{path}} = 0.01$. A

doi:10.6342/NTU202601285

derivation of (2.18) from (2.17) and an explicit comparison with the curved probability flow of DDPM are in Appendix B.3, §C.6.



FlowCast. Ribeiro and Pucer [18] introduced FlowCast, the first fully probabilistic application of I-CFM to precipitation nowcasting. FlowCast trains a single U-Net backbone — in the original paper an Earthformer-style cuboid-attention network operating in a VAE latent space — to regress the straight-line vector field $v_\theta(x_t, t, c)$ conditioned on a past-observation bundle c. Sampling uses a 10-step Euler integrator of $\dot{x}_t = v_\theta(x_t, t, c)$. In a direct ablation on the SEVIR benchmark, FlowCast at 1 ODE step beat DDIM at 100 steps on CRPS and CSI, and matched or exceeded the state-of-the-art probabilistic nowcasting baselines CasCast and PreDiff at a fraction of the inference cost. Two features make it the natural flow-matching baseline for this thesis: (i) the straight-line ODE is by construction the trajectory any one-step sampler is trying to approximate, so the performance gap between 1-step and 10-step sampling is much smaller than for curved diffusion ODEs; and (ii) training is simulation-free and requires no teacher — FlowCast learns its velocity field directly from data in a single training run. In Section 3.5 we adapt FlowCast to the StormCast residual pipeline as a baseline, against which the average-velocity MeanFlow head of this thesis — which generalises exactly this straight-line objective — is measured on the same 2022 Taiwan validation year.

MeanFlow: average velocity for one-step generation. Even a straight-line flow is sampled by numerically integrating an ODE, and the integration is what keeps the NFE budget at $K \approx 10$: the marginal velocity field that the network actually learns is curved (the average over crossing conditional paths), so each Euler step still pays a truncation error against that curvature. A separate line of work asks whether the integral can be re-

doi:10.6342/NTU202601285

moved from the sampler altogether. Consistency models [25, 27] train a network to map any point of a trajectory directly to its endpoint, but impose this as a constraint on the network's behaviour — pairs of network outputs are pulled together along discretised trajectories — which is known to need carefully tuned discretisation curricula and to train unstably. MeanFlow [5] reaches the same goal from a field-theoretic direction: it defines the average velocity $u(z, r, t)$ — the displacement of the flow between two times $r < t$ divided by the interval length — as a new ground-truth field induced by the instantaneous velocity v, and derives an exact identity relating the two, $u = v - (t - r)\,\frac{d}{dt}u$ in the paper's data-at-$t{=}0$ convention. Training regresses the network onto this identity using only the same conditional-velocity samples as CFM (the derivative term is evaluated by a single Jacobian–vector product and treated as a stop-gradient target), so no integral, teacher, curriculum, or auxiliary consistency constraint is needed; at $r = t$ the objective collapses to standard flow matching. Because the network outputs an average velocity, one evaluation transports a prior sample across the whole flow — 1–2 NFE sampling by construction. On ImageNet 256×256, a from-scratch MeanFlow model reaches an FID of 3.43 at 1 NFE, a 50–70% relative improvement over the best previous one-step diffusion/flow models and close to many-step baselines at 500 NFE [5]. Section 3.6 transcribes the identity into this thesis's noise-to-data convention and adapts MeanFlow to the StormCast residual pipeline as the generative residual head, trained under hyperparameters identical to FlowCast so that the average-velocity objective can be compared against its straight-line special case like for like.

doi:10.6342/NTU202601285

2.9 Flow Matching versus Diffusion: Why Straight-Line Paths Help

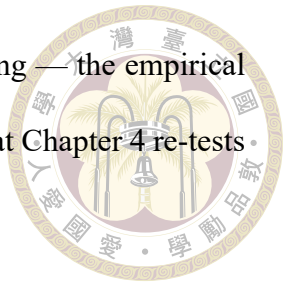
Both score-based diffusion and Conditional Flow Matching learn a deterministic ODE that carries a Gaussian prior sample to a data sample; the difference is the shape of the path that ODE follows, and that shape is what governs how few steps a sampler can use. It is worth stating the contrast explicitly, because it is the single fact that motivates the choice of FlowCast over diffusion in this thesis.

The diffusion path is curved. The EDM PF-ODE of Equation (2.6) has a velocity $-\sigma \nabla_x \log p_\sigma(x)$ that depends on the score of the noised marginal, which itself changes nonlinearly with σ. The trajectory $\sigma \mapsto x(\sigma)$ is therefore curved, and a low-order integrator (Euler, or even Heun) accumulates a truncation error that grows as the step size grows. This is why high-quality diffusion sampling sits in the 20–50-NFE band: below it, the integrator's curvature error becomes visible, and it shows up first on the sparse, heavy-tailed precipitation channel where the score is most ill-conditioned at small σ.

The I-CFM path is a straight line. Independent CFM (Section 2.8) defines its conditional probability path as the straight-line interpolation $x_t = (1-t)x_0 + t x_1$ between a prior sample x_0 and a data sample x_1, with the constant target velocity $u_t = x_1 - x_0$. Along a single such conditional path the trajectory has zero curvature, so an explicit Euler step incurs no truncation error on the conditional path; what error remains comes only from the fact that the learned marginal velocity field averages over many crossing conditional paths. In practice this makes the gap between 1-step and 10-step flow-matching sampling

doi:10.6342/NTU202601285

far smaller than the gap between 1-step and 25-step diffusion sampling — the empirical claim that Ribeiro and Pucer [18] verified on radar nowcasting and that Chapter 4 re-tests on the Taiwan RWRF domain.



No teacher, one training run. A third practical difference is organisational. The standard way to make diffusion cheap at inference is distillation: train a fast student to imitate a slow, pretrained teacher's sampling output (progressive distillation [20], consistency distillation [25, 27], adversarial distillation [21]). Distillation needs a converged teacher, a second training stage, and careful matching of the teacher's sampler and preconditioning. Rectified flow [12] showed that one can instead reshape the path itself so that straight-line integration suffices. FlowCast takes this route to its simplest conclusion: it trains the straight-line velocity field directly from data in a single run, with no teacher and no second stage. MeanFlow [5] — the method of this thesis — goes one step further along the same axis, learning the flow's average velocity so that the integral disappears from the sampler entirely while training remains single-stage and teacher-free. This thesis therefore measures a single-run average-velocity forecaster (with its straight-line special case as a flow-matching baseline) against the diffusion model it would otherwise have to distil, rather than comparing distillation schemes among themselves.

2.10 Prior Generative Models for Precipitation Nowcasting

Generative precipitation modelling has converged on the same conclusion from several directions. DGMR [17] used a conditional GAN and was the first to argue convinc-

doi:10.6342/NTU202601285

ingly that adversarial/generative training, not deterministic regression, is what produces skilful sharp radar nowcasts. Diffusion-based nowcasters and downscalers followed: GenCast [16] at global scale and StormCast [15] at regional convection-permitting scale both use score-based diffusion residuals, and NVIDIA's CorrDiff line applies an EDM super-resolution residual to km-scale downscaling. The common cost of all of these is the multi-step diffusion sampler. Flow matching is the most recent entrant: Ribeiro and Pucer [18] introduced FlowCast as the first fully-probabilistic Independent-CFM nowcaster and reported that it matched or beat state-of-the-art diffusion nowcasters (CasCast, PreDiff) at a small fraction of their NFE on the SEVIR radar benchmark. To the best of our knowledge, the present thesis is the first to adapt conditional flow matching to a regional convection-permitting forecaster (as opposed to a radar-to-radar nowcaster), to anchor it to a frozen NWP-style regression mean in the StormCast two-stage decomposition, to bring the average-velocity MeanFlow objective [5] into weather forecasting at all, and to report a three-way diffusion-versus-CFM-versus-MeanFlow comparison with precipitation-specific metrics on the Taiwan domain.

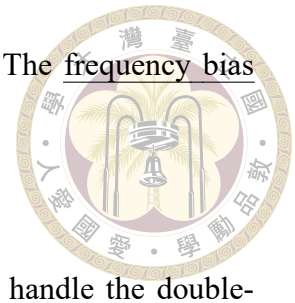
2.11 Evaluation of Precipitation Forecasts

Pixelwise RMSE alone is a poor measure of precipitation-forecast quality because it rewards smooth, low-amplitude predictions that miss extreme events. We adopt a suite of complementary metrics, each introduced briefly here and defined formally in Chapter 4.

Categorical skill scores. For a precipitation threshold τ, every pixel in the forecast and observation is classified as wet ($\geq \tau$) or dry, producing a 2×2 contingency table with hits H, misses M, false alarms F, and correct negatives. The Critical Success Index

doi:10.6342/NTU202601285

CSI $= H/(H + M + F)$ [22] is the standard operational score. The frequency bias $(H + F)/(H + M)$ diagnoses over- or under-prediction.



Fractions Skill Score. Roberts and Lean [19] introduced FSS to handle the double-penalty problem: a forecast that is slightly displaced in space but correct in structure should not be penalised as heavily as one that is wrong everywhere. FSS computes the fraction of pixels exceeding τ in a neighbourhood of radius n around each point and compares forecast and observation fractions.

Spectral diagnostics. The radially averaged logarithmic power spectral density (log-PSD) of the forecast and observation fields quantifies sharpness. A deterministic model that produces blurry outputs will have a log-PSD that drops off too quickly at high wavenumbers; a generative model that matches the data distribution will track the observation spectrum.

Rollout error growth. For an autoregressive forecaster, single-step error understates the operational risk. We therefore report errors at rollout horizons of 1, 3, 6, and 12 hours, which captures the cumulative effect of single-step residual-head error on autoregressive predictions.

Probabilistic scores (optional). For ensemble forecasts we additionally report the continuous ranked probability score (CRPS) and rank histograms for `qpepre`.

doi:10.6342/NTU202601285

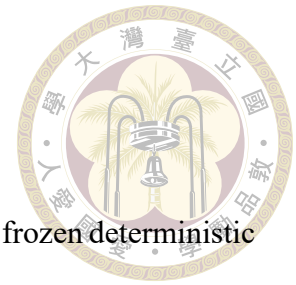
doi:10.6342/NTU202601285

Chapter 3 Method

This chapter describes what was actually built. Section 3.1 states the residual-forecasting problem in precise terms. Section 3.2 describes the Taiwan RWRF dataset and its normalisation. Section 3.3 documents the underlying data sources and the preprocessing pipeline that turns them into the training Zarr store. Section 3.4 recaps the frozen regression mean and the EDM diffusion residual that serves as our reference baseline. Section 3.5 describes the FlowCast Conditional Flow Matching residual — the straight-line flow-matching baseline that the method of this thesis generalises — which we develop first because MeanFlow is defined relative to it. Section 3.6, the core of the chapter, describes **MeanFlow**, the average-velocity flow-matching residual that is the method of this thesis: it pushes the sampling budget down to one or two network evaluations with no ODE integration, while keeping every other design choice identical to the FlowCast baseline. Section 3.7 describes the dataset-specific loss modifications for the precipitation channel, which are shared by the diffusion baseline and both flow-matching heads. Section 3.8 documents the training infrastructure.

doi:10.6342/NTU202601285

3.1 Problem Setup



We adopt the StormCast two-stage decomposition throughout. A frozen deterministic regression network F_ξ produces a conditional mean

$$M_t = F_\xi(X_{t-1}, S_t, I), \tag{3.1}$$

and a generative residual head models the residual $R_t = X_t - M_t$ conditioned on a bundle assembled from the previous state, the regression mean, and the invariants. The regression mean is the same frozen checkpoint for every method in this thesis, so the only object that varies is the generative residual head.

We compare three interchangeable residual heads, holding the dataset, the regression mean M_t, the backbone architecture, and the data scale σ_{data} fixed:

- **EDM diffusion (reference baseline).** A frozen-recipe EDMPrecond SongUNet D_ψ models the residual as a score-based diffusion and samples it by integrating the PF-ODE from σ_{max} to σ_{min} with a Heun sampler ($N = 18$ steps, $2N - 1 = 35$ NFE per hourly step). This is the model we set out to accelerate, and the reference against which the flow-matching heads are measured; it is the residual head of the original StormCast, save for the trimmed conditioning bundle that it shares with them (Section 3.5).

- **FlowCast (flow-matching baseline).** A FlowCastPrecond SongUNet velocity field v_θ models the residual as a straight-line Conditional Flow Matching ODE and samples it with K explicit Euler steps (K NFE per hourly step), where we sweep K from 1 to 50 (the head-to-head scoreboard reports $K \in \{10, 15, 20\}$). It is the straight-

doi:10.6342/NTU202601285

line special case that MeanFlow generalises, retained as the natural flow-matching comparator.

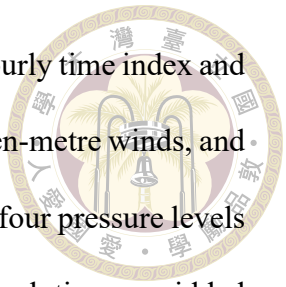
- **MeanFlow (the method of this thesis).** A MeanFlowPrecond SongUNet u_θ models the average velocity of the same flow over an interval $[r, t]$ [5], so that one network evaluation transports the state across the whole interval with no ODE integration. Sampling uses K average-velocity segments at exactly K NFE per hourly step, with $K \in \{1, 2\}$ as the headline operating points (Section 3.6). Training hyperparameters are kept identical to the FlowCast baseline, and MeanFlow's objective reduces exactly to FlowCast's at zero interval length, making MeanFlow-versus-FlowCast an A/B test on the training objective alone.

In all cases the full prediction at each step is $\hat{X}_t = M_t + \hat{R}_t$, and the model is rolled out autoregressively, with the previous generated state feeding the next step's regression input. The central empirical question of the thesis is whether the single-run, teacher-free MeanFlow head matches or beats the diffusion head at a substantially smaller NFE (and wall-clock) budget — $K \in \{1, 2\}$ network evaluations against the diffusion sampler's 35, with the FlowCast baseline ($K \approx 10$) marking the straight-line flow-matching reference point in between — all at a matched training-sample budget so that the comparison reflects the generative trajectory rather than training compute. Neither flow head distils D_ψ or queries it during training; the three heads are trained independently on the same data.

3.2 Dataset: Taiwan RWRF

The training data live in a Zarr store under the project directory (see Appendix A.5). The store has three top-level groups.

doi:10.6342/NTU202601285



LowRes conditioning S_t. Shape $(T, 24, 224, 128)$, where T is the hourly time index and the 24 channels are mean sea-level pressure, two-metre temperature, ten-metre winds, and the specific humidity, temperature, and zonal and meridional winds at four pressure levels (1000 hPa, 850 hPa, 500 hPa and 250 hPa). These channels are low-resolution re-gridded ERA5 fields that serve as synoptic-scale boundary conditioning.

HighRes target X_t. Shape $(T, 4, 224, 128)$, with four channels: two-metre temperature (`t2m`), ten-metre zonal and meridional winds (`u10`, `v10`), and hourly accumulated precipitation (`qpepre`). The horizontal grid covers Taiwan from 21.6° to 25.6°N and 119.75° to 122.25°E at approximately 2 km resolution. The dataset splits into 21,216 training hours (2019-08 through 2021-12) and 8,760 validation hours (calendar year 2022).

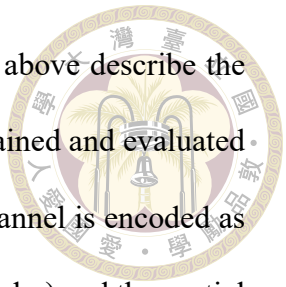
Invariants I. Shape $(2, 224, 128)$, channels land–sea mask and orography. These are broadcast across time and concatenated into the conditioning bundle.

Normalisation. Per-channel means and standard deviations are computed on the training split and stored in the Zarr metadata. The HighRes statistics are

$$\mu_{\text{HR}} = (295.98,\ -1.58,\ -2.65,\ 0.182), \qquad \sigma_{\text{HR}} = (5.61,\ 3.84,\ 5.66,\ 9.12), \tag{3.2}$$

which show that `qpepre` has a near-zero mean with standard deviation an order of magnitude larger than its mean — a direct fingerprint of its heavy-tailed, sparse distribution. Training and validation losses are computed on normalised fields; metrics in Chapter 4 are reported on denormalised fields with physical units.

doi:10.6342/NTU202601285



Cleaned variant used for the reported experiments. The shapes above describe the original store. All FlowCast-versus-diffusion runs in this thesis are trained and evaluated on a cleaned re-export of the same data, in which the precipitation channel is encoded as $\log(1+x)$ before standardisation (the `qpepre_log1p` option of the loader) and the spatial domain is centre-cropped to a U-Net-friendly 192×96 grid. The cleaned store covers 18,765 training hours (2019-08-01 through 2021-12-31) and 8,759 validation hours (calendar year 2022). The $\log(1+x)$ encoding is the single most important preprocessing choice for the precipitation channel: it compresses `qpepre`'s dynamic range from a raw standard deviation of $9.12\,\mathrm{mm\,h^{-1}}$ to roughly 0.37 in log space, which makes the standardised residual far better-conditioned for every residual head (Section 4.7). Because the spatial crop changes the field of view, RMSE/CRPS magnitudes are not directly comparable to the legacy 224×128 baseline of Section 4.9; we flag this wherever the two are tabulated together. The regression mean and all three residual heads are trained on this same cleaned, log1p-encoded store, which is what the "train on the same qpepre encoding" gotcha in Section 4.12 refers to.

3.3 Data Sources and Preprocessing Pipeline

Section 3.2 described the Zarr store consumed by the training loop. This section describes where the underlying fields come from and how the preprocessing pipeline in `data_preprocessing/` assembles them. The design follows a low-resolution guidance, high-resolution learning strategy: a coarse global reanalysis provides synoptic-scale conditioning, and a convection-permitting regional model plus a radar-derived precipitation product provides the fine-scale target.

doi:10.6342/NTU202601285

3.3.1 Raw data sources

ERA5 (low-resolution input, S_t). The ECMWF ERA5 reanalysis [6] at approximately 25 km spatial resolution supplies the synoptic-scale environmental state. From ERA5 we take four surface variables (mean sea-level pressure `mslp`, two-metre temperature `t2m`, and ten-metre winds `u10`, `v10`) and five upper-air variables (specific humidity `q`, temperature `t`, zonal wind `u`, meridional wind `v`, and geopotential height `z`) at four pressure levels $\{1000, 850, 500, 250\}$ hPa, for a total of $4 + 5 \times 4 = 24$ channels. These are the 24 LowRes channels listed in Section 3.2.

RWRF (high-resolution target, part of X_t). The Radar-assimilated Weather Research and Forecasting (RWRF) numerical model provides convection-permitting ($\approx$2 km) analyses over Taiwan. We use hourly RWRF output (`wrfout_d01_*_interp`) for the three near-surface dynamical channels: `t2m`, `u10`, `v10`. RWRF contains the fine-scale dynamic and thermodynamic structures that the complex Taiwan terrain imprints on near-surface weather, which is exactly what a regional convective-scale forecaster must learn.

QPEPRE (high-resolution target, precipitation channel). Quantitative Precipitation Estimation (QPEPRE) is a radar-based hourly-rainfall product distributed as fixed-grid text files (`qpepre_YYYYMMDDHHMM-YYYYMMDDHHMM_1_h.txt`), one per hour. Unlike the RWRF fields, which are model outputs, QPEPRE is an observational product and therefore serves as a more faithful ground truth for surface rainfall than any model-forecast precipitation field. It occupies the fourth HighRes channel, `qpepre`.

doi:10.6342/NTU202601285

Table 3.1: Raw data sources used for training. Low-resolution ERA5 provides synoptic conditioning; RWRF and QPEPRE together supply the convection-permitting target.

	ERA5	**RWRF**	**QPEPRE**
Type	Reanalysis	Numerical model	Radar observation
Resolution	≈25 km	≈2 km	≈2 km
Role	LowRes input S_t	HighRes target X_t	HighRes target X_t
Channels used	24 (4 sfc + 5 × 4 levels)	3 (`t2m`, `u10`, `v10`)	1 (`qpepre`)
Native format	NetCDF	NetCDF (`wrfout`)	ASCII text

Invariants I. Two static fields are extracted once from an RWRF file and broadcast across time: the land–sea mask `lsm` and orography `orog`. These let the network condition on terrain without re-reading static fields at every time step.

Table 3.1 summarises the three sources.

3.3.2 Preprocessing pipeline

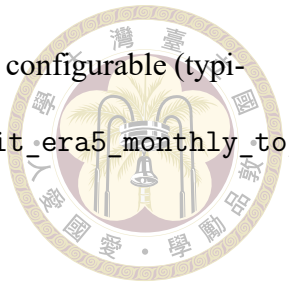
The project ships two preprocessing stages under `data_preprocessing/`:

1. `nc_month_to_day/` — split monthly ERA5 NetCDF files into single-hour files.
2. `nc_to_zarr/` — harmonise ERA5, RWRF, and QPEPRE onto a common Taiwan grid and write a StormCast-style Zarr store.

Stage 1: monthly-to-hourly split. ERA5 is distributed as monthly NetCDF files, each containing an hourly time axis. The script `split_era5_monthly_to_daily.py` walks a root directory, opens every file matching a configured pattern, decodes the `time` axis with `netCDF4.num2date`, and writes one file per hour with a canonical `{variable}_YYYYMMDDTHH.nc` filename. The output uses uniform `hours since 1970-01-01 00:00:00` time units so that downstream indexing does not have to handle per-file calendar differences. The split

doi:10.6342/NTU202601285

runs in parallel via a `multiprocessing.Pool` whose worker count is configurable (typically 20 processes on a workstation). Driver: `nc_month_to_day/split_era5_monthly_to_daily.py`; configuration: `nc_month_to_day/config.yaml`.



Stage 2: unified NetCDF-to-Zarr assembly. The main pipeline is `nc_to_zarr/nc_to_zarr_pipeline` driven by `config.yaml`. It is parameterised by four paths and a domain description:

```
era5-path, rwrf-path, qpepre-path, output-path
domain-size: [224, 128]
lon-bounds: [119.75, 122.25]
lat-bounds: [21.6, 25.6]
train-ranges / valid-ranges
```

For each requested split (train, valid, or combined), the pipeline performs the following steps.

1. **Time-stamp generation.** The configured date ranges are expanded into a dense hourly list between the start and end of each range.

2. **File indexing.** Three in-memory indices are built — one per source — that map timestamps to absolute file paths. For ERA5 this is a nested {variable → {date-key → path}} dictionary built by a regex on filenames; for RWRF it is {folder-name → path}; for QPEPRE it is {time-key → path} derived from the `qpepre_{start}-{end}_1_h.txt` filename convention. When multiprocessing is enabled the indices are serialised to a single JSON file `.shared_index.json` in the output directory so that worker processes can mmap-read the index instead of receiving a pickled copy.

doi:10.6342/NTU202601285

3. **Combined validity precomputation.** Before processing anything, the pipeline walks the timestamp list once and flags every hour for which any of the required ERA5 variables, RWRF file, or QPEPRE file is missing. Only timesteps that pass this joint check are processed, and the Zarr store carries an explicit per-timestamp `valid` mask so that downstream code can skip gaps without re-scanning the raw files.

4. **Per-timestep field extraction.**

   - ERA5: open the NetCDF, select the target time index, select the correct pressure level from the level dimension (for upper-air variables), crop to the latitude/longitude bounds, and resample to the configured 224×128 target grid.
   - RWRF: read `XLAT` and `XLONG` to cache the native latitude/longitude, find the target `Time` index from the `Times` character variable, slice each requested variable to the domain, optionally apply per-variable transforms, and resample to the target grid.
   - QPEPRE: load the text file, parse it into a {lat, lon, rain} scatter, grid it onto the native QPEPRE resolution, and resample to the 224×128 target grid. Because QPEPRE is already an observational product, we use it directly as the `qpepre` HighRes channel rather than the RWRF precipitation field.

5. **Grid harmonisation.** All three sources are resampled to a shared $H \times W = 224 \times 128$ grid covering lat $[21.6, 25.6]$°N, lon $[119.75, 122.25]$°E at approximately 2 km spacing. This guarantees that the LowRes, HighRes, and invariant arrays share a pixel-aligned coordinate system, which is essential because the diffusion network conditions on LowRes and invariants by channel-wise concatenation.

doi:10.6342/NTU202601285

6. **Streaming statistics.** As each training sample is filled into the data cube, a `StreamingStats` accumulator updates per-channel running means and standard deviations in single-pass Welford form. After the train split finishes, the stats are written to `stats/means.npy` and `stats/stds.npy` alongside the Zarr store. These are the normalisation statistics quoted in Equation 3.2.

7. **Zarr writing.** The assembled cube is written as an `xarray.Dataset` with dimensions `(time, channel, y, x)`, chunked along time and space for efficient random access, and consolidated (`zarr.consolidate_metadata`) so that the training loader can open the store with a single metadata read. Invariants are written once, without a time dimension, as a separate Zarr store under `invariants/invariants.zarr`.

Output layout. A successful run produces the tree

```
<output-path>/
  LowRes/
    stormcast_test_train.zarr      # (T_train, 24, 224, 128)
    stormcast_test_valid.zarr      # (T_valid, 24, 224, 128)
    stats/{means.npy,stds.npy,channels.txt}
  HighRes/
    stormcast_test_train.zarr      # (T_train, 4,  224, 128)
    stormcast_test_valid.zarr      # (T_valid, 4,  224, 128)
    stats/{means.npy,stds.npy,channels.txt}
  invariants/
    invariants.zarr                # (2, 224, 128)
```


doi:10.6342/NTU202601285

which matches the layout documented in `zarr_metadata.json` and consumed by `data_loader_rwrf_era`. For the experiment used in this thesis the pipeline produces $T_{\text{train}} = 21{,}216$ hourly samples (2019-08-01 through 2021-12-31) and $T_{\text{valid}} = 8{,}760$ hourly samples (calendar year 2022).

Why this matters here. Three features of this pipeline are load-bearing for the work in this thesis:

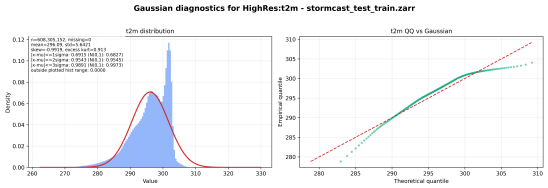
- Pixel-aligned multi-source grids. The diffusion baseline and FlowCast both consume conditioning and target on the same grid, so residual losses can be computed pixel-wise without any on-the-fly re-gridding.
- Explicit `valid` mask. Roughly 10–15 % of hours fail the joint-validity check (usually a missing QPEPRE file). Both loaders and metrics respect the mask, so downstream scores are never contaminated by NaN-filled cells.
- Frozen training-split statistics. Because the Zarr store commits to a single pair of $(\mu_{\text{HR}}, \sigma_{\text{HR}})$ at write time, all residual heads see identical normalised inputs — a prerequisite for the σ_{data}-matching discussion in Section 3.4.

3.3.3 Target Distributions: Sparsity, Gaussianity, and the Diffusion Prior

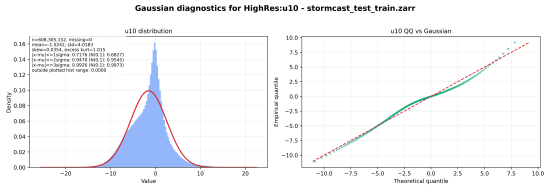
Diffusion models place a Gaussian prior on their target: the forward process is $x_\sigma = x_0 + \sigma\,\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$, and the denoiser is trained under the implicit assumption that the marginals of x_0 are reasonably well-behaved relative to that noise scale. How closely each of the four HighRes channels actually conforms to this assumption directly shapes how the

doi:10.6342/NTU202601285

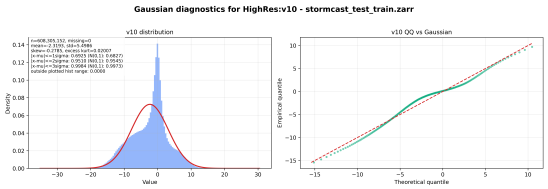
generative residual head must be regularised. Figure 3.1 shows per-channel histograms (with a fitted Gaussian overlay) and Gaussian QQ plots computed over the full training split ($n = 6.08 \times 10^8$ pixels per channel). The streaming statistics are summarised in Table 3.2.



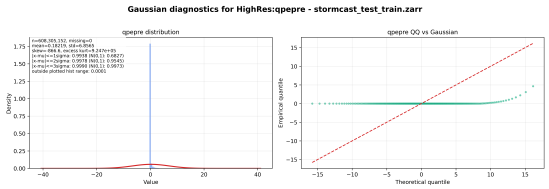
(a) `t2m` — 2 m temperature (K).



(b) `u10` — 10 m zonal wind ($\mathrm{m\,s^{-1}}$).



(c) `v10` — 10 m meridional wind ($\mathrm{m\,s^{-1}}$).



(d) `qpepre` — radar precipitation ($\mathrm{mm\,h^{-1}}$).

Figure 3.1: Per-channel distribution and Gaussian QQ plots for the four HighRes target channels over the 2.5-year training split. Left subpanels: empirical density (red) with a Gaussian fit of matched (μ, σ) (blue). Right subpanels: empirical quantiles against theoretical Gaussian quantiles; the red dashed line is $y = x$ for a perfect Gaussian. Generated by `data_preprocessing/inspect_tools/plot_high_res_vars.py`.

Table 3.2: Summary statistics of the four HighRes target channels over the training split. A standard Gaussian has skew $= 0$, excess kurt $= 0$, and concentrates $68.3/95.5/99.7\,\%$ of its mass within $1/2/3\sigma$.

Channel	Mean	Std	Skewness	Excess kurt.	% in 1σ	% in 3σ
`t2m`	296.09	5.64	-0.99	0.91	69.2	98.9
`u10`	-1.52	4.02	0.04	1.01	71.8	99.3
`v10`	-2.32	5.50	-0.28	0.02	69.3	99.8
`qpepre`	0.18	6.86	-866.6	9.25×10^5	99.4	99.9

The three near-surface state channels are approximately Gaussian. `u10` is the closest: almost zero skewness, mild leptokurtosis, and a QQ plot that tracks the diagonal out to roughly $\pm 4\sigma$ before fanning into the heavy tails associated with gust events. `v10` is almost symmetric with essentially zero excess kurtosis. `t2m` is the least symmetric of the three: left-skewed (skew $= -0.99$) with a secondary mode around 280 K corresponding to winter cold surges over northern Taiwan and the surrounding sea. For all three channels the

doi:10.6342/NTU202601285

empirical $1\sigma/2\sigma/3\sigma$ coverage agrees with the Gaussian reference to within a percentage point, which is exactly the regime for which the EDM noise schedule with $\sigma_{\text{data}} = 0.5$ was designed: the generative residual head sees each channel as a mild deformation of a unit-variance Gaussian after normalisation, and the standard ℓ_2 objective is a reasonable fit.

`qpepre` is a different object entirely. Its marginal is dominated by an atom at zero — the overwhelming majority of pixel-hours are dry — with a long, very sparse tail of heavy rainfall events. The raw statistics make this quantitative: the coverage inside $\pm 1\sigma$ is $99.4\,\%$ rather than the Gaussian $68.3\,\%$, which is the signature of a near-delta distribution whose standard deviation is inflated entirely by the tail. The empirical moments reported in Table 3.2 (skewness ≈ -867, excess kurtosis $\approx 10^6$) are formally enormous because the raw `qpepre` field uses a -9999 sentinel for missing observations that is only filtered at the next pipeline stage; the physical tail above zero is still extreme (excursions beyond $100\,\text{mm}\,\text{h}^{-1}$ occur during typhoon landfalls), but the numbers headline the practical message: the `qpepre` histogram looks nothing like a Gaussian, and the QQ plot sits on a flat shelf near zero that peels almost vertically at the upper end.

This mismatch has three consequences for residual generative modelling that recur throughout the rest of this thesis:

1. Score matching on raw `qpepre` is ill-posed at small σ. When the target distribution is almost a point mass at zero, the diffusion score $\nabla_x \log p_\sigma(x)$ diverges as $\sigma \to \sigma_{\text{min}}$ — one reason the EDM baseline needs many small-σ steps to render precipitation. Learning the residual relative to the regression mean M_t partially absorbs this, but the residual inherits the same atom-plus-tail shape, and the $\log(1+x)$ encoding of

doi:10.6342/NTU202601285

Section 3.7 is what finally compresses its dynamic range. The straight-line FlowCast velocity field has no score term and is correspondingly better-behaved on this channel.

2. Pixel-wise losses undersell the tail. Because dry pixels vastly outnumber rainy ones, an ℓ_2 loss converges to a "slightly wet everywhere" solution that scores well on RMSE but destroys the convective structure we actually care about. This is the motivation for the radial log-PSD spectral regulariser applied specifically to `qpepre` in the FlowCast loss (Section 3.7), and for the precipitation-specific metrics (CSI, FSS, HSS, frequency-of-exceedance at fixed rainfall thresholds) reported in Chapter 4.

3. The channel budget matters. Because only one of the four output channels is pathological, per-channel loss weights β_k are a natural knob: the three near-Gaussian fields can be trained with a standard denoising objective while `qpepre` receives heavier regularisation and, if needed, an asinh or $\log(1+x)$ transform to compress its dynamic range before the noise is added. Having only four target channels (versus the 99 of global StormCast) makes such per-channel bespoke treatment computationally trivial.

In short, Figure 3.1 is the quantitative justification for every "qpepre is the hard channel" design decision that follows: three channels live comfortably inside the Gaussian prior that diffusion assumes, and one does not.

3.4 Regression Mean and the EDM Diffusion Baseline

The reference model is the two-stage StormCast described in Section 2.3, with the EDM hyperparameters of Section 2.6. It supplies both the frozen regression mean that

doi:10.6342/NTU202601285

FlowCast reuses unchanged and the diffusion residual that FlowCast is compared against.

- **Regression F_ξ.** A `StormCastUNet` taking (X_{t-1}, S_t, I) as input and predicting M_t. The checkpoint is used without modification or further training, and is shared by the diffusion baseline, FlowCast, and MeanFlow so that all three residual heads model a residual relative to the same mean.

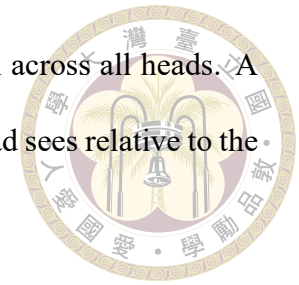
- **Diffusion baseline D_ψ.** An EDMPrecond SongUNet trained to denoise the residual $R_t = X_t - M_t$ conditioned on $c_t = \text{concat}(X_{t-1}, S_t, M_t, I)$. EDM hyperparameters: $\sigma_{\text{min}} = 0.002$, $\sigma_{\text{max}} = 80$, $\sigma_{\text{data}} = 0.5$, $\rho = 7$; sampling is the Heun integrator of Section 2.6. The baseline is trained from scratch on the same cleaned dataset as FlowCast and evaluated at the same $\approx$ 2 M-training-sample budget (the nearest saved checkpoint; see Section 4.1).

Both networks are instantiated through the factory `get_preconditioned_architecture` in `stormcast/utils/nn.py`, which selects the EDM or FlowCast preconditioning wrapper by name. The conditioning bundle is assembled by `build_network_condition_and_target`, which handles the channel-wise concatenation and the one-hot time-offset encoding used in the rollout.

Why the baseline matters for a fair comparison. The value of the comparison in Chapter 4 rests on holding everything except the generative trajectory fixed. The diffusion baseline, FlowCast, and MeanFlow share the dataset, the frozen F_ξ, the SongUNet backbone (`channel_mult` $= [1, 2, 2, 2, 2]$), and the data scale $\sigma_{\text{data}} = 0.5$ that anchors the residual statistics. The EDM data scale matters here for a subtle reason: both flow heads standardise their residual by exactly the same σ_{data} (Sections 3.5 and 3.6), so the per-channel loss

doi:10.6342/NTU202601285

scaling and the regression-mean anchoring are numerically identical across all heads. A mismatched σ_{data} would silently shift the residual distribution one head sees relative to the others and make any quality difference uninterpretable.



3.5 The FlowCast Baseline (Conditional Flow Matching)

FlowCast is the straight-line flow-matching baseline of this thesis, and we develop it first because the MeanFlow method of Section 3.6 is defined relative to it — MeanFlow reuses FlowCast's conditioning, standardised-residual pipeline, backbone, and losses unchanged, and its objective reduces exactly to FlowCast's at zero interval length. FlowCast replaces the diffusion baseline's curved PF-ODE with a simulation-free straight-line flow: a single network is trained from scratch to regress the target vector field $u_t = x_1 - x_0$ of Independent CFM (Section 2.8), and sampling is a handful of Euler steps. FlowCast is not a distillation method — it never queries the EDM diffusion model and uses no teacher at all — it is a one-training-run residual head that learns its velocity field directly from data. The same StormCast residual decomposition is preserved: the deterministic regression mean M_t is produced by the frozen F_ξ, and FlowCast learns only the residual $R_t = X_t - M_t$ conditioned on a bundle c_t assembled by `build_network_condition_and_target` in `stormcast/utils/nn.py`.

Conditioning bundle. FlowCast and the EDM diffusion baseline are conditioned on exactly the same channel bundle. Both share the config setting

```
regression_conditions: ["state", "background", "invariant"]
 diffusion_conditions: ["state", "regression", "invariant"]
```


doi:10.6342/NTU202601285

which is identical in `stormcast/config/model/flowcast.yaml` and `stormcast/config/model/diffusion.yaml`: the frozen regression network F_ξ is fed the full (X_{t-1}, S_t, I), but the generative network — diffusion or flow — is conditioned only on

$$c_t = \text{concat}\big(X_{t-1},\ M_t,\ I\big). \tag{3.3}$$

The low-resolution background S_t is deliberately omitted from the direct conditioning path of every head (a deviation from the original StormCast, which concatenates S_t alongside M_t); the synoptic-scale signal in S_t enters the residual only through the regression mean $M_t = F_\xi(X_{t-1}, S_t, I)$, which already summarises what the residual needs to know about it. Trimming S_t removes 24 input channels at no measured cost, and — crucially for a controlled comparison — the trim is applied identically to all heads (the MeanFlow config carries the same two condition lists verbatim). Keeping `"regression"` in the diffusion-conditions list anchors every residual to the same M_t, so the only thing that differs between the EDM, FlowCast, and MeanFlow residuals is the generative trajectory, not what they condition on. This is what makes the head-to-head comparison of Chapter 4 fair.

Standardised-residual pipeline. FlowCast operates on the residual in a standardised space. The loss (`stormcast/utils/flowcast_loss.py`) divides the raw residual by $\sigma_{\text{data}} = 0.5$ before building the flow trajectory, and the sampler (`stormcast/utils/nn.py:flowcast_model_forward`) multiplies the integrated state by σ_{data} on output. Keeping σ_{data} matched to the EDM baseline means the regression-mean anchoring statistics and the per-channel loss scaling are identical across the residual heads. Concretely, with $x_1 = R_t/\sigma_{\text{data}}$ and $x_0 \sim \mathcal{N}(0, I)$, the I-CFM interpolated state is

$$x_t = (1-t)\,x_0 + t\,x_1 + \sigma_{\text{path}}\,\eta, \qquad t \sim \mathcal{U}(t_\varepsilon, 1-t_\varepsilon),\ \eta \sim \mathcal{N}(0, I), \tag{3.4}$$

doi:10.6342/NTU202601285

with $\sigma_{\text{path}} = 0.01$ from Tong et al. [28] and $t_\varepsilon = 10^{-5}$ clamping the endpoints. The small $\sigma_{\text{path}}\eta$ perturbation thickens the otherwise-Dirac conditional path $p_t(x_t \mid x_0, x_1)$ into a narrow Gaussian tube, separating it from the deterministic-interpolant ($\sigma_{\text{path}} = 0$) limit. The "Independent" in I-CFM refers instead to the coupling: x_0 and x_1 are drawn independently [28], as opposed to the minibatch optimal-transport coupling of OT-CFM.

Velocity predictor. The FlowCast network is the `FlowCastPrecond` module at `stormcast/utils/flowcast_precond.py`, instantiated through the factory call `get_preconditioned_architectur…`) at `stormcast/utils/nn.py`. The factory wraps a SongUNet with `channel_mult=[1, 2, 2, 2, 2]`, identical to the EDM baseline's backbone. Unlike `EDMPrecond` there are no $c_{\text{skip}}/c_{\text{in}}/c_{\text{out}}$ scalings — the network output is the velocity estimate:

$$v_\theta(x_t, t, c_t) = \text{SongUNet}\big(\text{concat}[x_t, c_t],\ t \cdot \tau\big), \tag{3.5}$$

where $\tau = 1000$ (the `time_scale` field in `stormcast/config/model/flowcast.yaml`) rescales the flow time $t \in [0, 1]$ before it enters the SongUNet's positional-embedding layer. The SongUNet positional embedding was designed for diffusion-timestep-valued inputs with dynamic range of order 10^2–10^3; feeding raw $t \in [0, 1]$ would collapse the embedding to a nearly-constant signal. $\tau = 1000$ restores enough dynamic range for the embedding to carry useful time information without changing the SongUNet itself, so no architectural changes from the diffusion baseline's backbone are needed.

Loss. The training loss is the standardised-space I-CFM regression

$$\mathcal{L}_{\text{FC}} = \mathbb{E}\big[\, \|v_\theta(x_t, t, c_t) - (x_1 - x_0)\|^2_{2,\beta} \,\big] + \lambda_{\text{spec}}\, \mathcal{L}_{\text{spec}}\big(\hat{x}_1,\ x_1\big), \tag{3.6}$$

doi:10.6342/NTU202601285

Algorithm 1 FlowCast training step (I-CFM on the standardised residual)

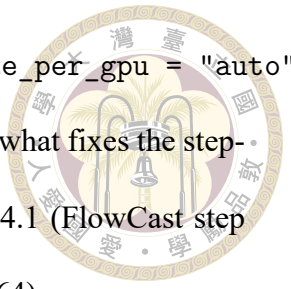
Require: frozen F_ξ, FlowCast velocity field v_θ, schedule $(\sigma_{\text{data}}, \sigma_{\text{path}}, t_\varepsilon)$, channel weights β, spectral weight λ_{spec}

sample mini-batch $\{(X_{t-1}, S_t, X_t)\}$; form $M_t \leftarrow F_\xi(X_{t-1}, S_t, I)$, $R_t \leftarrow X_t - M_t$
build conditioning $c_t \leftarrow \text{concat}(X_{t-1}, M_t, I)$ ▷ note: S_t reaches v_θ only via M_t
standardise: $x_1 \leftarrow R_t / \sigma_{\text{data}}$
sample $x_0 \sim \mathcal{N}(0, I)$, $t \sim \mathcal{U}(t_\varepsilon, 1 - t_\varepsilon)$, $\eta \sim \mathcal{N}(0, I)$
build interpolant: $x_t \leftarrow (1 - t)\, x_0 + t\, x_1 + \sigma_{\text{path}}\, \eta$
target field: $u_t \leftarrow x_1 - x_0$
predict: $\hat{v} \leftarrow v_\theta(x_t, t, c_t)$
pointwise: $\mathcal{L}_{\text{pw}} \leftarrow \|\hat{v} - u_t\|^2_{2,\beta}$
implied clean: $\hat{x}_1 \leftarrow x_t + (1 - t)\, \hat{v}$
spectral: $\mathcal{L}_{\text{spec}} \leftarrow \frac{1}{|\mathcal{K}|} \sum_\kappa \left| \log P_{\hat{x}_1^{\text{qpepre}}}(\kappa) - \log P_{x_1^{\text{qpepre}}}(\kappa) \right|$
$\mathcal{L} \leftarrow \mathcal{L}_{\text{pw}} + \lambda_{\text{spec}}\, \mathcal{L}_{\text{spec}}$
AdamW step on θ; EMA update $\theta^- \leftarrow 0.999\, \theta^- + 0.001\, \theta$

where $\|\cdot\|_{2,\beta}$ is the per-channel-weighted L2 norm from Section 3.7 with $\beta = (1, 1, 1, 2)$ (the `channel_weights` field), $\hat{x}_1 = x_t + (1 - t)\, v_\theta$ is the flow's implied clean-residual estimate at the current time, and $\mathcal{L}_{\text{spec}}$ is the radial log-PSD regulariser of Section 3.7 with `spectral_channels` $= \{\texttt{qpepre}\}$ and $\lambda_{\text{spec}} = 0.1$. Operating the spectral term on the implied-clean field $\hat{x}_1$ rather than on the raw velocity keeps the spectrum interpretable. Algorithm 1 summarises the per-step update.

Optimisation. Following the FlowCast paper we use the AdamW optimiser with $\beta_{\text{Adam}} = (0.9, 0.999)$, learning rate 5×10^{-4}, and weight decay 10^{-4} on all parameters. The learning-rate schedule is a 4,000-step linear warmup (~1% of the training budget) into a cosine decay to $0.01 \times 5 \times 10^{-4} = 5 \times 10^{-6}$ over the remaining steps, implemented inline in `trainer_flowcast.py`. Gradients are globally clipped at 1.0, and per-parameter NaN gradients are sanitised with `torch.nan_to_num` (a safety net borrowed from the EDM trainer). Training runs for a total of 400,000 optimiser steps at a global batch size of 96 (set by the launcher `stormcast/zettabyte_scripts/train_flowcast.sh`, which overrides the config default of 64), split across the available GPUs through PyTorch's

doi:10.6342/NTU202601285



`DistributedDataParallel`; the per-GPU micro-batch is `batch_size_per_gpu = "auto"`, which the trainer resolves to $96/W$ for W workers. This batch size is what fixes the step-to-sample conversion of the matched-budget comparison in Section 4.1 (FlowCast step $20{,}833 \approx 2$ M samples at batch 96, versus EDM step $31{,}250$ at batch 64).

EMA shadow network. A second `FlowCastPrecond` instance is kept as an exponential-moving-average shadow of the online weights, with a fixed decay of 0.999. The shadow is updated after every optimiser step by `ExponentialMovingAverage.update` and copied into the shadow network by `apply_shadow` (`stormcast/utils/ema.py`). A single fixed decay suffices throughout training because the FlowCast target — the straight-line velocity field — is stationary; there is no shrinking time discretisation for the decay to track. The shadow weights, not the online weights, are used at validation and inference time, and are saved to `ema_state.pt` alongside each regular checkpoint. Loading the raw `checkpoint_*.pt` instead of `ema_state.pt` yields visibly noisier samples.

Sampling. Generation integrates the learned ODE $\dot{x}_t = v_\theta(x_t, t, c_t)$ from $t = 0$ to $t = 1$ with a fixed-step solver. The implementation `flowcast_model_forward` in `stormcast/utils/nn.py` exposes two choices. The default is explicit Euler with K steps:

$$x_{t+\Delta t} = x_t + \Delta t \; v_\theta(x_t, t, c_t), \qquad \Delta t = 1/K, \qquad x_{t=0} \sim \mathcal{N}(0, I), \tag{3.7}$$

which costs K NFEs. The alternative `solver="midpoint"` is a single-stage Runge–Kutta second-order scheme:

$$v_{1/2} = v_\theta(x_t, t, c_t), \quad \tilde{x} = x_t + \tfrac{1}{2}\Delta t \, v_{1/2}, \quad x_{t+\Delta t} = x_t + \Delta t \, v_\theta(\tilde{x}, t + \tfrac{1}{2}\Delta t, c_t), \tag{3.8}$$

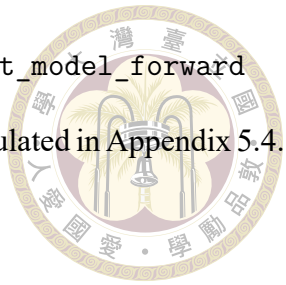
doi:10.6342/NTU202601285

which costs $2K$ NFEs but is globally second-order accurate ($O(\Delta t^2)$ global error, versus explicit Euler's $O(\Delta t)$). The sampler finishes by de-standardising: $\hat{R}_t = \sigma_{\text{data}}\, x_{t=1}$ and $\hat{X}_t = M_t + \hat{R}_t$. Validation inside the training loop uses $K = \texttt{valid\_num\_steps} = 10$ Euler steps, while the main experiments report the scoreboard at $K \in \{10, 15, 20\}$ and sweep K from 1 to 50, placing each cost point against the EDM Heun baseline on the NFE/wall-clock–quality trade-off of Chapter 4.

Why a straight-line flow is the right residual head. Three properties of I-CFM make FlowCast attractive against the diffusion baseline rather than merely a faster imitation of it. First, the straight-line probability path is by construction what a one-step deterministic generator is trying to approximate, so the FlowCast $K = 1$ and $K = 2$ numbers are a strong reference point for the best a few-step ODE sampler trained on the same data can do — and the gap to the $K = 10$ sampler is small (Chapter 4), unlike the steep degradation a diffusion sampler suffers below ≈ 20 NFE. Second, the training is teacher-free and single-stage, so FlowCast avoids the second training run, the sampler/preconditioning matching, and the EMA-coupling subtleties that a distillation of the diffusion baseline would require. Third, and specific to precipitation, Ribeiro and Pucer [18] report that I-CFM is noticeably more stable than score matching on sparse heavy-tailed fields — a claim directly relevant to the `qpepre` channel and one we re-test on the Taiwan domain.

Implementation pointers. Entry point `stormcast/train_flowcast.py`; training loop `stormcast/utils/trainer_flowcast.py` (`flowcast_training_loop`); loss `stormcast/utils/flowcast_loss.py` (`FlowCastLoss`); preconditioner `stormcast/utils/flowcast_precond.py` (`FlowCastPrecond`); EMA helper `stormcast/utils/ema.py` (`ExponentialMovingAverage`); configs `stormcast/config/flowcast.yaml`, `stormcast/config/model/flowcast.yaml`,

doi:10.6342/NTU202601285



`stormcast/config/training/flowcast.yaml`; sampler `flowcast_model_forward` in `stormcast/utils/nn.py`. The numerical hyperparameters are tabulated in Appendix 5.4.

3.6 The MeanFlow Residual (Average-Velocity Flow Matching)

FlowCast still pays for its samples with $K \approx 10$ Euler steps, because what it learns is the instantaneous velocity of a marginal field whose trajectories are curved (Section 2.8): every Euler step incurs truncation error against that curvature, and the error only vanishes as K grows. MeanFlow [5] removes the integration from the sampler instead of refining it. It trains the network to predict the average velocity of the flow over a whole interval $[r, t]$, so that a single evaluation transports the state across the interval exactly — no ODE solver, and therefore no discretisation error to amortise. We adapt MeanFlow to the StormCast residual pipeline as the generative residual head of this thesis: it reuses the conditioning bundle c_t, the standardised-residual convention, the SongUNet backbone, the per-channel loss weights, and the spectral regulariser of the FlowCast baseline unchanged, and differs from it only in the training objective and the (correspondingly cheaper) sampler. MeanFlow is likewise teacher-free and trained from scratch in a single run.

Average velocity. Work in the standardised residual space of Section 3.5: $x_1 = R_t/\sigma_{\text{data}}$ is the data endpoint, $x_0 \sim \mathcal{N}(0, I)$ the prior, and the linear path $z_s = (1-s)\,x_0 + s\,x_1$ runs from noise at $s = 0$ to data at $s = 1$ with conditional velocity $v = x_1 - x_0$. (MeanFlow uses the pure interpolant, $\sigma_{\text{path}} = 0$: the identity below differentiates along the exact trajectory, so the Gaussian thickening used by FlowCast is dropped.) For two times $r \le t$ the average

doi:10.6342/NTU202601285

velocity is the displacement over $[r, t]$ divided by the interval length,

$$u(z_r, r, t) \triangleq \frac{1}{t-r}\int_r^t v(z_\tau, \tau)\, d\tau, \qquad \lim_{t\to r} u(z_r, r, t) = v(z_r, r), \tag{3.9}$$

where $v(\cdot, \tau)$ is the marginal velocity field of Section 2.8. Like v, the field u is a property of the underlying flow, not of any network. Its defining property is that it makes transport across an interval a single algebraic step,

$$z_t = z_r + (t-r)\, u(z_r, r, t), \tag{3.10}$$

exactly, with no solver: if u is known, one evaluation at $(r, t) = (0, 1)$ maps noise to data. The cost has been moved from inference into training: Eq. (3.9) contains an intractable path integral, so u cannot be regressed directly.

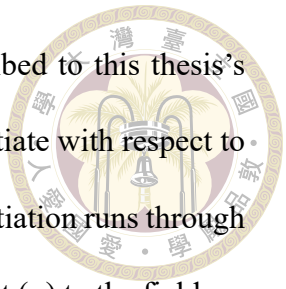
The MeanFlow identity. The integral is eliminated by differentiation rather than quadrature. Rewrite Eq. (3.9) as $(t-r)\, u(z_r, r, t) = \int_r^t v\, d\tau$ and differentiate with respect to the state time r, holding t fixed, with z_r moving along the trajectory ($dz_r/dr = v(z_r, r)$, $dt/dr = 0$). The product rule on the left and the fundamental theorem of calculus on the right give $-u + (t-r)\,\frac{d}{dr}u = -v(z_r, r)$, i.e.

$$\boxed{u(z_r, r, t) \;=\; v(z_r, r) \;+\; (t-r)\,\frac{d}{dr}\,u(z_r, r, t),} \tag{3.11}$$

with the total derivative expanding into partials along the trajectory tangent,

$$\frac{d}{dr}\,u(z_r, r, t) \;=\; v(z_r, r)\,\partial_z u \;+\; \partial_r u. \tag{3.12}$$

doi:10.6342/NTU202601285



Equation (3.11) is the MeanFlow identity of Geng et al. [5] transcribed to this thesis's noise-to-data convention: Geng et al. place data at $s = 0$ and differentiate with respect to t, obtaining $u = v - (t - r)\,\frac{d}{dt}u$; here data sits at $s = 1$, so the differentiation runs through the lower limit and the sign flips. The identity relates the field we want (u) to the field we can sample (v) using only quantities available at a single point of a single trajectory — the path integral is gone. At $t = r$ it collapses to $u = v$, recovering plain flow matching as a boundary condition.

Training objective. A network $u_\theta(z_r, r, t, c_t)$ is encouraged to satisfy Eq. (3.11) by regressing onto the identity's right-hand side with the network's own derivative bootstrapped in:

$$\mathcal{L}_{\text{MF}} = \mathbb{E}\Big[\,\text{sg}(w)\,\big\|u_\theta(z_r, r, t, c_t) - \text{sg}(u_{\text{tgt}})\big\|_{2,\beta}^2\,\Big] + \lambda_{\text{spec}}\,\mathcal{L}_{\text{spec}}, \qquad u_{\text{tgt}} = v + (t-r)\big(v\,\partial_z u_\theta + \partial_r u_\theta\big), \tag{3.13}$$

where $v = x_1 - x_0$ is the conditional I-CFM velocity standing in for the marginal field (the same substitution FlowCast makes, valid in expectation), sg is stop-gradient, and $\|\cdot\|_{2,\beta}$ is the channel-weighted norm of Section 3.7 with the same $\beta = (1, 1, 1, 2)$ as FlowCast. Three implementation choices follow Geng et al. [5]:

- **JVP bootstrap with stop-gradient.** The derivative in u_{tgt} is evaluated by a single forward-mode Jacobian–vector product through u_θ with tangent $(v, 1, 0)$ for (z, r, t) — one extra forward-mode pass, no higher-order gradients, because the whole target is stop-gradient (`torch.func.jvp` under `no_grad`). A network achieving zero loss satisfies the identity, and hence the definition Eq. (3.9).

- **Two-time sampling with partial collapse.** Each sample draws two i.i.d. times from

doi:10.6342/NTU202601285

$\mathcal{U}(t_\varepsilon, 1 - t_\varepsilon)$ and sorts them into $r \leq t$. Only a fraction $\rho_{\text{MF}} = 0.25$ of the batch keeps $r < t$ and trains on the bootstrapped target; the remaining 75% collapses $t \leftarrow r$, for which $u_{\text{tgt}} = v$ and the objective reduces exactly to the FlowCast I-CFM regression. The collapsed subset pins the boundary condition $u(\cdot, r, r) = v$ that the $r < t$ subset propagates outward; Geng et al. [5] find $\rho_{\text{MF}} = 0.25$ optimal, and the JVP overhead is confined to a quarter of the batch. The prediction pass itself runs once over the full batch, which keeps distributed training simple.

- **Adaptive weighting.** The per-sample weight $w = (\|\Delta\|_{2,\beta}^2 + \varepsilon_w)^{-p}$ with $p = 1$, $\varepsilon_w = 10^{-3}$, applied with stop-gradient, implements the adapted-loss-weight scheme that Geng et al. [5] found necessary for stable bootstrapped training (their Tab. 1e); $p = 0$ recovers plain weighted MSE.

The spectral term $\mathcal{L}_{\text{spec}}$ is the radial log-PSD regulariser of Section 3.7, applied — as in FlowCast — on the implied clean residual $\hat{x}_1 = z_r + (1 - r)\, u_\theta$, but restricted to the collapsed ($t = r$) subset of the batch where that extrapolation is the flow-matching one and the construction stays directly comparable to Eq. (3.6). Algorithm 2 summarises the update.

Network parameterisation: two times into one backbone. The network is the `MeanFlowPrecond` module (`stormcast/utils/meanflow_precond.py`), wrapping the same SongUNet backbone (`channel_mult` $= [1, 2, 2, 2, 2]$) as the EDM and FlowCast heads. The only new requirement is injecting two times instead of one. The state time r takes FlowCast's route: it is scaled by $\tau = 1000$ and fed through the SongUNet's diffusion-time positional embedding (Eq. (3.5)). The interval length $\Delta = t - r$ enters through the SongUNet's otherwise-unused augmentation-label pathway as a fixed 32-dimensional sin / cos Fourier feature

doi:10.6342/NTU202601285

Algorithm 2 MeanFlow training step (average-velocity matching on the standardised residual)

Require: frozen F_ξ, average-velocity network u_θ, scale σ_{data}, clamp t_ε, MeanFlow ratio ρ_{MF}, adaptive (p, ε_w), channel weights β, spectral weight λ_{spec}

sample mini-batch $\{(X_{t-1}, S_t, X_t)\}$; form $M_t \leftarrow F_\xi(X_{t-1}, S_t, I)$, $R_t \leftarrow X_t - M_t$
build conditioning $c_t \leftarrow \text{concat}(X_{t-1}, M_t, I)$; standardise $x_1 \leftarrow R_t / \sigma_{\text{data}}$
sample $x_0 \sim \mathcal{N}(0, I)$; set $v \leftarrow x_1 - x_0$
sample (r, t): two i.i.d. $\mathcal{U}(t_\varepsilon, 1 - t_\varepsilon)$ draws, sorted so $r \le t$; with prob. $1 - \rho_{\text{MF}}$ collapse $t \leftarrow r$
interpolate state: $z_r \leftarrow (1 - r)\, x_0 + r\, x_1$
if $r < t$: $\frac{d}{dr} u \leftarrow \text{jvp}\big(u_\theta(\cdot, \cdot, \cdot, c_t);\ (z_r, r, t);\ (v, 1, 0)\big)$ ▷ forward-mode, no grad
    $u_{\text{tgt}} \leftarrow v + (t - r)\, \frac{d}{dr} u$ **else**: $u_{\text{tgt}} \leftarrow v$
predict (single forward over the full batch): $\hat{u} \leftarrow u_\theta(z_r, r, t, c_t)$
per-sample error: $\Delta^2 \leftarrow \|\hat{u} - \text{sg}(u_{\text{tgt}})\|^2_{2,\beta}$; weight $w \leftarrow (\Delta^2 + \varepsilon_w)^{-p}$
pointwise: $\mathcal{L}_{\text{pw}} \leftarrow \text{mean}\big(\text{sg}(w)\, \Delta^2\big)$
spectral (collapsed subset only): $\hat{x}_1 \leftarrow z_r + (1 - r)\, \hat{u}$; $\mathcal{L}_{\text{spec}}$ per Section 3.7
$\mathcal{L} \leftarrow \mathcal{L}_{\text{pw}} + \lambda_{\text{spec}}\, \mathcal{L}_{\text{spec}}$
AdamW step on θ; EMA update $\theta^- \leftarrow 0.999\, \theta^- + 0.001\, \theta$

vector with log-spaced frequencies in $[1, 10^3]$:

$$u_\theta(z, r, t, c_t) = \text{SongUNet}\big(\text{concat}[z, c_t],\ r \cdot \tau,\ \text{FF}(t - r)\big). \tag{3.14}$$

This is the (r, Δ) variant of the two-time embeddings ablated by Geng et al. [5], who find time-plus-interval conditioning the most effective; the added parameters (one linear map from the 32 Fourier features) are negligible against the 78.7 M backbone. At $\Delta = 0$ the parameterisation reduces to FlowCast's up to a constant augmentation vector, which is what makes the per-parameter comparison between the two students clean.

Sampling. Generation replaces numerical integration with Eq. (3.10) applied over K equal segments of $[0, 1]$ (`meanflow_model_forward` in `stormcast/utils/nn.py`):

$$z_{(i+1)/K} = z_{i/K} + \tfrac{1}{K}\, u_\theta\big(z_{i/K},\ \tfrac{i}{K},\ \tfrac{i+1}{K},\ c_t\big), \qquad i = 0, \ldots, K - 1, \qquad z_0 \sim \mathcal{N}(0, I), \tag{3.15}$$

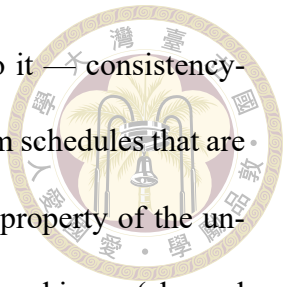
doi:10.6342/NTU202601285

at exactly one NFE per segment. $K = 1$ is the pure one-step sampler $\hat{x}_1 = x_0 + u_\theta(x_0, 0, 1, c_t)$; $K = 2$ trades one extra evaluation for a mid-trajectory correction and is the validation default. Unlike the FlowCast Euler sampler, increasing K here does not reduce a truncation error — each segment is already exact under a perfect u — it instead shortens the intervals over which the learned u_θ must be accurate. The sampler finishes identically to FlowCast: $\hat{R}_t = \sigma_{\text{data}}\, x_{t=1}$ and $\hat{X}_t = M_t + \hat{R}_t$.

Optimisation and checkpoints. Everything FlowCast-side is kept: AdamW with $\beta_{\text{Adam}} = (0.9, 0.999)$, learning rate 5×10^{-4}, weight decay 10^{-4}, a 4,000-step linear warmup into cosine decay, global gradient clipping at 1.0, and a fixed-decay 0.999 EMA shadow whose weights are saved to `ema_state.pt` and used by the in-training validation sampler. The production run trains on 7 GPUs at a global batch of 112; its step-to-sample conversion places the Chapter 4 checkpoint (step 20,000, ≈ 2.24 M samples) just above the ≈ 2 M-sample anchor shared by the EDM and FlowCast checkpoints — within 12%, which is small against the snapshot noise quantified in Section 4.1. In the head-to-head harness the FlowCast and MeanFlow students are loaded from their online checkpoints in exactly the same way, so the A/B comparison is symmetric end to end.

Why average velocity for this residual. First, it attacks the only cost FlowCast leaves on the table. The straight-line path made each Euler step cheap to take; MeanFlow removes the need to take many of them, compressing the residual head to 1–2 NFE per forecast hour — the regime where large ensembles between radar volumes stop being a budget question. Second, it is the right shape for a controlled study: because the $t = r$ subset of the objective is FlowCast's objective, MeanFlow is a strict generalisation trained under identical hyperparameters, and any skill difference in Chapter 4 is attributable to the

doi:10.6342/NTU202601285



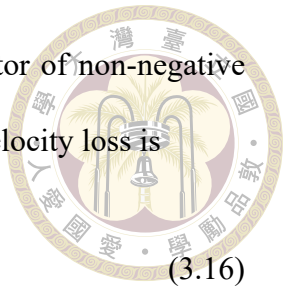
average-velocity bootstrap alone. Third, the principled alternative to it — consistency-style models — impose network-behaviour constraints with curriculum schedules that are notoriously unstable on heavy-tailed targets; MeanFlow's target is a property of the underlying field, needs no curriculum, and trains with the same loss machinery (channel weights, spectral term) already validated on `qpepre`.

Implementation pointers. Entry point `stormcast/train_meanflow.py`; training loop `stormcast/utils/trainer_meanflow.py`; loss `stormcast/utils/meanflow_loss.py` (`MeanFlowLoss`); preconditioner `stormcast/utils/meanflow_precond.py` (`MeanFlowPrecond`); configs `stormcast/config/meanflow.yaml`, `stormcast/config/model/meanflow.yaml`, `stormcast/config/training/meanflow.yaml`; sampler `meanflow_model_forward` in `stormcast/utils/nn.py`. The numerical hyperparameters are tabulated in Appendix 5.4.

3.7 Loss Design for the Precipitation Channel

The four-channel RWRF target makes per-channel loss engineering cheap, and we use this budget to attack the hardest channel, `qpepre`, from two directions. Both terms below are shared by the EDM baseline, FlowCast, and MeanFlow where applicable; the two flow heads use the channel-weighted L2 form on their velocity regression plus the spectral term on the implied clean residual (for MeanFlow, on the $t = r$ subset of the batch; Section 3.6).

doi:10.6342/NTU202601285



Channel-weighted L2. Let $\beta = (\beta_{\text{t2m}}, \beta_{\text{u10}}, \beta_{\text{v10}}, \beta_{\text{qpepre}})$ be a vector of non-negative per-channel weights. The weighted L2 norm used by the FlowCast velocity loss is

$$\|a\|_{2,\beta}^2 = \sum_{k=1}^{4} \beta_k \, \|a_k\|^2, \tag{3.16}$$

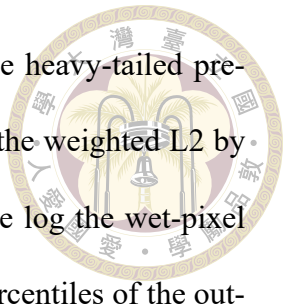
where the sum runs over the four output channels and the norm inside each term is the ordinary spatial L2. The default configuration uses $\beta = (1, 1, 1, w_{\text{pr}})$, with the precipitation weight w_{pr} exposed as the last entry of `channel_weights` in `stormcast/config/model/flowcast.yaml`. The headline FlowCast run uses $w_{\text{pr}} = 2$; Section 4.8 sweeps $w_{\text{pr}} \in [1.0, 2.4]$ to locate the precipitation/wind trade-off. A robust Pseudo-Huber variant $\rho_{c_{\text{h}},\beta}(a, b) = \sum_k \beta_k(\sqrt{\|a_k - b_k\|^2 + c_{\text{h}}^2} - c_{\text{h}})$ with a small c_{h} is also available for bounded gradients on extreme tail events, but is not used in the headline run.

Radial log-PSD regulariser. Because RMSE-like losses reward smooth, low-amplitude precipitation outputs, we add a spectral-matching regulariser that penalises the L1 distance between the radial log-PSD of the predicted and target `qpepre` fields:

$$\mathcal{L}_{\text{spec}} = \frac{1}{|\mathcal{K}|} \sum_{\kappa \in \mathcal{K}} \left| \log P_{\text{pred}}(\kappa) - \log P_{\text{target}}(\kappa) \right|, \tag{3.17}$$

where $P(\kappa)$ is the radially averaged 2D power spectral density at wavenumber κ and $\mathcal{K}$ is the set of non-zero radial wavenumbers on the spatial grid. The regulariser is active only on channels listed in `spectral_channels` and weighted by `spectral_weight` $= \lambda_{\text{spec}}$ (default 0.1). By default we apply it to `qpepre` only, on the flow's implied clean residual $\hat{x}_1$ rather than the raw velocity (Algorithm 1).

doi:10.6342/NTU202601285



Monitoring for mode collapse. Even with the above terms, sparse heavy-tailed precipitation is an invitation to mode collapse: the model can minimise the weighted L2 by producing a slightly-wet constant field. During training we therefore log the wet-pixel fraction at thresholds $\{0.1, 1.0, 5.0\}$ mm h^{-1} and the 95th and 99th percentiles of the output distribution, and compare them to the same quantities on the training batch. A large divergence is a collapse warning.

3.8 Training Infrastructure

All runs use PyTorch with `torchrun --standalone --nnodes=1 --nproc_per_node=N` for multi-GPU training. Configuration is managed with Hydra; experiment overrides are passed on the command line as `key=value` pairs. Checkpoints are written every M iterations (default $M = 1000$) under

```
StormCast_{method}/{experiment}/0/checkpoints{_method}/
checkpoint_*.pt
```

together with CSV loss logs and, every V iterations, validation heatmap and power-spectrum images for the four output channels. Auto-resume restores both the online checkpoint and, for FlowCast and MeanFlow, the EMA shadow (`ema_state.pt`) by a `checkpoint_latest.pt` convention in `trainer_flowcast.py` / `trainer_meanflow.py`; the EDM baseline restores from `checkpoints_diffusion/`. A run can therefore be restarted with the same command line and will pick up from the last written checkpoint.

doi:10.6342/NTU202601285

Chapter 4 Experiments and Results

This chapter presents the empirical evaluation of the MeanFlow residual of Chapter 3 against its two reference heads — the EDM diffusion baseline it sets out to accelerate and the FlowCast straight-line flow-matching baseline it generalises. Section 4.1 defines the experimental protocol and the matched-sample-budget convention. Section 4.2 defines the evaluation metrics. Section 4.3 reports the headline comparison on the 2022 validation year. Section 4.4 reports the NFE / wall-clock–versus–quality trade-off for both flow heads, where MeanFlow's one-to-two-evaluation operating point is the central result. Section 4.5 drills into precipitation skill by threshold; Section 4.6 reports autoregressive rollout error growth. Sections 4.7 and 4.8 report the `log1p`-encoding and precipitation-weight ablations. Section 4.9 places the heads against a legacy old-StormCast baseline. Section 4.10 presents qualitative case studies, Section 4.11 catalogues failure modes, and Section 4.12 concludes with a discussion.

4.1 Experimental Protocol

Validation split. All reported metrics are computed on the cleaned 2022 full-year RWRF validation store (8,759 hourly samples; Section 3.2). This contrasts with the stale `valid_dates` shipped in `stormcast/config/inference/stormcast.yaml`, which we override ex-

doi:10.6342/NTU202601285

plicitly in every run.

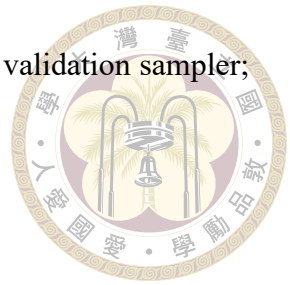
Matched training-sample budget. The fair-comparison unit is training samples seen, not optimiser steps, because the residual heads use different global batch sizes. The EDM baseline trains at batch 64, FlowCast at batch 96, and MeanFlow at batch 112, so ≈ 2 M samples corresponds to step $31{,}250$ for EDM and step $20{,}833$ for FlowCast; we evaluate each head at the nearest saved checkpoint to this anchor (EDM `EDMPrecond.0.31000`, FlowCast `FlowCastPrecond.0.20000`, MeanFlow `MeanFlowPrecond.0.20000`). The MeanFlow checkpoint lands at ≈ 2.24 M samples, 12 % above the anchor — small against the 10–30 % snapshot noise below, but noted for completeness. Holding training samples fixed means any quality difference reflects the generative trajectory and not a training-compute advantage.

Single-snapshot noise. Each validation point is a single snapshot. Adjacent validation logs drift by 10–30 %, so differences below ≈ 5 % on a single metric should not be over-interpreted; we call out only gaps that are large relative to this noise floor.

NFE budgets and samplers. The EDM baseline samples with its native 18-step Heun schedule (two model evaluations per step except the last, $2N - 1 = 35$ NFE per hourly step). FlowCast samples with K explicit Euler steps, K NFE; we report $K \in \{10, 15, 20\}$ in the rollout scoreboard and sweep K from 1 to 50 in the inference-step study of Section 4.4. MeanFlow samples with K average-velocity segments (Eq. (3.15)), also exactly K NFE; its headline operating points are $K \in \{1, 2\}$, and Section 4.4 sweeps it over the same 1–50 grid as FlowCast. The scoreboard harness loads the two flow students from their online checkpoints in the same way, so the FlowCast-versus-MeanFlow comparison

doi:10.6342/NTU202601285

is symmetric end to end (the EMA shadow is used by the in-training validation sampler; Section 3.5).



Rollouts and ensembling. The scoreboard metrics are computed from autoregressive rollouts driven by `compare_diffusion_vs_flowcast.py`, which denormalises every output (undoing standardisation and the `log1p` encoding) so that `qpepre` is scored in $\mathrm{mm\,h^{-1}}$ regardless of the on-disk encoding. CRPS uses a **ten-member** per-hour ensemble of independent noise seeds — matching the ensemble-size bracket of operational convection-permitting ensemble systems (e.g. HREF at ≈ 10 members, MOGREPS-UK at 18) — and deterministic and categorical scores use the ensemble mean. Ten initial times are spaced evenly across the validation year, each rolled out to $+12\,\mathrm{h}$.

Hardware. Training: multi-GPU single node. Evaluation: single GPU. The "Time/Seq." column in the scoreboards is the wall-clock cost of generating one rollout sequence on the evaluation GPU and is the operationally relevant cost figure.

4.2 Evaluation Metrics

All metrics are computed on denormalised fields with physical units (K, $\mathrm{m\,s^{-1}}$, $\mathrm{mm\,h^{-1}}$).

RMSE and MAE. For channel k,

$$\mathrm{RMSE}_k = \sqrt{\frac{1}{|\mathcal{T}|\,HW} \sum_{t \in \mathcal{T}} \sum_{i,j} \left(\hat{X}^{(k)}_{t,i,j} - X^{(k)}_{t,i,j}\right)^2}, \tag{4.1}$$

with MAE defined analogously.

doi:10.6342/NTU202601285

CRPS. The continuous ranked probability score evaluates the full predictive distribution of the small per-hour ensemble against the observation; lower is better. It rewards a forecast that is both sharp and calibrated, and is our primary probabilistic score.

Categorical precipitation scores. For threshold τ (mm h^{-1}), classify each pixel as wet ($\geq \tau$) or dry. With hits H, misses M, false alarms F:

$$\mathrm{CSI}(\tau) = \frac{H}{H+M+F}, \quad \mathrm{FAR}(\tau) = \frac{F}{H+F}, \quad \mathrm{HSS}(\tau) = \frac{2(H\,\mathrm{CN} - M\,F)}{(H+M)(M+\mathrm{CN}) + (H+F)(F+\mathrm{CN})}, \tag{4.2}$$

where CN is the correct-negative count. We report $\tau \in \{0.1, 1.0, 5.0, 10.0, 16.0\}$ mm h^{-1}; "-M" suffixes denote the mean over thresholds and "-P16" the $\tau = 16$ mm h^{-1} heavy-rain point. CSI and HSS reward correct detection (higher is better); FAR penalises over-prediction (lower is better).

Fractions Skill Score. With neighbourhood radius n,

$$\mathrm{FSS}(\tau, n) = 1 - \frac{\sum_{i,j}(p^{\mathrm{fc}}_{i,j} - p^{\mathrm{ob}}_{i,j})^2}{\sum_{i,j}(p^{\mathrm{fc}}_{i,j})^2 + \sum_{i,j}(p^{\mathrm{ob}}_{i,j})^2}, \tag{4.3}$$

where $p_{i,j}$ is the fraction of τ-exceeding pixels in an $n \times n$ box centred at (i, j). FSS mitigates the double-penalty problem and is reported at the heavy-rain point $\tau = 16$ mm h^{-1} for $n \in \{3, 7, 15\}$.

Radial log-PSD. We compute the 2D FFT of each `qpepre` field, square its magnitude, radially average to obtain $P(\kappa)$, and compare $\log P(\kappa)$ of forecast and observation. A model that blurs precipitation has a log-PSD that decays too fast at high wavenumbers.

doi:10.6342/NTU202601285

Rollout metrics. Per-channel RMSE is additionally reported as a function of lead time $\ell \in \{1, \ldots, 12\}$ h to expose autoregressive error growth.

4.3 MeanFlow versus Diffusion and FlowCast: Headline Comparison

Table 4.1 is the central result of the thesis: MeanFlow (at one and two average-velocity segments) set against its two reference heads — the EDM diffusion baseline and the FlowCast straight-line baseline (at three Euler-step counts) — on the 2022 validation year, all at the matched ≈ 2 M-sample budget, sharing the same frozen regression mean and cleaned log1p dataset. Figure 4.1 visualises the same scoreboard.

Table 4.1: Headline three-way comparison on the 2022 validation year (cleaned 192×96, log1p qpepre, ≈ 2 M training samples; ten-member ensembles over ten initial times, $+12$ h rollouts). Precipitation scores are in mm h^{-1} after denormalisation. "-M" = mean over thresholds, "-P16" = 16 mm h^{-1}. ↑/↓ indicate the better direction; **bold** marks the best in each column. Time/Seq. is the wall-clock cost of one ten-member rollout sequence on a single NVIDIA RTX A6000; wall-clock on the shared GPU varies at the ± 10 % level across sessions (the MeanFlow rows were timed in a later session whose diffusion leg ran 8 % faster than tabulated), so speedup factors are quoted to two significant figures at most.

Method	Time/Seq. (s)	CRPS↓	CSI-M↑	FSS-P16↑	HSS-M↑	FAR-M↓	RMSE u10↓	RMSE v10↓	RMS qpepre
EDM diffusion (35 NFE)	21.78	0.6783	**0.1498**	**0.0013**	0.1804	0.4727	1.5626	1.9574	0.942
FlowCast (K=10)	5.88	0.6833	0.1364	0.0000	0.1717	0.4188	**1.4935**	**1.6423**	0.895
FlowCast (K=15)	8.35	**0.6740**	0.1435	0.0000	0.1805	0.4244	1.5203	1.6710	**0.889**
FlowCast (K=20)	10.28	0.6856	0.1435	0.0000	0.1769	0.5436	1.5226	1.7316	0.901
MeanFlow (K=1)	**1.83**	0.8485	0.1449	**0.0013**	0.1832	**0.3861**	1.6540	2.1740	0.926
MeanFlow (K=2)	2.35	0.7265	0.1492	0.0000	**0.1852**	0.5241	1.5350	1.7902	0.940

Read. At $K = 10$ FlowCast runs in 5.88 s per ten-member sequence versus 21.78 s for the 35-NFE EDM Heun sampler — a **3.7× speedup** — and at that budget it matches or beats the diffusion baseline on most columns: CRPS is within the snapshot noise (0.6833 vs 0.6783, a 0.7 % gap), and FlowCast is clearly better on u10 RMSE (-4 %), v10 RMSE

doi:10.6342/NTU202601285

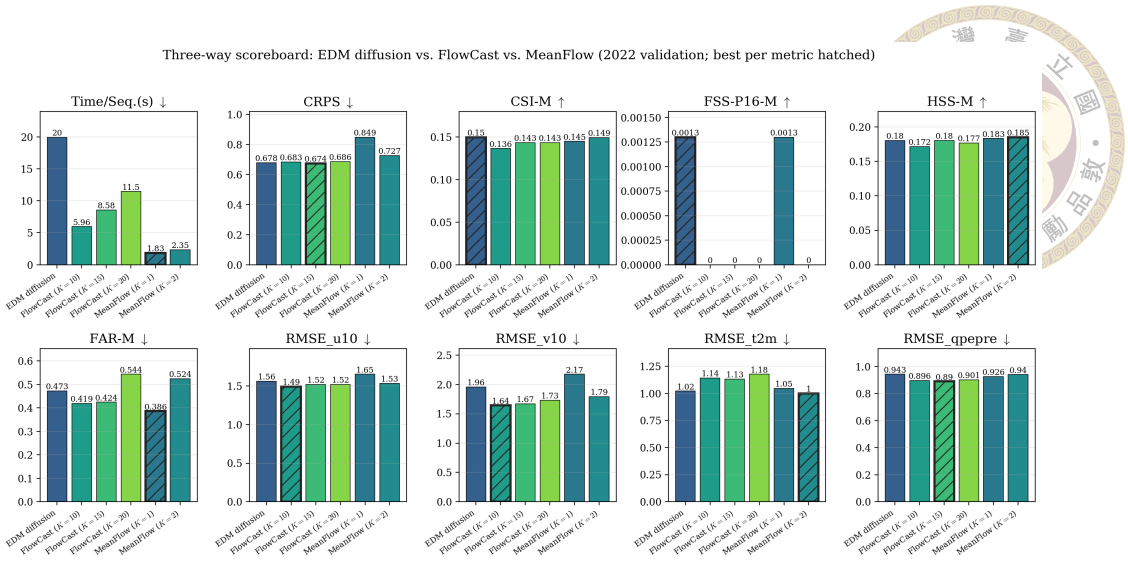


Figure 4.1: Per-metric comparison of the EDM diffusion baseline, FlowCast at $K \in \{10, 15, 20\}$, and MeanFlow at $K \in \{1, 2\}$ on the 2022 validation year (ten-member ensembles). FlowCast attains its skill at roughly a quarter of the diffusion wall-clock cost, MeanFlow at roughly a tenth.

($-16\,\%$), `qpepre` RMSE ($-5\,\%$), and the false-alarm ratio (FAR-M 0.42 vs 0.47). The diffusion baseline retains an edge on the hit-based categorical score CSI-M, on the (near-degenerate; Section 4.5) heavy-rain FSS, and on `t2m` RMSE (diffusion 1.0208 vs FlowCast 1.1410; the one channel where FlowCast trails, omitted from Table 4.1 for width but shown in Figure 4.2 (a)). Pushing FlowCast to $K = 15$ buys the best CRPS (0.6740) and best `qpepre` RMSE (0.8897) in the table at $8.35\,\mathrm{s}$, still $2.6\times$ cheaper than diffusion; beyond that, $K = 20$ shows no further gain and a FAR regression, consistent with the straight-line ODE being essentially converged by $\approx$ 10–15 steps.

Read (MeanFlow rows). MeanFlow compresses the cost axis by another large factor: $2.35\,\mathrm{s}$ at $K = 2$ — $\approx 9\times$ cheaper than the diffusion sampler and $2.5\times$ cheaper than FlowCast's $K = 10$ — and $1.83\,\mathrm{s}$ at $K = 1$. At $K = 2$ it posts the best `t2m` RMSE of any head (0.9999 vs the diffusion baseline's 1.0208; Figure 4.2 (a)) — repairing the one channel FlowCast loses — together with the best HSS-M in the table (0.1852) and a CSI-M (0.1492) at the diffusion baseline's level, i.e. it does not inherit FlowCast's conservative

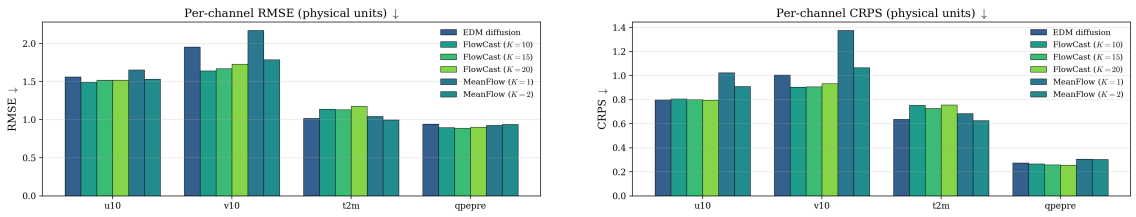
doi:10.6342/NTU202601285

hit-rate deficit. Its concessions are probabilistic and wind-deterministic: the 12-hour rollout CRPS (0.7265) trails the ODE heads by $\approx$ 6–7 %, FAR-M at $K = 2$ (0.52) sits at FlowCast's K=20 level rather than its K=10 level, and v10 RMSE (1.79) lands between FlowCast (1.64) and diffusion (1.96). The $K = 1$ row makes the mechanism legible: deterministic and categorical scores stay close to $K = 2$ (and FAR-M 0.3861 is the lowest in the table — a narrow ensemble raises few alarms), but CRPS degrades to 0.8485, essentially the legacy baseline's level. This is the under-dispersion failure mode of Section 4.11 compounding autoregressively: the same checkpoint's single-step calibration is ahead of FlowCast at every step count (Section 4.4), so the rollout gap is accumulated spread deficit, not single-step quality.

Interpretation. The categorical picture is a coherent trade rather than a wash: FlowCast issues fewer false alarms (lower FAR) but also slightly fewer hits (lower CSI), i.e. it is the more conservative precipitation forecaster, while the diffusion baseline is more aggressive in placing wet pixels and is rewarded by CSI but penalised by FAR. Notably, the FAR gap is much smaller at the ten-member standard than in small-ensemble pilots (0.42 vs 0.47, where four-member runs showed 0.43 vs 0.70): averaging more members suppresses the diffusion head's speculative wet pixels, so part of its apparent false-alarm problem was an ensemble-size artefact. Which trade is preferable is an operational choice; the unambiguous wins for FlowCast are the wind fields, precipitation RMSE, and a $\sim$3.7$\times$ lower inference cost at statistically indistinguishable CRPS. MeanFlow occupies a third corner of the trade space: diffusion-level hit rates and the best temperature at a tenth of the diffusion cost, paid for with a single-snapshot-scale rollout-CRPS gap whose origin (narrowed ensemble spread under autoregression) is diagnosed in Sections 4.4 and 4.11; restoring rollout spread at $K \leq 2$ is listed as future work in Chapter 5.

doi:10.6342/NTU202601285

Per-channel RMSE and CRPS. Figures 4.2 (a) and 4.2 (b) break the deterministic and probabilistic errors out by channel. FlowCast's advantage is largest and consistent across both metrics on the meridional wind `v10` and on the precipitation channel. On `u10` the two metrics disagree — FlowCast has the lower RMSE but the diffusion baseline the lower CRPS at $K \leq 15$ (at $K = 20$ the two are tied) — and on `t2m` the diffusion baseline leads both flow-ODE settings, FlowCast's temperature residual being marginally noisier at this budget. MeanFlow redraws the `t2m` column: at $K = 2$ its temperature RMSE ($0.9999\,\mathrm{K}$) is the best of all heads, $2\,\%$ below the diffusion baseline and $12\,\%$ below FlowCast — evidence that FlowCast's `t2m` weakness belongs to its learned trajectory rather than to the shared backbone, loss weights, or residual pipeline, since MeanFlow shares all three.



(a) Per-channel RMSE (physical units). (b) Per-channel CRPS.

Figure 4.2: Per-channel deterministic (RMSE) and probabilistic (CRPS) errors for the diffusion baseline, FlowCast, and MeanFlow on the 2022 validation year. FlowCast wins on the winds and precipitation; MeanFlow at $K{=}2$ takes `t2m` from the diffusion baseline while staying competitive elsewhere.

4.4 Inference Cost: NFE and Wall-Clock Trade-off

The defining property of a straight-line flow is that it integrates accurately in very few steps; the defining property of an average-velocity model is that it does not integrate at all. The three FlowCast rows of Table 4.1 already show that quality is essentially flat across $K \in \{10, 15, 20\}$ while wall-clock rises linearly, so FlowCast's operational sweet spot sits at the low end of that range; the MeanFlow rows reach the table's deterministic and

doi:10.6342/NTU202601285

categorical skill class at $K \in \{1, 2\}$ (with the rollout-CRPS concession analysed above). Figure 4.3 extends both sweeps over the full $K \in \{1, \ldots, 50\}$ range on the same matched checkpoints: FlowCast's precipitation CRPS saturates by $K \approx 8$–10, while MeanFlow starts below FlowCast's saturated level already at $K = 1$–2 — the two heads pay for quality in different currencies, integration steps versus a harder training objective, and MeanFlow's currency is the cheaper one at inference time. Wall-clock per sequence is linear in K for both, so the few-step regime is simultaneously the accurate and the cheap one. Per NFE at the $K = 10$ operating point, FlowCast costs $\approx 0.59\,\mathrm{s}$ per network evaluation (single forward pass, ten members); because a fixed per-sequence overhead is amortised over more steps, the effective per-NFE cost falls toward $\approx 0.51\,\mathrm{s}$ at $K = 20$. Either way it undercuts the diffusion Heun sampler's $\approx 0.62\,\mathrm{s}$ per evaluation (35 evaluations per hourly step), so FlowCast is cheaper both because it uses fewer steps and because each step is a single forward pass with no second-order correction.

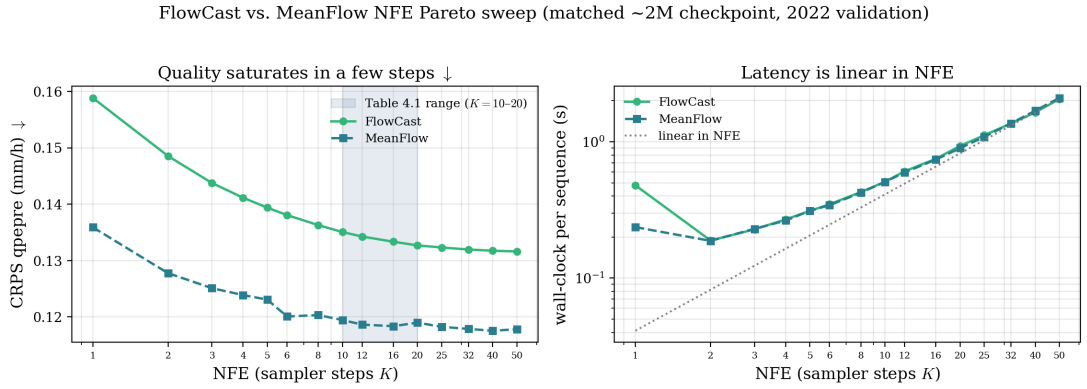


Figure 4.3: FlowCast and MeanFlow quality and cost as the sampler step count K is swept, holding each matched ≈ 2 M-sample checkpoint fixed (2022 validation, single generative step). (a) FlowCast's precipitation CRPS is essentially converged by $K \approx 8$–10; MeanFlow's curve (dashed) sits below FlowCast's at every K, and its 1–2-segment values already undercut FlowCast at any step count in the sweep — the average-velocity objective shifts the whole quality–NFE frontier left. The shaded band marks the $K \in \{10, 20\}$ FlowCast range tabulated in Table 4.1. (b) wall-clock per sequence grows linearly in K for both heads (dotted reference anchored at the most-amortised point), so the few-step regime is also the cheap regime. The head-to-head against the diffusion baseline is in Table 4.1; this sweep is on a smaller sequence subset, so absolute magnitudes are not directly comparable to the scoreboard.

doi:10.6342/NTU202601285

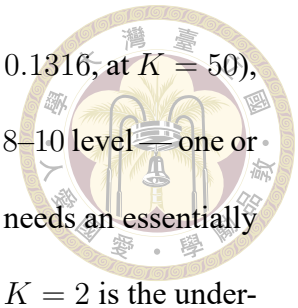
Table 4.2 lists the full FlowCast sweep; Table 4.3 repeats it for the MeanFlow head. Two systematic trends sit underneath FlowCast's CRPS saturation. First, CRPS and ensemble-mean RMSE move in opposite directions with K: CRPS improves monotonically (calibration sharpens as the ODE is integrated more accurately) while qpepre RMSE of the ensemble mean is actually best at $K = 1$ and drifts up by $\approx 2\,\%$ at $K = 50$ — coarse integration acts as an implicit smoother, pulling members toward a common (RMSE-friendly, structure-poor) field. Second, FAR-M rises steadily with K (≈ 0.11–0.12 at $K \leq 3$ to ≈ 0.25 at $K = 50$): better-integrated members place more, sharper wet pixels and incur more false alarms, converging toward the diffusion baseline's behaviour. Both trends reinforce the operational conclusion already drawn from Figure 4.3: past $K \approx 10$ the only things still changing are wall-clock and the conservative-to-aggressive trade-off, not headline skill.

Table 4.2: Full FlowCast NFE sweep behind Figure 4.3 (matched ≈ 2 M-sample checkpoint; 12 sequences $\times$ 10 members, single $+1$ h generative step; qpepre in mm h^{-1}). **Bold** = best per quality column. The K=1 wall-clock carries one-off CUDA warm-up and is not representative; the timing column is best read from $K \geq 2$, where it is linear in K.

K (NFE)	Time/Seq. (s)	CRPS qpepre↓	RMSE qpepre↓	CSI-M↑	FAR-M↓
1	0.39	0.1588	**0.5637**	0.2499	0.1206
2	0.18	0.1486	0.5643	0.2509	**0.1105**
3	0.23	0.1438	0.5649	0.2519	0.1108
4	0.27	0.1411	0.5657	0.2507	0.1179
5	0.31	0.1394	0.5663	0.2529	0.1188
6	0.34	0.1381	0.5666	0.2527	0.1259
8	0.41	0.1363	0.5675	0.2526	0.1350
10	0.48	0.1351	0.5679	0.2523	0.1862
12	0.56	0.1343	0.5686	0.2533	0.1945
16	0.73	0.1333	0.5699	0.2569	0.2173
20	0.87	0.1327	0.5708	0.2631	0.2130
25	1.07	0.1323	0.5723	**0.2657**	0.2141
32	1.35	0.1320	0.5737	0.2653	0.2253
40	1.70	0.1318	0.5749	0.2653	0.2298
50	2.05	**0.1316**	0.5764	0.2648	0.2485

Three readings of Table 4.3 are specific to the average-velocity objective. First, the frontier shift visible in Figure 4.3 is large: at $K = 2$ MeanFlow's precipitation CRPS

doi:10.6342/NTU202601285



(0.1278) is better than FlowCast's at any step count in Table 4.2 (best 0.1316, at $K = 50$), and even the single-evaluation value (0.1359) sits at FlowCast's $K \approx 8$–10 level — one or two average-velocity evaluations deliver what the straight-line ODE needs an essentially converged integration to reach. Second, the gap between $K = 1$ and $K = 2$ is the under-dispersion failure mode of Section 4.11 in numerical form: CRPS improves by 6 % from one segment to two while the ensemble-mean RMSE barely moves ($0.5522 \rightarrow 0.5505$) — the second evaluation buys back ensemble spread, not accuracy. Third, beyond $K = 2$ the same two systematic trends as FlowCast set in, only flatter: CRPS drifts down by a further ≈ 8 % to its floor near $K = 40$ while wall-clock grows by an order of magnitude, and FAR-M rises from 0.17–0.19 at $K \leq 2$ toward ≈ 0.29 at $K = 50$ — more segments place more, sharper wet pixels, the same conservative-to-aggressive drift. The recommended operating point therefore stays at $K = 2$. (The isolated FAR/CSI dip at $K = 4$ is a single-configuration outlier of this small sweep subset and is not interpreted.)

Table 4.3: Full MeanFlow NFE sweep behind Figure 4.3 (matched ≈ 2 M-sample checkpoint; 12 sequences $\times$ 10 members, single $+1$ h generative step; `qpepre` in mm h^{-1}). **Bold** = best per quality column. As in Table 4.2, the K=1 wall-clock carries one-off CUDA warm-up and is not representative; the timing column is best read from $K \geq 2$, where it is linear in K.

K (NFE)	Time/Seq. (s)	CRPS qpepre↓	RMSE qpepre↓	CSI-M↑	FAR-M↓
1	0.24	0.1359	0.5522	0.2720	**0.1657**
2	0.19	0.1278	0.5505	0.2777	0.1924
3	0.23	0.1251	0.5531	0.2662	0.1826
4	0.26	0.1239	0.5549	0.2541	0.2510
5	0.31	0.1231	**0.5497**	0.2834	0.2078
6	0.34	0.1201	0.5507	0.2802	0.2481
8	0.42	0.1203	0.5507	0.2759	0.2315
10	0.51	0.1195	0.5506	0.2929	0.2488
12	0.60	0.1187	0.5523	0.3086	0.2846
16	0.74	0.1184	0.5522	0.2953	0.2640
20	0.90	0.1190	0.5521	0.2879	0.2461
25	1.09	0.1183	0.5512	0.2959	0.2584
32	1.36	0.1179	0.5515	0.3066	0.2926
40	1.69	**0.1176**	0.5538	0.3021	0.2827
50	2.10	0.1178	0.5524	**0.3161**	0.2894

doi:10.6342/NTU202601285

The practical consequence is the one that motivated the thesis (Section 1.4). A 50-member, 24-hour ensemble rollout costs, per forecast cycle, 50×24 sequence-steps; at ≈ 0.59 s per NFE and $K = 10$, FlowCast brings the generative head's contribution down by the same $\approx 3.7\times$ factor seen in the single-sequence timing, moving a configuration that the 35-NFE diffusion sampler places outside the between-radar-volume budget back inside it. MeanFlow at $K = 2$ divides the generative evaluation count by a further $5\times$: the same 50-member cycle needs $50 \times 24 \times 2 = 2{,}400$ network evaluations against $42{,}000$ for the diffusion sampler, cheap enough that the ensemble size itself — not the sampler — becomes the budget knob. Crucially, these speedups are obtained without a second distillation training stage: they are intrinsic to the straight-line and average-velocity models.

4.5 Precipitation Skill by Threshold

Mean categorical scores hide the threshold dependence that matters operationally. Figure 4.4 plots CSI, FAR, and HSS against the rainfall threshold, and Figure 4.5 the heavy-rain FSS at three neighbourhood sizes.

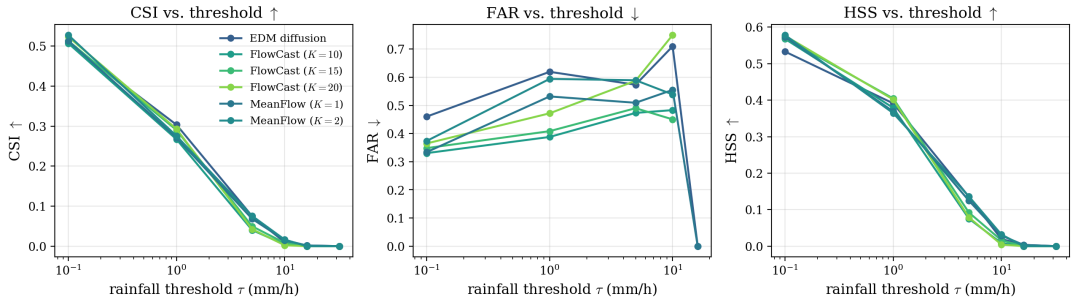


Figure 4.4: CSI, FAR, and HSS as a function of rainfall threshold for the diffusion baseline, FlowCast at $K \in \{10, 15, 20\}$, and MeanFlow at $K \in \{1, 2\}$ (ten-member ensemble means). The flow heads hold a FAR advantage at every threshold; the diffusion baseline pulls ahead on CSI at the heavier 5–10 mm h^{-1} thresholds.

doi:10.6342/NTU202601285

Light to moderate rain. At $\tau = 0.1\,\mathrm{mm\,h^{-1}}$ the two heads are close on CSI (diffusion 0.509, FlowCast-$K20$ 0.524, FlowCast-$K10$ 0.507), but FlowCast's false-alarm ratio is substantially lower (0.33 vs 0.46). The same ordering holds at $\tau = 1.0\,\mathrm{mm\,h^{-1}}$ (CSI: diffusion 0.303 vs FlowCast-$K15$ 0.290; FAR: FlowCast-$K10$ 0.39 vs diffusion 0.62). This is the quantitative form of the "more conservative" behaviour noted above: FlowCast does not paint speculative light rain across dry pixels.

Heavy rain. At $\tau = 5.0$ and $10.0\,\mathrm{mm\,h^{-1}}$ the diffusion baseline leads on CSI (0.075 vs 0.040 at $5\,\mathrm{mm\,h^{-1}}$), reflecting its more aggressive placement of intense cells. FlowCast's heavy-rain hit rate at this ≈ 2 M-sample budget is its weakest point and a clear target for the longer-training and tail-loss experiments listed in the project to-do. At the heavy-rain point itself the comparison degenerates: with ten-member ensemble means, almost no pixel of either head exceeds $16\,\mathrm{mm\,h^{-1}}$, so FSS-P16 is zero for FlowCast and ≤ 0.002 for diffusion at every neighbourhood size (Figure 4.5). This is an ensemble-mean artefact — averaging members suppresses extremes — rather than a statement about individual members, and it is precisely why per-member (probabilistic) heavy-rain scoring over larger ensembles is listed as future work in Chapter 5.

4.6 Autoregressive Rollout

Single-step skill understates operational risk because errors compound under autoregression. Figure 4.6 plots per-channel RMSE against lead time out to 12 hours.

Read. Both heads show the expected monotone error growth. For `v10` FlowCast stays below the diffusion baseline at every lead time (e.g. $1.98\,\mathrm{m\,s^{-1}}$ vs $2.58\,\mathrm{m\,s^{-1}}$ at $10\,\mathrm{h}$);

doi:10.6342/NTU202601285

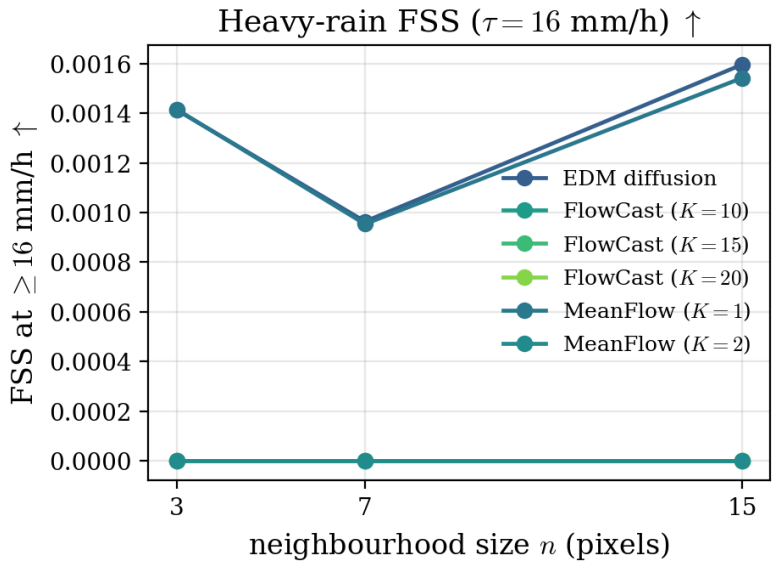


Figure 4.5: FSS at $\tau = 16\,\mathrm{mm\,h^{-1}}$ for neighbourhood sizes $n \in \{3, 7, 15\}$, computed on ten-member ensemble means, for all three heads. The score is near-zero across the board: averaging ten members suppresses $\geq 16\,\mathrm{mm\,h^{-1}}$ extremes almost entirely, so this threshold mainly diagnoses ensemble-mean smoothing rather than model skill. Heavy-rain assessment at the member level requires probabilistic scoring (future work).

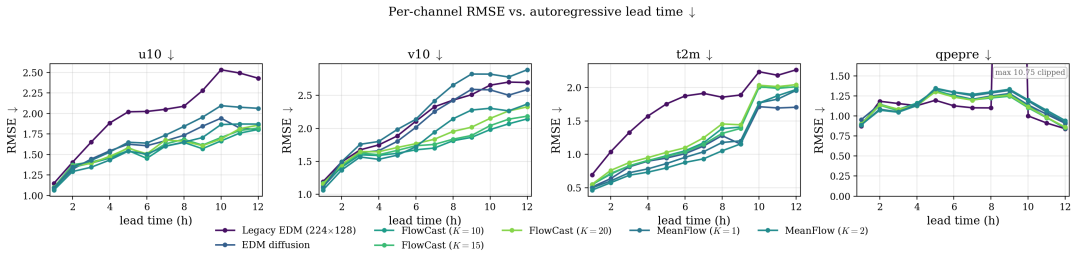


Figure 4.6: Per-channel RMSE as a function of autoregressive lead time (1–12 h) for the legacy baseline, the EDM diffusion baseline, FlowCast, and MeanFlow at $K \in \{1, 2\}$. FlowCast tracks or improves on the diffusion baseline for the winds and precipitation across the horizon; MeanFlow at $K{=}2$ holds the lowest `t2m` error of the cleaned heads, and neither flow head destabilises under rollout.

doi:10.6342/NTU202601285

for u10 the two are close, with FlowCast slightly ahead at long lead times (1.67 m s^{-1} vs 1.94 m s^{-1} at 10 h). The qpepre rollout RMSE is comparable between heads and does not diverge — the conservative FlowCast precipitation does not destabilise under rollout. The one adverse signal is t2m: FlowCast's temperature error grows faster after ≈ 8 h (reaching ≈ 2.0 K at 10–12 h versus ≈ 1.7 K for diffusion), consistent with its weaker single-step t2m number. No run exhibited the unphysical blow-up that aggressive few-step diffusion samplers are prone to, which we attribute to the straight-line ODE's stability.

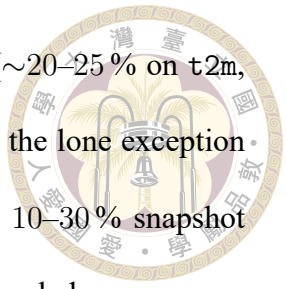
4.7 Ablation: log1p Precipitation Encoding

The single most consequential preprocessing decision is whether to store qpepre as raw mm h^{-1} or as $\log(1 + x)$ before standardisation. Table 4.4 reports the within-architecture A/B test (single-step standardised validation RMSE at ≈ 2 M samples) for the EDM and FlowCast heads. MeanFlow is absent from this table by necessity: only a log1p MeanFlow was trained, so its raw-mm/h cell does not exist. Because the mechanism identified below is head-agnostic — the badly-scaled raw channel contaminates the shared U-Net trunk — and MeanFlow shares that trunk, loss weighting, and the FlowCast objective on 75 % of its batch, we adopt log1p for MeanFlow without a separate ablation.

Table 4.4: Single-step standardised validation RMSE at ≈ 2 M samples, log1p versus raw-mm/h qpepre encoding. Values are dimensionless standard scores. **Bold** = better within each architecture pair on the non-qpepre channels. The qpepre column is not directly comparable across the encoding boundary (different units in standardised space); see the note.

Run	RMSE t2m	RMSE u10	RMSE v10	RMSE qpepre†
diffusion, log1p	**0.0994**	**0.3523**	**0.2762**	0.7865
diffusion, raw mm/h	0.1245	0.3955	0.3115	0.4156
FlowCast, log1p	**0.0835**	**0.2931**	0.2463	0.3851
FlowCast, raw mm/h	0.1118	0.3079	**0.2331**	0.3471

doi:10.6342/NTU202601285



Read. The `log1p` encoding improves both heads on `t2m` and `u10` (∼20–25 % on `t2m`, ∼5–11 % on `u10`) and improves the diffusion baseline's `v10` as well; the lone exception is FlowCast's `v10`, where the two encodings differ by less than the 10–30 % snapshot noise and so are effectively tied. The mechanism is that the badly-scaled raw `qpepre` channel (standard deviation $9.12\,\mathrm{mm\,h^{-1}}$) propagates through the shared U-Net trunk and degrades the other channels; compressing it to a standard deviation of ≈ 0.37 in log space removes that pressure. The net benefit is similar for both architectures, so the encoding choice is essentially independent of the head, and we adopt `log1p` as canonical throughout. The qpepre column (†) cannot be compared directly across the encoding boundary: in standardised space the two encodings carry different units, and inverting `log1p` on a scalar RMSE is invalid because `expm1` does not commute with the root-mean-square. The $\mathrm{mm\,h^{-1}}$-correct comparison is the denormalised scoreboard of Section 4.3, where `compare_diffusion_vs_flowcast.py` inverts the encoding per pixel before scoring.

4.8 Ablation: Precipitation Channel Weight

The precipitation weight w_{pr} (the last entry of `channel_weights`) trades precipitation skill against wind/temperature skill. Table 4.5 reports the full eight-point FlowCast sweep $w_{\mathrm{pr}} \in [1.0, 2.4]$ (single-step standardised validation RMSE, log1p, ≈ 2 M samples). Because `channel_weights` is applied only inside the training loss and not at evaluation, these RMSEs are directly comparable across rows without rescaling by w_{pr}. The sweep is FlowCast-only: w_{pr} is a training-time parameter, so a MeanFlow sweep would require seven additional training runs. MeanFlow instead adopts the sweep's winning all-round setting $w_{\mathrm{pr}} = 2.0$ unchanged, which keeps the MeanFlow-versus-FlowCast headline comparison an A/B on the objective alone.

doi:10.6342/NTU202601285

Table 4.5: FlowCast precipitation-weight sweep, single-step standardised validation RMSE (log1p, $\approx$ 2 M samples). **Bold** = column minimum. $w_{\text{pr}} = 2.0$ is the headline run.

w_{pr}	RMSE t2m	RMSE u10	RMSE v10	RMSE qpepre
1.0	0.0994	0.3188	0.2326	0.5464
1.2	0.1032	0.2912	0.2110	**0.3771**
1.4	0.0890	**0.2733**	**0.2050**	0.3996
1.6	0.0930	0.3282	0.2391	0.6796
1.8	0.0983	0.2838	0.2075	0.4831
2.0 (headline)	**0.0835**	0.2931	0.2463	0.3851
2.2	0.0990	0.3178	0.2499	0.5178
2.4	0.0991	0.3075	0.2355	0.5515

Read. There is no clean monotone trend: both $w_{\text{pr}} = 1.4$ and $w_{\text{pr}} = 2.0$ beat their neighbours, while $w_{\text{pr}} = 1.6$ and $w_{\text{pr}} \geq 2.2$ are poor across the board. $w_{\text{pr}} = 1.4$ is the per-channel sweet spot for the wind fields (lowest `u10` and `v10` RMSE in the sweep), whereas $w_{\text{pr}} = 2.0$ is best on `t2m`; on `qpepre` the minimum is a near-tie among $w_{\text{pr}} \in \{1.2, 1.4, 2.0\}$ that sits inside the 10–30 % single-snapshot noise, so we do not read fine ordering between those three. The robust conclusions are (i) some up-weighting of `qpepre` helps but the floor-to-ceiling spread is large and real (e.g. $w_{\text{pr}} = 1.4$ at 0.3996 vs $w_{\text{pr}} = 1.6$ at 0.6796 is a $\sim$70 % jump on `qpepre` RMSE); (ii) $w_{\text{pr}} \approx 2.0$ is a reasonable all-round default that wins `t2m` and is competitive on `qpepre`, which is why the canonical run adopts it; and (iii) the wind/precipitation trade-off is genuine and would justify dropping to $w_{\text{pr}} = 1.4$ if the winds were the priority.

4.9 Comparison Against the Legacy Baseline

To place the cleaned-pipeline heads in context, Table 4.6 adds the legacy old-StormCast EDM baseline (the original 224×128 raw-qpepre model on the legacy dataset). Because the legacy grid and encoding differ, this is a pipeline-level rather than a controlled head-to-

doi:10.6342/NTU202601285

head comparison, and the absolute magnitudes are not directly comparable to the cleaned rows (the field of view differs by the centre crop).

Table 4.6: Legacy old-StormCast EDM baseline versus the cleaned-pipeline diffusion, FlowCast, and MeanFlow heads. Precipitation in mm h^{-1}. The legacy leg uses a different grid/encoding, so absolute values are comparable only in magnitude (see Section 3.2).

Method	Time/Seq. (s)	CRPS↓	CSI-M↑	FAR-M↓	RMSE qpepre↓
Legacy EDM (raw, 224×128)	49.55	0.8762	0.0614	0.6923	1.1736
Cleaned EDM (35 NFE)	21.78	**0.6783**	**0.1498**	0.4727	0.9429
Cleaned FlowCast (K=10)	5.88	0.6833	0.1364	**0.4188**	**0.8959**
Cleaned MeanFlow (K=2)	**2.35**	0.7265	0.1492	0.5241	0.9404

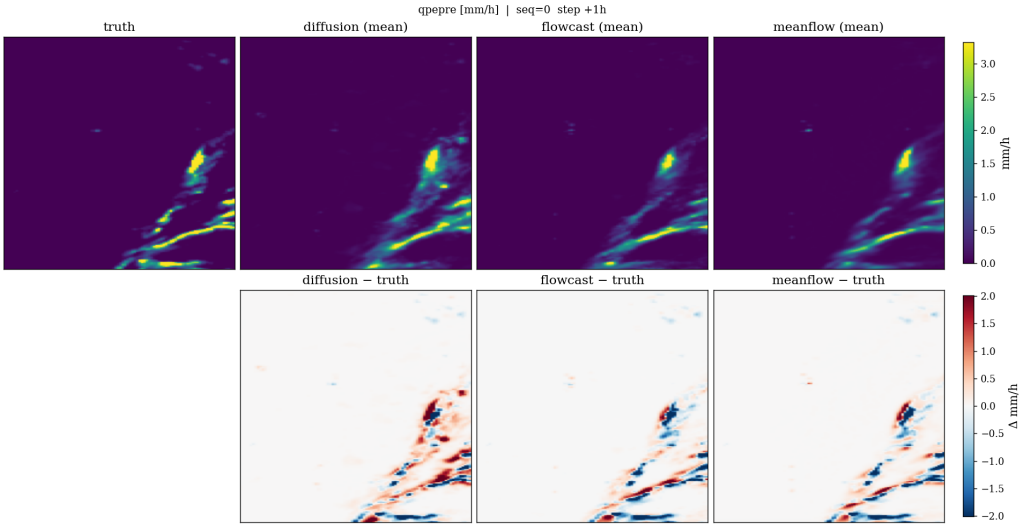
Read. The cleaning + `log1p` + retraining pipeline is a large step up over the legacy baseline on every metric (CRPS $0.88 \rightarrow 0.68$, CSI-M $0.06 \rightarrow 0.15$), independent of the generative head; on top of that, FlowCast adds the inference-cost and false-alarm wins at near-parity on CRPS (0.6833 vs 0.6783, inside the snapshot noise), and MeanFlow compresses the cost axis to 2.35 s — a $21\times$ end-to-end speedup over the legacy system and $\approx 9\times$ over the cleaned diffusion baseline — while holding diffusion-level CSI; its rollout CRPS (0.7265) gives back part of FlowCast's parity but remains far below the legacy level. The legacy-to-cleaned and cleaned-EDM-to-flow gaps are therefore largely orthogonal: most of the headline improvement is the pipeline, and the flow heads' specific contribution is to deliver comparable skill at a fraction of the cost without a teacher — FlowCast at near-CRPS-parity, MeanFlow at the lowest cost.

4.10 Qualitative Case Studies

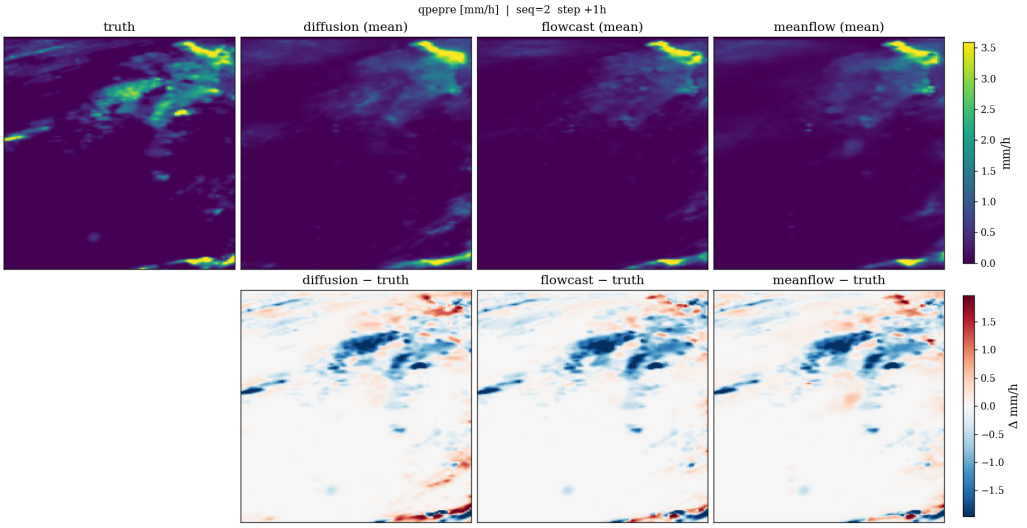
Figure 4.7 shows side-by-side denormalised precipitation panels (truth, diffusion, FlowCast, MeanFlow) for representative 2022 validation hours, and Figure 4.8 the same for the zonal wind. The qualitative reading matches the metrics: both flow heads produce

doi:10.6342/NTU202601285

sharp, spatially coherent precipitation with fewer speculative light-rain pixels in nominally dry regions (the FAR advantage made visible), while the diffusion baseline paints more widespread weak precipitation. MeanFlow's two-evaluation fields are not visibly smoother than FlowCast's ten-step fields — the average-velocity sampler does not pay the blurring penalty that an aggressively truncated ODE integration would. On the wind field the flow heads' structure is visibly closer to the RWRF target.



(a) Validation sequence A.



(b) Validation sequence B.

Figure 4.7: Denormalised hourly precipitation (mm h^{-1}): RWRF/QPEPRE truth, EDM diffusion, FlowCast, and MeanFlow, for two 2022 validation sequences. The flow heads keep nominally dry regions dry while preserving convective structure; MeanFlow achieves this at two network evaluations.

doi:10.6342/NTU202601285

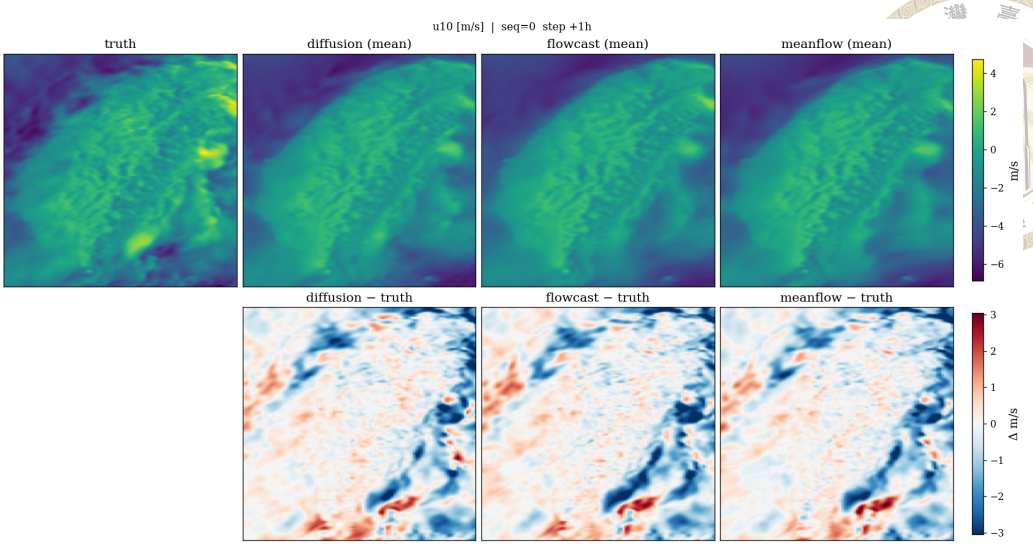
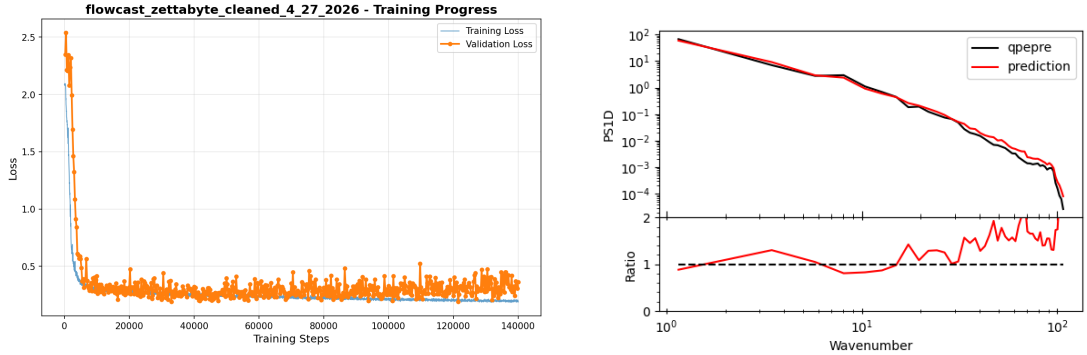


Figure 4.8: Denormalised 10 m zonal wind `u10` ($\mathrm{m\,s^{-1}}$): truth, diffusion, FlowCast, and MeanFlow. The flow heads' fields are closer to the RWRF target, consistent with their lower wind RMSE.

Spectral sharpness. Figure 4.9 (b) shows a representative FlowCast `qpepre` radial log-PSD against the target during training: the spectral-regularised FlowCast tracks the observation spectrum into the high-wavenumber band rather than rolling off early, which is the spectral signature of a sharp (non-blurry) generative forecast.



(a) FlowCast training/validation loss.

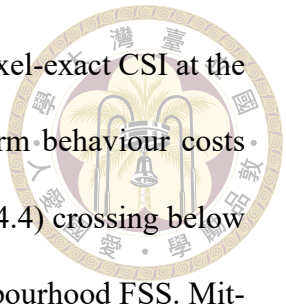
(b) `qpepre` radial log-PSD.

Figure 4.9: (a) FlowCast loss curves over training; (b) radial log power-spectral density of a validation `qpepre` sample versus the target. The spectral-matching regulariser keeps the predicted spectrum from decaying too fast at high wavenumbers.

4.11 Failure Modes

We document four failure modes observed during development.

doi:10.6342/NTU202601285



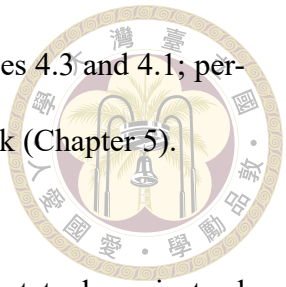
Heavy-rain under-detection. The flow heads' weakest metric is pixel-exact CSI at the heavier thresholds (5–10 mm h^{-1}): their conservative, low-false-alarm behaviour costs hits on intense cells. Diagnosis: the per-threshold CSI curve (Figure 4.4) crossing below the diffusion baseline above 5 mm h^{-1}, partly recovered by the neighbourhood FSS. Mitigation candidates (deferred to future work): longer training, a higher w_{pr} paired with the spectral term, or an explicit tail-quantile loss on `qpepre`.

`t2m` rollout drift. FlowCast's temperature error grows faster than the diffusion baseline's after $\approx$ 8 rollout hours (Figure 4.6). This is the autoregressive signature of its weaker single-step `t2m` number. Notably, MeanFlow does not inherit this drift despite sharing FlowCast's backbone and loss weights — its `t2m` rollout error stays at or below the diffusion baseline's — which suggests the drift is a property of the learned trajectory rather than of the architecture. Mitigation for FlowCast: a per-channel weight that does not starve `t2m`, or a short teacher-forcing warmup.

One-evaluation under-dispersion (MeanFlow). At $K = 1$ MeanFlow's deterministic skill is intact but its CRPS lags its own $K = 2$ setting consistently: a single average-velocity evaluation maps each noise draw almost straight onto the conditional mean, narrowing ensemble spread and costing calibration. One extra segment ($K = 2$) restores most of the spread at one additional NFE, which is why $K = 2$ is the recommended MeanFlow operating point and the validation default. The effect compounds under autoregression: in the single-step sweep the K=1-to-K=2 CRPS gap is 6 % (Table 4.3), but over 12-hour rollouts it widens to 17 % (0.8485 vs 0.7265, Table 4.1) because each hour's narrowed ensemble seeds the next; the residual $\approx$ 6 % rollout-CRPS gap that remains at $K = 2$ against the ODE heads has the same origin, even though MeanFlow's single-step calibration is the

doi:10.6342/NTU202601285

best in the study (Section 4.4). Diagnosis: the CRPS columns of Tables 4.3 and 4.1; per-member rank histograms and rollout spread restoration are future work (Chapter 5).



EMA dependence. Loading the online `checkpoint_*.pt` training-state dump instead of the proper weights produces visibly noisier flow-head samples. This is not a model failure but an evaluation pitfall. The in-training validation samplers use the EMA shadow (`ema_state.pt`); the head-to-head harness of this chapter loads the online `.mdlus` checkpoints for FlowCast and MeanFlow symmetrically, so the A/B comparison between the two students is unaffected by the convention, and the EMA-versus-online gap itself shrinks as training proceeds.

4.12 Discussion

The headline question of the thesis is whether a single-run, teacher-free average-velocity residual can replace the EDM diffusion residual in a regional convection-permitting forecaster without losing the features that matter for Taiwan precipitation — and do so cheaply enough to deploy. The evidence at the matched ≈ 2 M-sample budget is a qualified yes: at $K \in \{1, 2\}$ network evaluations — roughly a tenth of the diffusion sampler's cost — MeanFlow reaches the diffusion baseline's deterministic and categorical skill class, posts the best `t2m` accuracy of any head, and delivers a single-step precipitation CRPS below the FlowCast baseline at any step count, conceding only a spread-driven rollout-CRPS gap (largest at $K = 1$, mostly closed by $K = 2$). The FlowCast straight-line baseline charts the intermediate point on the same trajectory axis: at $K = 10$ Euler steps it matches the diffusion baseline's CRPS, beats it on wind and precipitation RMSE, cuts its false-alarm ratio by $\approx 11\,\%$ (FAR-M 0.42 vs 0.47), and runs $\approx 3.7\times$ faster — confirming that the

doi:10.6342/NTU202601285

straight-line path already recovers most of the diffusion skill before the average-velocity jump removes the last of the integration cost.

Three caveats bound these conclusions. First, all numbers are at a single ≈ 2 M-sample snapshot with 10–30 % validation noise; the FlowCast canonical run continues to ≈ 13.4 M samples (step 140,000) and a final comparison at converged budgets is part of the remaining work (Chapter 5 and the project to-do). Second, both heads are anchored to the same frozen regression mean and the same cleaned log1p dataset, so the comparison is clean but specific to this Taiwan RWRF setup; transfer to other domains is untested. Third, the precipitation-encoding gotcha of Section 4.7 means the regression mean and residual head must always share the `log1p` encoding — mixing encodings silently corrupts the residual distribution. The next chapter summarises what we have learned and lays out the concrete experiments and data still needed to turn this provisional result into a defensible thesis claim.

doi:10.6342/NTU202601285

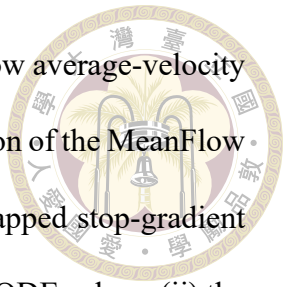
doi:10.6342/NTU202601285

Chapter 5 Conclusion and Future Work

5.1 Summary of Contributions

This thesis asked whether a single-run, teacher-free average-velocity residual can replace the EDM diffusion residual in a regional, convection-permitting StormCast forecaster over the Taiwan domain — delivering the speed a curved diffusion ODE cannot, without sacrificing the sharpness and probabilistic calibration that motivated generative forecasting in the first place — and how few network evaluations such a residual ultimately needs. Starting from the StormCast two-stage decomposition (a frozen deterministic regression mean plus a generative residual head) trained on the Taiwan RWRF dataset, we adapted **MeanFlow** — which learns the average velocity over a whole time interval and transports the state across it in one or two network evaluations, with no integration at all — as the generative residual head. As baselines we retained the EDM diffusion residual it sets out to accelerate and **FlowCast**, a Conditional-Flow-Matching head that learns the straight-line instantaneous velocity field and samples it with a handful of explicit Euler steps and that MeanFlow generalises exactly. All three were compared head-to-head on the full 2022 hourly validation year.

doi:10.6342/NTU202601285

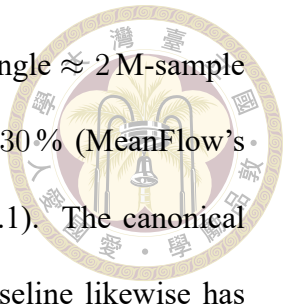


The concrete contributions are: (i) a single-training-run MeanFlow average-velocity residual pipeline for StormCast — to our knowledge the first application of the MeanFlow objective to a weather forecaster — trained with the identity-bootstrapped stop-gradient loss, sampling the residual in one or two network evaluations with no ODE solver; (ii) the supporting engineering and two reference heads (EDM diffusion and the FlowCast I-CFM baseline) anchored to exactly the same frozen regression mean, backbone, and σ_{data}, so the comparison isolates the generative trajectory and the MeanFlow-versus-FlowCast leg is an A/B on the training objective alone (MeanFlow's loss reducing to FlowCast's at zero interval length); (iii) a precipitation-aware loss — channel-weighted L2 plus a radial log-PSD regulariser on `qpepre` — together with `log1p`-encoding and precipitation-weight ablations; (iv) a precipitation-aware evaluation suite spanning deterministic (RMSE/MAE), probabilistic (CRPS), categorical (CSI/FSS/HSS/FAR), spectral, and rollout metrics; and (v) the resulting finding that, at a matched ≈ 2 M-sample budget, MeanFlow reaches the diffusion baseline's deterministic and categorical skill class at $K \in \{1, 2\}$ evaluations — roughly a tenth of the diffusion sampler's cost — with the best `t2m` of all heads and a single-step precipitation CRPS below FlowCast's at any step count, conceding only a spread-driven rollout-CRPS gap ($\approx 6\,\%$ at $K = 2$, larger at $K = 1$); the FlowCast baseline meanwhile matches the diffusion CRPS, beats it on wind and precipitation RMSE and on false-alarm ratio, and runs $\approx 3.7\times$ faster at 10 Euler steps, marking the intermediate point on the same trajectory axis.

5.2 Limitations

Several limitations should be kept in mind when interpreting these results.

doi:10.6342/NTU202601285



Single training snapshot. The headline comparison is taken at a single ≈ 2 M-sample checkpoint per head, where single-snapshot validation noise is 10–30 % (MeanFlow's checkpoint sits 12 % above the anchor, inside that noise; Section 4.1). The canonical FlowCast run continues to ≈ 13.4 M samples, and the diffusion baseline likewise has more training available; a converged-budget comparison is not yet in hand.

Single regression mean and single domain. All three heads are anchored to one frozen regression mean trained on one regional domain (Taiwan RWRF). We cannot assess transfer to other convection-permitting domains (HRRR/CONUS StormCast) without retraining, nor disentangle how much of the flow heads' behaviour depends on the specific quality of this regression mean.

Heavy-rain detection. The flow heads' conservative, low-false-alarm behaviour costs pixel-exact hits at the heavier rainfall thresholds. The neighbourhood FSS recovers part of this, suggesting a displacement rather than a pure miss, but the deterministic heavy-rain CSI is a genuine weakness at the current budget — and at the heaviest threshold, ten-member ensemble-mean scoring degenerates for every head (averaging suppresses extremes), so the experiment cannot yet rank heads there at all.

Limited probabilistic evaluation. CRPS is computed from ten-member per-hour ensembles — the operational convection-permitting EPS bracket — but a large-ensemble calibration study (rank histograms, spread–skill at 50–100 members), which a cheap few-step sampler makes affordable, has not been run.

doi:10.6342/NTU202601285

Under-dispersion at low NFE (MeanFlow). At $K = 1$ MeanFlow's deterministic skill is intact but its ensemble spread narrows — a single average-velocity jump lands each noise draw nearly on the conditional mean — so its CRPS lags its own $K = 2$ setting on every channel, and the deficit compounds under autoregression (a 6 % single-step gap widens to 17 % over 12-hour rollouts; a residual $\approx$ 6 % remains at $K = 2$ against the ODE heads even though MeanFlow's single-step calibration is the best in the study; Section 4.11). The second evaluation restores most of the spread, which is why $K = 2$ is the recommended operating point; a per-member rank-histogram diagnosis has not yet been run.

No distilled-diffusion comparator. We compare the flow heads against the full 35-NFE diffusion sampler, not against a distilled few-step diffusion student. MeanFlow demonstrates that the 1–2-NFE regime is reachable in a single run with no teacher; whether distillation could carry the diffusion baseline's heavy-rain CSI into that same regime is an open question this thesis does not answer.

5.3 Future Work

The most promising directions, several of which are scoped concretely in the accompanying project to-do, are:

1. **Train to convergence and re-score.** Run all three heads to a matched, converged sample budget and repeat the full metric suite, smoothing over several validation snapshots to beat down the 10–30 % noise. This is the single most important step to turn the provisional headline into a defensible claim.

doi:10.6342/NTU202601285

2. **Isolate the loss recipe.** Re-train the EDM diffusion baseline with the same rain-aware additions the flow heads use (channel weight w_{pr}, spectral log-PSD term), splitting the recipe's contribution from the generative objective's. The speed and NFE results are untouched by this experiment; the skill gaps are its target.

3. **Close the heavy-rain gap.** Test an explicit tail-quantile or focal loss on `qpepre`, a higher w_{pr} paired with the spectral term, and a stronger `asinh`/`log1p` dynamic-range transform, targeting the 5–10 mm h^{-1} CSI deficit without inflating false alarms.

4. **Restore low-NFE rollout spread.** MeanFlow under-disperses at $K = 1$ and the deficit compounds over autoregressive rollouts (a residual CRPS gap survives even at $K = 2$); candidate fixes — a stochastic single-segment sampler, temperature on the noise draw, per-step noise re-injection during rollout, or a small spread-aware loss term — would make the true 1–2-NFE regime probabilistically competitive end to end, diagnosed with per-member rank histograms.

5. **Large-ensemble calibration.** Exploit the flow heads' low per-sample cost to run 50–100-member ensembles and report CRPS, rank histograms, and spread–skill — the uncertainty-quantification payoff that cheap sampling unlocks. At MeanFlow's two evaluations, a 100-member 24-hour cycle costs 4,800 network evaluations — about a ninth of what the 35-NFE diffusion sampler needs for a 50-member one — so ensemble size, not the sampler, becomes the budget knob.

6. **Transfer to other domains.** Apply the same single-run flow recipes (FlowCast and MeanFlow) to another convection-permitting regression mean (e.g. an HRRR-trained StormCast) to measure how much of the pipeline is Taiwan-specific.

doi:10.6342/NTU202601285

5.4 Closing Remarks

Generative modelling is almost certainly the right inductive bias for short-range convective-scale forecasting; the obstacle to deploying it has been the multi-step diffusion sampler. This thesis shows that, on a real regional dataset, a straight-line flow-matching residual can recover most of the diffusion baseline's skill — and exceed it on winds, precipitation RMSE, and false alarms — at roughly a quarter of the inference cost, trained in a single run with no teacher; and that an average-velocity residual trained under the same recipe compresses the sampler further still, to one or two network evaluations — a tenth of the diffusion cost — with the study's best single-step precipitation calibration and temperature accuracy, trading a modest spread-driven rollout-calibration gap. The path a generative model takes from noise to weather turned out to be a dial worth turning twice: first straightened, then jumped. The remaining gaps (heavy-rain hits, FlowCast's `t2m` drift, one-evaluation spread, convergence at a larger budget) are concrete and addressable. Making sharp, probabilistic, regional convective forecasting fast enough to run between radar volumes is within reach, and single-run flow residuals — straight-line or average-velocity — are a credible path to it.

doi:10.6342/NTU202601285

References

[1] M. Andrychowicz, L. Espeholt, D. Li, S. Merchant, A. Merose, F. Zyda, S. Agrawal, and N. Kalchbrenner. Deep learning for day forecasts from sparse observations. arXiv preprint arXiv:2306.06079, 2023.

[2] K. Bi, L. Xie, H. Zhang, X. Chen, X. Gu, and Q. Tian. Accurate medium-range global weather forecasting with 3D neural networks. Nature, 619:533–538, 2023.

[3] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud. Neural ordinary differential equations. In Advances in Neural Information Processing Systems (NeurIPS), 2018.

[4] L. Espeholt, S. Agrawal, C. K. Sønderby, M. Kumar, J. Heek, C. Bromberg, C. Gazen, R. Carver, M. Andrychowicz, J. Hickey, A. Bell, and N. Kalchbrenner. Deep learning for twelve hour precipitation forecasts. Nature Communications, 13, 2022.

[5] Z. Geng, M. Deng, X. Bai, J. Z. Kolter, and K. He. Mean flows for one-step generative modeling. arXiv preprint arXiv:2505.13447, 2025.

[6] H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, J. Nicolas, C. Peubey, R. Radu, D. Schepers, A. Simmons, C. Soci, S. Abdalla, X. Abellan, G. Balsamo, P. Bechtold, G. Biavati, J. Bidlot, M. Bonavita, G. D.

doi:10.6342/NTU202601285

Chiara, P. Dahlgren, D. Dee, M. Diamantakis, R. Dragani, J. Flemming, R. Forbes, M. Fuentes, A. Geer, L. Haimberger, S. Healy, R. J. Hogan, E. Hólm, M. Janisková, S. Keeley, P. Laloyaux, P. Lopez, C. Lupu, G. Radnoti, P. de Rosnay, I. Rozum, F. Vamborg, S. Villaume, and J.-N. Thépaut. The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society, 146(730):1999–2049, 2020.

[7] J. Ho and T. Salimans. Classifier-free diffusion guidance. arXiv preprint arXiv:2207.12598, 2022.

[8] J. Ho, A. Jain, and P. Abbeel. Denoising diffusion probabilistic models. In Advances in Neural Information Processing Systems (NeurIPS), 2020.

[9] T. Karras, M. Aittala, T. Aila, and S. Laine. Elucidating the design space of diffusion-based generative models. In Advances in Neural Information Processing Systems (NeurIPS), 2022.

[10] R. Lam, A. Sanchez-Gonzalez, M. Willson, P. Wirnsberger, M. Fortunato, F. Alet, S. Ravuri, T. Ewalds, Z. Eaton-Rosen, W. Hu, A. Merose, S. Hoyer, G. Holland, O. Vinyals, J. Stott, A. Pritzel, S. Mohamed, and P. Battaglia. Learning skillful medium-range global weather forecasting. Science, 382(6677):1416–1421, 2023.

[11] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In International Conference on Learning Representations (ICLR), 2023.

[12] X. Liu, C. Gong, and Q. Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In International Conference on Learning Representations (ICLR), 2023.

doi:10.6342/NTU202601285

[13] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu. DPM-Solver: A fast ODE solver for diffusion probabilistic model sampling in around 10 steps. In Advances in Neural Information Processing Systems (NeurIPS), 2022.

[14] J. Pathak, S. Subramanian, P. Harrington, S. Raja, A. Chattopadhyay, M. Mardani, T. Kurth, D. Hall, Z. Li, K. Azizzadenesheli, P. Hassanzadeh, K. Kashinath, and A. Anandkumar. FourCastNet: A global data-driven high-resolution weather model using adaptive Fourier neural operators. arXiv preprint arXiv:2202.11214, 2022.

[15] J. Pathak, Y. Cohen, P. Garg, P. Harrington, N. Brenowitz, D. Durran, M. Mardani, A. Vahdat, S. Xu, K. Kashinath, and M. Pritchard. Kilometer-scale convection allowing model emulation using generative diffusion modeling. arXiv preprint arXiv:2408.10958, 2024.

[16] I. Price, A. Sanchez-Gonzalez, F. Alet, T. R. Andersson, A. El-Kadi, D. Masters, T. Ewalds, J. Stott, S. Mohamed, P. Battaglia, R. Lam, and M. Willson. Probabilistic weather forecasting with machine learning. Nature, 2024.

[17] S. Ravuri, K. Lenc, M. Willson, D. Kangin, R. Lam, P. Mirowski, M. Fitzsimons, M. Athanassiadou, S. Kashem, S. Madge, R. Prudden, A. Mandhane, A. Clark, A. Brock, K. Simonyan, R. Hadsell, N. Robinson, E. Clancy, A. Arribas, and S. Mohamed. Skilful precipitation nowcasting using deep generative models of radar. Nature, 597:672–677, 2021.

[18] B. P. Ribeiro and J. F. Pucer. FlowCast: Advancing precipitation nowcasting with conditional flow matching. arXiv preprint, 2025.

[19] N. M. Roberts and H. W. Lean. Scale-selective verification of rainfall accumulations

doi:10.6342/NTU202601285

from high-resolution forecasts of convective events. Monthly Weather Review, 136 (1):78–97, 2008.

[20] T. Salimans and J. Ho. Progressive distillation for fast sampling of diffusion models. In International Conference on Learning Representations (ICLR), 2022.

[21] A. Sauer, D. Lorenz, A. Blattmann, and R. Rombach. Adversarial diffusion distillation. arXiv preprint arXiv:2311.17042, 2023.

[22] J. T. Schaefer. The critical success index as an indicator of warning skill. Weather and Forecasting, 5(4):570–575, 1990.

[23] C. K. Sønderby, L. Espeholt, J. Heek, M. Dehghani, A. Oliver, T. Salimans, S. Agrawal, J. Hickey, and N. Kalchbrenner. MetNet: A neural weather model for precipitation forecasting. arXiv preprint arXiv:2003.12140, 2020.

[24] J. Song, C. Meng, and S. Ermon. Denoising diffusion implicit models. In International Conference on Learning Representations (ICLR), 2021.

[25] Y. Song and P. Dhariwal. Improved techniques for training consistency models. arXiv preprint arXiv:2310.14189, 2023.

[26] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-based generative modeling through stochastic differential equations. In International Conference on Learning Representations (ICLR), 2021.

[27] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever. Consistency models. In International Conference on Machine Learning (ICML), 2023.

[28] A. Tong, K. Fatras, N. Malkin, G. Huguet, Y. Zhang, J. Rector-Brooks, G. Wolf, and

doi:10.6342/NTU202601285

Y. Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. Transactions on Machine Learning Research, 2024.

doi:10.6342/NTU202601285

doi:10.6342/NTU202601285

Appendix A — Hyperparameters and Training Configuration

A.1 EDM Diffusion Baseline Hyperparameters

The EDM diffusion baseline uses the following parameters. FlowCast and MeanFlow share the data scale σ_{data} verbatim so that the standardised-residual statistics and per-channel loss scaling are identical across all three heads (Section 3.4):

Parameter	Value
$\sigma_{\min}$	0.002
$\sigma_{\max}$	80
σ_{data}	0.5
ρ	7
Sampler	Heun
N (steps)	18 ($2N-1 = 35$ NFE)
Global batch size	64
Optimiser	AdamW

doi:10.6342/NTU202601285

A.2 FlowCast (Conditional Flow Matching)

Hyperparameters copied from `stormcast/config/model/flowcast.yaml` and `stormcast/config/training/flowcast.yaml`:

doi:10.6342/NTU202601285

Parameter	Value
σ_{data}	0.5
σ_{path} (I-CFM path std)	0.01
t_ε (time clamp)	10^{-5}
Time embedding scale τ	1000
`diffusion_conditions`	`[state, regression, invariant]`
`regression_conditions`	`[state, background, invariant]`
Backbone `channel_mult`	$[1, 2, 2, 2, 2]$
Optimiser	AdamW
Adam (β_1, β_2)	$(0.9, 0.999)$
Learning rate	5×10^{-4}
Weight decay	1×10^{-4}
Warmup steps (linear)	4,000
Min LR ratio (cosine floor)	10^{-2}
Total training steps	4×10^5
Global batch size	96
Gradient clip (global L2)	1.0
EMA decay (fixed)	0.999
Channel weights β	$(1, 1, 1, w_{\text{pr}})$, $w_{\text{pr}} = 2$ (headline)
Spectral weight λ_{spec}	0.1
Spectral channels	{`qpepre`}
Validation sampler	Euler, $K = 10$
Inference solvers	`euler` / `midpoint`

doi:10.6342/NTU202601285

A.3 MeanFlow (Average-Velocity Flow Matching)

Hyperparameters copied from `stormcast/config/model/meanflow.yaml` and `stormcast/config/training/meanflow.yaml`. Everything shared with FlowCast (optimiser, schedule, conditioning, backbone, channel/spectral weights, EMA) is kept identical so the FlowCast-versus-MeanFlow comparison isolates the objective; only the rows below the divider are MeanFlow-specific:

doi:10.6342/NTU202601285

Parameter	Value
σ_{data}	0.5
t_ε (time clamp)	10^{-5}
State-time embedding scale τ (on r)	1000
`diffusion_conditions`	`[state, regression, invariant]`
`regression_conditions`	`[state, background, invariant]`
Backbone `channel_mult`	$[1, 2, 2, 2, 2]$
Optimiser / Adam (β_1, β_2)	AdamW / $(0.9, 0.999)$
Learning rate / weight decay	5×10^{-4} / 1×10^{-4}
Warmup steps (linear) / min-LR ratio	4,000 / 10^{-2}
Gradient clip (global L2)	1.0
EMA decay (fixed)	0.999
Channel weights β	$(1, 1, 1, 2)$
Spectral weight λ_{spec} / channels	0.1 / {`qpepre`}
(r, t) sampler	two i.i.d. $\mathcal{U}(t_\varepsilon, 1 - t_\varepsilon)$, sorted $r \leq t$
MeanFlow ratio ρ_{MF} ($r < t$ fraction)	0.25
Adaptive weight exponent p	1.0
Adaptive weight stabiliser ε_w	10^{-3}
Interval embedding (on $t - r$)	32-dim sin / cos Fourier, freqs $\in [1, 10^3]$
σ_{path}	0 (pure interpolant)
Global batch size	112 (7 GPUs)
Validation sampler	average-velocity, $K = 2$
Inference NFE (headline)	$K \in \{1, 2\}$

doi:10.6342/NTU202601285

A.4 Ablation Grids

The two ablations of Chapter 4 vary a single knob each, holding the rest of the FlowCast config fixed (the w_{pr} and encoding grids are FlowCast/EDM-only; MeanFlow adopts the winning settings, Sections 4.7–4.8):

Ablation	Values swept
Precipitation encoding (Section 4.7)	`log1p` / raw mm h^{-1}
Precipitation weight w_{pr} (Section 4.8)	$\{1.0, 1.2, 1.4, 1.6, 1.8, 2.0, 2.2, 2.4\}$
Inference steps K, both flow heads (Section 4.4)	$\{1, 2, 3, 4, 5, 6, 8, 10, 12, 16, 20, 25, 32, 40, 50\}$; scoreb

A.5 Per-Channel Dataset Statistics

High-resolution target statistics computed on the 2019-08 – 2021-12 training split (original store; the cleaned store re-derives statistics after the `log1p` encoding of `qpepre`):

Channel	Mean	Std
`t2m` (K)	295.98	5.61
`u10` (m s^{-1})	-1.58	3.84
`v10` (m s^{-1})	-2.65	5.66
`qpepre` (mm h^{-1}, raw)	0.182	9.12
`qpepre` ($\log(1+x)$ space)	—	≈ 0.37

doi:10.6342/NTU202601285

Appendix B — Reproducibility

B.1 Code Availability

The full training and evaluation code for this thesis lives in the `Stormcast-Distilation` repository. The key entry points are:

- `stormcast/train.py` — regression and EDM diffusion baseline training (Hydra config selects the mode).
- `stormcast/train_flowcast.py` — FlowCast (Conditional Flow Matching) training.
- `stormcast/train_meanflow.py` — MeanFlow (average-velocity flow matching) training.
- `stormcast/utils/trainer_flowcast.py` / `trainer_meanflow.py` — the two students' training loops with EMA.
- `stormcast/utils/flowcast_loss.py` — `FlowCastLoss` (I-CFM + per-channel weights + radial log-PSD).
- `stormcast/utils/meanflow_loss.py` — `MeanFlowLoss` (MeanFlow identity with JVP-bootstrapped stop-gradient target, adaptive weighting, same per-channel

doi:10.6342/NTU202601285

weights + log-PSD term on the $t = r$ subset).

- `stormcast/utils/flowcast_precond.py` / `meanflow_precond.py` — `FlowCastPrecond` velocity wrapper and `MeanFlowPrecond` two-time average-velocity wrapper.
- `stormcast/utils/nn.py` — `get_preconditioned_architecture`, `build_network_condition` and the samplers `diffusion_model_forward` / `flowcast_model_forward` / `meanflow_model_for`
- `experiment_scripts/compare_diffusion_vs_flowcast.py` — the head-to-head rollout scoreboard harness used for Chapter 4 (the MeanFlow leg is enabled with `--meanflow-checkpoint`).
- `experiment_scripts/flowcast_nfe_sweep.py` — the NFE Pareto sweep behind Figure 4.3 and Table 4.2; `--method meanflow` runs the identical sweep for the MeanFlow head.
- `experiment_scripts/thesis_style.py` — the single source of truth for figure colour: `viridis` for all two-dimensional fields (matching the trainer-time validation plots), a diverging map for signed error fields, and a categorical per-method palette sampled from the viridis ramp.
- `experiment_scripts/make_thesis_figures.py` / `make_thesis_qualitative.sh` / `export_results_md.py` — regenerate every result figure of Chapter 4 and the consolidated `results.md` (see the regeneration commands below).

B.2 Dataset and Checkpoints

Zarr store. The cleaned Taiwan RWRF dataset used for the reported experiments is the local Zarr store

doi:10.6342/NTU202601285

exp_3_…_full_cleaned_4_27_2026/

with sub-groups LowRes/, HighRes/, and invariants/, a 192×96 grid, and log1p-encoded qpepre (Section 3.2).

Checkpoints (cleaned, log1p, ≈ 2 M-sample anchor).

- Regression mean: StormCastUNet.0.8000.mdlus (shared by all three heads).
- EDM diffusion baseline: EDMPrecond.0.31000.mdlus.
- FlowCast: FlowCastPrecond.0.20000.mdlus; the sibling ema_state.pt holds the EMA shadow used by the in-training validation sampler. The canonical FlowCast run continues to FlowCastPrecond.0.140000.mdlus.
- MeanFlow: MeanFlowPrecond.0.20000.mdlus (≈ 2.24 M samples at batch 112) with its sibling ema_state.pt. The Chapter 4 harness loads the FlowCast and MeanFlow online checkpoints symmetrically.

B.3 Commands

All commands below launch from the stormcast/ directory. The shipped configs point at stale cluster paths and outdated valid_dates; override dataset.location, the regression weights, and dataset.valid_dates=['2022/01/01','2022/12/31'] on the local workstation.

Regression mean (pre-train once).

doi:10.6342/NTU202601285

```
python train.py \
    dataset.location=.../zarr_..._cleaned_4_27_2026 \
    training.experiment_name=regression_main
```

EDM diffusion baseline.

```
python train.py --config-name=diffusion \
    model.regression_weights=.../StormCastUNet.0.8000.mdlus \
    dataset.location=.../zarr_..._cleaned_4_27_2026 \
    training.experiment_name=diffusion_main
```

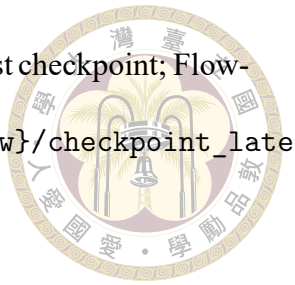
FlowCast.

```
python train_flowcast.py \
    model.regression_weights=.../StormCastUNet.0.8000.mdlus \
    dataset.location=.../zarr_..._cleaned_4_27_2026 \
    training.experiment_name=flowcast_main
```

MeanFlow.

```
python train_meanflow.py \
    model.regression_weights=.../StormCastUNet.0.8000.mdlus \
    dataset.location=.../zarr_..._cleaned_4_27_2026 \
    training.experiment_name=meanflow_main
```


doi:10.6342/NTU202601285



Auto-resume. Re-running the same command picks up from the latest checkpoint; FlowCast and MeanFlow restore both `checkpoints_{flowcast,meanflow}/checkpoint_latest.pt` and `ema_state.pt`.

Head-to-head evaluation. The scoreboard, per-threshold, FSS, CRPS, rollout, and qualitative-panel outputs of Chapter 4 are produced by

```
bash experiment_scripts/run_main_experiment.sh
```

which runs `compare_diffusion_vs_flowcast.py` (with the MeanFlow leg at `MEANFLOW_NFES="1 2"` by default) and writes `scoreboard.{md,csv}`, `per_threshold.csv`, `fss_p16.csv`, `rmse_per_step.csv`, `crps_per_channel.csv`, and a `panels/` tree in physical units (qpepre in mm h^{-1} after inverse-`log1p`).

NFE Pareto sweeps. The data behind Figure 4.3 and Tables 4.2–4.3 come from

```
python experiment_scripts/flowcast_nfe_sweep.py \
    --flowcast-checkpoint .../FlowCastPrecond.0.20000.mdlus \
    --nfes 1 2 3 4 5 6 8 10 12 16 20 25 32 40 50 \
    --n-sequences 12 --n-steps 1 --ensemble 10 --seed 0
python experiment_scripts/flowcast_nfe_sweep.py --method meanflow \
    --meanflow-checkpoint .../MeanFlowPrecond.0.20000.mdlus \
    --nfes 1 2 3 4 5 6 8 10 12 16 20 25 32 40 50 \
    --n-sequences 12 --n-steps 1 --ensemble 10 --seed 0
```

The matched ≈ 2 M-sample checkpoints (step 20,000) are passed explicitly — the script's FlowCast default points at the longer-trained step-140,000 checkpoint, which would not

doi:10.6342/NTU202601285

be comparable to the Chapter 4 scoreboard. `run_flowcast_nfe_sweep.sh` wraps both invocations (`METHODS="flowcast meanflow"`).

Figure and results regeneration. Every result figure of Chapter 4 and the consolidated results document are regenerated end-to-end by two commands:

```
bash experiment_scripts/make_thesis_figures.sh      # CPU only
bash experiment_scripts/make_thesis_qualitative.sh  # one GPU, ~2 min
```

The first rebuilds the data-driven figures (scoreboard, per-channel RMSE/CRPS, CSI/FAR/HSS by threshold, heavy-rain FSS, rollout RMSE, NFE Pareto with both flow heads) directly from the harness CSVs — no inference — writing them into the thesis `figures/results/` tree, and then re-exports `experiment_scripts/results.md`, a single Markdown document collecting every scoreboard with its provenance. The second re-runs the cleaned-leg models for one generative step and re-renders the qualitative truth/diffusion/FlowCast/MeanFlow field panels. All plotting reads its colours from `thesis_style.py`, so the entire figure set stays on one palette by construction.

doi:10.6342/NTU202601285

Appendix C — Detailed Derivations

This appendix collects line-by-line derivations of the identities the method of this thesis rests on, without passing through to the literature. The goal is that any reader with a working knowledge of multivariate Gaussians and basic Itô calculus can reconstruct, from scratch, the score–denoiser (Tweedie) identity and the probability-flow ODE that define the EDM diffusion baseline, the EDM preconditioning and noise schedule, the Conditional Flow Matching theorem and its straight-line Independent-CFM instantiation (the FlowCast baseline), and finally the MeanFlow average-velocity identity at the centre of the thesis. The discrete-time DDPM/DDIM machinery is omitted, as neither MeanFlow nor the continuous-time EDM baseline relies on it; the condensed summary in Chapter 2 suffices for context. Notation in each section is local; all symbols have the same meaning as in Chapters 2–3 unless otherwise specified.

C.1 Tweedie's Formula: Score and Denoiser

Let $x_0 \sim p_{\text{data}}$ and $x = x_0 + \sigma\epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$. Write $p_\sigma(x) = \int p_{\text{data}}(x_0)\, \mathcal{N}(x; x_0, \sigma^2 I)\, dx_0$. Differentiating under the integral,

$$\nabla_x p_\sigma(x) = \int p_{\text{data}}(x_0)\, \nabla_x \mathcal{N}(x; x_0, \sigma^2 I)\, dx_0 = \int p_{\text{data}}(x_0)\, \frac{x_0 - x}{\sigma^2}\, \mathcal{N}(x; x_0, \sigma^2 I)\, dx_0.$$

doi:10.6342/NTU202601285

Dividing by $p_\sigma(x)$,

$$\nabla_x \log p_\sigma(x) = \frac{1}{\sigma^2}\,\mathbb{E}[x_0 - x \mid x] = \frac{\mathbb{E}[x_0 \mid x] - x}{\sigma^2}.$$

Identifying the MMSE denoiser $D^\star(x;\sigma) = \mathbb{E}[x_0 \mid x]$ gives Tweedie's formula

$$\boxed{\nabla_x \log p_\sigma(x) = \frac{D^\star(x;\sigma) - x}{\sigma^2}.} \tag{C.1}$$

Because the Bayes-optimal denoiser is the conditional expectation, score matching and denoising regression are equivalent as learning objectives: fitting D_θ to minimise $\mathbb{E}\|D_\theta(x;\sigma) - x_0\|^2$ simultaneously learns an estimate of the score via (C.1).

C.2 Probability-Flow ODE from the VE-SDE

The variance-exploding forward SDE is

$$dx = g(\sigma)\,dW, \qquad g(\sigma) = \sqrt{\tfrac{d\sigma^2}{dt}},$$

where we indexed time by σ. The corresponding Fokker–Planck equation for the marginal $p_t(x) = p_\sigma(x)$ is

$$\frac{\partial p_\sigma}{\partial t} = \frac{1}{2}\,g(\sigma)^2\,\Delta_x p_\sigma.$$

For the deterministic PF-ODE $dx/dt = f(x,\sigma)$ the continuity equation is

$$\frac{\partial p_\sigma}{\partial t} = -\nabla_x \cdot (f\,p_\sigma).$$

doi:10.6342/NTU202601285

Requiring both to produce the same marginals and rearranging,

$$-\nabla_x \cdot (f\, p_\sigma) = \tfrac{1}{2} g^2\, \Delta_x p_\sigma = \tfrac{1}{2} g^2\, \nabla_x \cdot (\nabla_x p_\sigma) = \tfrac{1}{2} g^2\, \nabla_x \cdot (p_\sigma \nabla_x \log p_\sigma),$$

which is solved by $f = -\frac{1}{2} g^2\, \nabla_x \log p_\sigma$. Switching to σ as the time variable ($d/dt = (d\sigma/dt)\, d/d\sigma$ and $g^2 = 2\sigma\, d\sigma/dt$) yields the clean form

$$\boxed{\frac{dx}{d\sigma} = -\sigma\, \nabla_x \log p_\sigma(x).} \tag{C.2}$$

Combining (C.2) with Tweedie (C.1) gives the denoiser-parameterised PF-ODE

$$\frac{dx}{d\sigma} = \frac{x - D^\star(x;\sigma)}{\sigma}, \tag{C.3}$$

which is the form that first-order (Euler / DDIM) and second-order (Heun) integrators actually evaluate at inference.

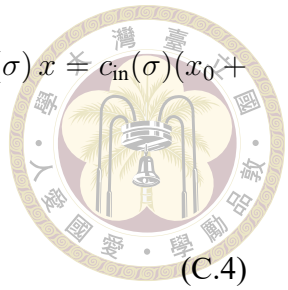
C.3 EDM Preconditioning Derivation

Karras et al. [9] parameterise the denoiser through an inner network F_θ with skip, input, output, and noise scalings:

$$D_\theta(x;\sigma) = c_{\text{skip}}(\sigma)\, x + c_{\text{out}}(\sigma)\, F_\theta\big(c_{\text{in}}(\sigma)\, x,\ c_{\text{noise}}(\sigma)\big).$$

We derive $(c_{\text{skip}}, c_{\text{in}}, c_{\text{out}})$ from three requirements.

doi:10.6342/NTU202601285



Requirement 1: unit-variance input to F_θ. The network input $c_{\text{in}}(\sigma)\, x = c_{\text{in}}(\sigma)(x_0 + \sigma\epsilon)$ has variance $c_{\text{in}}(\sigma)^2(\sigma_{\text{data}}^2 + \sigma^2)$. Setting this to 1 gives

$$c_{\text{in}}(\sigma) = \frac{1}{\sqrt{\sigma^2 + \sigma_{\text{data}}^2}}. \tag{C.4}$$

Requirement 2: unit-variance target for F_θ. The squared loss $\|D_\theta(x;\sigma) - x_0\|^2$ can be written in terms of the network's effective target:

$$F^\star(x;\sigma) \;=\; \frac{x_0 - c_{\text{skip}}(\sigma)\, x}{c_{\text{out}}(\sigma)}.$$

For the F_θ-regression loss to remain well-conditioned across σ, $\text{Var}(F^\star) = 1$:

$$\begin{aligned}\text{Var}(F^\star) &= \frac{1}{c_{\text{out}}^2}\,\text{Var}(x_0 - c_{\text{skip}}\, x) \\ &= \frac{1}{c_{\text{out}}^2}\Big[(1 - c_{\text{skip}})^2\sigma_{\text{data}}^2 + c_{\text{skip}}^2\sigma^2\Big].\end{aligned}$$

Setting this equal to 1 is one equation in two unknowns $(c_{\text{skip}}, c_{\text{out}})$; we need the third requirement to pin them down.

Requirement 3: minimise the error amplification c_{out}. Requirement 2 fixed c_{out}^2 as a function of c_{skip}; Karras et al. [9] pin c_{skip} down by minimising c_{out}, so the inner network's prediction error is amplified as little as possible on the way to D_θ:

$$c_{\text{out}}^2 = (1 - c_{\text{skip}})^2\sigma_{\text{data}}^2 + c_{\text{skip}}^2\sigma^2.$$

Differentiating in c_{skip} and setting to zero,

$$-2(1 - c_{\text{skip}})\sigma_{\text{data}}^2 + 2c_{\text{skip}}\sigma^2 = 0 \implies c_{\text{skip}}(\sigma) = \frac{\sigma_{\text{data}}^2}{\sigma^2 + \sigma_{\text{data}}^2}.$$

doi:10.6342/NTU202601285

Substituting back into the unit-variance requirement,

$$c_{\text{out}}^2 = \left(\frac{\sigma^2}{\sigma^2+\sigma_{\text{data}}^2}\right)^2 \sigma_{\text{data}}^2 + \left(\frac{\sigma_{\text{data}}^2}{\sigma^2+\sigma_{\text{data}}^2}\right)^2 \sigma^2 = \frac{\sigma^2\sigma_{\text{data}}^2}{\sigma^2+\sigma_{\text{data}}^2},$$

so

$$c_{\text{skip}}(\sigma) = \frac{\sigma_{\text{data}}^2}{\sigma^2+\sigma_{\text{data}}^2}, \qquad c_{\text{out}}(\sigma) = \frac{\sigma\,\sigma_{\text{data}}}{\sqrt{\sigma^2+\sigma_{\text{data}}^2}}. \tag{C.5}$$

Noise scaling. $c_{\text{noise}}(\sigma) = \frac{1}{4}\log\sigma$ is an empirically-motivated choice to equalise the dynamic range of the time-embedding input across $\sigma \in [\sigma_{\min}, \sigma_{\max}]$ (the log compresses orders of magnitude; the $\frac{1}{4}$ puts typical values in a band that matches the embedding frequencies).

Loss weighting. The EDM loss is

$$\mathcal{L}_{\text{EDM}} = \mathbb{E}_\sigma\Big[\lambda(\sigma)\,\|D_\theta(x_0+\sigma\epsilon;\sigma) - x_0\|^2\Big].$$

Substituting $D_\theta = c_{\text{skip}}x + c_{\text{out}}F_\theta(\cdot)$ and the target $F^\star$, the inner squared error is $c_{\text{out}}^2\,\|F_\theta - F^\star\|^2$. To make the F_θ-regression effectively unit-weighted, we choose

$$\lambda(\sigma) = \frac{1}{c_{\text{out}}(\sigma)^2} = \frac{\sigma^2+\sigma_{\text{data}}^2}{\sigma^2\,\sigma_{\text{data}}^2}. \tag{C.6}$$

With the Karras log-normal sampling $\log\sigma \sim \mathcal{N}(P_{\text{mean}}, P_{\text{std}}^2)$, the training-time distribution of σ combined with (C.6) concentrates the learning signal at mid-range σ, which is where the denoiser has the highest leverage on the final sample quality. This concludes the derivation; the five preconditioning scalars $(c_{\text{skip}}, c_{\text{in}}, c_{\text{out}}, c_{\text{noise}}, \lambda)$ of Equation (2.10) of the main text are now all justified from first principles.

doi:10.6342/NTU202601285

C.4 Karras ρ-Spaced Noise Schedule

The sampling schedule $\{\sigma_i\}$ is chosen to equalise truncation error per step of a first-order ODE integrator of Equation (C.3). Under the ansatz

$$\sigma_i = \left(\sigma_{\max}^{1/\rho} + \tfrac{i}{N-1}(\sigma_{\min}^{1/\rho} - \sigma_{\max}^{1/\rho})\right)^{\rho},$$

the change of variables $u = \sigma^{1/\rho}$ makes the index i linear in u, and the per-step truncation error of Euler integration is approximately constant in u for ρ tuned to the curvature of the trajectory. Karras et al. [9] show empirically that $\rho = 7$ gives the lowest terminal FID among $\rho \in \{1, 3, 5, 7, 9\}$ across diverse datasets; we adopt it unchanged for the StormCast diffusion baseline, following Pathak et al. [15].

C.5 CFM Theorem: Conditional = Marginal Gradient

Fix a Gaussian conditional probability path $p_t(x \mid z)$ indexed by a latent $z \sim q(z)$, with corresponding conditional vector field $u_t(x \mid z)$. The marginal flow-matching loss is

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, p_t(x)}\left[\|v_\theta(x,t) - u_t(x)\|^2\right],$$

where $p_t(x) = \mathbb{E}_{q(z)}[p_t(x \mid z)]$ and $u_t(x) = \mathbb{E}[u_t(x \mid z) \mid x, t]$ is the marginal vector field. This loss is intractable because $u_t(x)$ requires marginalising over z. The conditional flow-matching loss replaces the marginal target with the conditional one:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, z, p_t(x|z)}\left[\|v_\theta(x,t) - u_t(x \mid z)\|^2\right].$$

doi:10.6342/NTU202601285

Theorem. $\nabla_\theta \mathcal{L}_{\text{FM}}(\theta) = \nabla_\theta \mathcal{L}_{\text{CFM}}(\theta)$.

Proof. Expanding the squared norms and dropping θ-independent terms,

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,p_t(x)}\big[\|v_\theta(x,t)\|^2 - 2\, v_\theta(x,t)^\top u_t(x)\big] + \text{const},$$

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,z,p_t(x|z)}\big[\|v_\theta(x,t)\|^2 - 2\, v_\theta(x,t)^\top u_t(x \mid z)\big] + \text{const}.$$

The first term is identical after applying the law of total expectation, $\mathbb{E}_{z,p_t(x|z)} = \mathbb{E}_{p_t(x)}$. For the second term,

$$\begin{aligned}\mathbb{E}_{t,z,p_t(x|z)}\big[v_\theta(x,t)^\top u_t(x \mid z)\big] &= \mathbb{E}_t\, \mathbb{E}_{p_t(x)}\big[v_\theta(x,t)^\top\, \mathbb{E}[u_t(x \mid z) \mid x,t]\big] \\ &= \mathbb{E}_{t,p_t(x)}\big[v_\theta(x,t)^\top u_t(x)\big],\end{aligned}$$

using $u_t(x) = \mathbb{E}[u_t(x \mid z) \mid x,t]$. Hence $\mathcal{L}_{\text{FM}} = \mathcal{L}_{\text{CFM}} + \text{const}$, and their gradients coincide. □

The upshot is that training against the conditional target is an unbiased (and computable) estimator of gradients of the marginal objective, and the latter is exactly the flow-matching objective whose ODE transports $p_0 \to p_1$.

C.6 Independent CFM: Straight-Line Path and Target

Take $z = (x_0, x_1)$ with $x_0 \sim \mathcal{N}(0, I)$ and $x_1 \sim p_{\text{data}}$, independent. Define the Gaussian probability path

$$p_t(x \mid x_0, x_1) = \mathcal{N}\big((1-t)x_0 + t x_1,\ \sigma^2_{\text{path}} I\big), \qquad t \in [0,1].$$

doi:10.6342/NTU202601285

Target vector field. Under the flow $\phi_t(x \mid x_0, x_1) = (1-t)x_0 + tx_1 + \sigma_{\text{path}}\,\eta$ with η fixed, the associated vector field is

$$u_t(x \mid x_0, x_1) = \frac{d}{dt}\,\phi_t = x_1 - x_0,$$

which is independent of x (and of t) — a straight-line constant-velocity interpolation from noise to data. Plugging into $\mathcal{L}_{\text{CFM}}$,

$$\mathcal{L}_{\text{I-CFM}}(\theta) = \mathbb{E}_{t,x_0,x_1,\eta}\big[\|v_\theta((1-t)x_0 + tx_1 + \sigma_{\text{path}}\eta,\ t) - (x_1 - x_0)\|^2\big]. \tag{C.7}$$

Why $\sigma_{\text{path}} > 0$. At $\sigma_{\text{path}} = 0$ the conditional path is a Dirac at each time, supported on the line segment between x_0 and x_1. In the marginalised view this is a measure-zero set, so the score of the marginal $p_t(x)$ is undefined on that set and the empirical loss gradient is extremely high-variance in high dimensions (Rectified-Flow pathology; see 12). Taking $\sigma_{\text{path}} > 0$ convolves the path with a spherical Gaussian of width σ_{path}, which regularises the target field onto a thin tube around the interpolant and smooths the loss landscape. Tong et al. [28] report $\sigma_{\text{path}} \in [10^{-3}, 10^{-1}]$ all work in practice; FlowCast and our adaptation use $\sigma_{\text{path}} = 0.01$.

Compared to diffusion. The straight-line ODE of I-CFM and the PF-ODE of a score-based diffusion both transport $\mathcal{N}(0, I) \to p_{\text{data}}$, but their trajectories differ. The diffusion PF-ODE is curved: its Jacobian sees the local score, which in turn depends on the local data density, and its integration error grows quickly as the step count shrinks. The I-CFM ODE has constant velocity $x_1 - x_0$ along each conditional trajectory, so the marginal field is the closest approximation (in L^2) to a linear transport available — which is exactly why a single Euler step is already competitive.

doi:10.6342/NTU202601285

C.7 FlowCast's Standardised-Residual Adaptation

The adaptation used in Chapter 3 applies I-CFM to the StormCast residual pipeline with the following changes to the textbook I-CFM of (C.7):

1. **Residual target.** x_1 is the standardised residual R_t/σ_{data} rather than the raw field. Standardisation absorbs the per-channel scale so the noise $x_0 \sim \mathcal{N}(0, I)$ is comparable in magnitude to the data target.

2. **Condition bundle.** v_θ is conditioned on $c_t = \text{concat}(X_{t-1}, M_t, I)$ (the low-resolution background S_t is omitted from the direct path and reaches the velocity field only through the regression mean M_t; see Section 3.5). This keeps the FlowCast residual anchored to exactly the same M_t as the EDM diffusion baseline. No extra VAE encoder is inserted, unlike the original FlowCast which works in latent space.

3. **Time rescaling.** The flow time $t \in [0, 1]$ is multiplied by $\tau = 1000$ before entering the SongUNet time embedding, because the positional-embedding frequencies of the pretrained architecture were tuned for the EDM $\log \sigma$ range.

4. **Channel-weighted + spectral loss.** The pointwise MSE is the channel-weighted L2 norm $\|\cdot\|_{2,\beta}$ of Section 3.7, and an additional radial log-PSD regulariser is applied to the implied clean residual $\hat{x}_1 = x_t + (1-t)\, v_\theta$ on the `qpepre` channel. Applying the regulariser on $\hat{x}_1$ rather than on v_θ keeps the spectrum interpretable as a physical field's spectrum.

5. **Endpoint clamp.** Training samples $t \sim \mathcal{U}(t_\varepsilon,\, 1 - t_\varepsilon)$ with $t_\varepsilon = 10^{-5}$. At $t = 0$ or $t = 1$ the conditional path collapses to a point mass and the Monte-Carlo loss

doi:10.6342/NTU202601285

gradient is unbounded; the clamp prevents numerical blow-ups at negligible bias cost.

The resulting training loop differs from I-CFM only in these five bookkeeping details; the underlying theorem (§C.5) and path derivation (§C.6) carry over unchanged, and the sampler (3.7) is the explicit Euler integrator of the straight-line ODE.

C.8 The MeanFlow Average-Velocity Identity

This is the derivation behind the method of this thesis: the MeanFlow identity [5], transcribed to the noise-at-0, data-at-1 convention used throughout (Section 3.6). It turns an intractable path integral into a pointwise relation the network can be regressed against.

Setup. Work in the standardised residual space of §C.7: $x_1 = R_t/\sigma_{\text{data}}$ is the data endpoint, $x_0 \sim \mathcal{N}(0, I)$ the prior, and the linear interpolant $z_s = (1-s)\,x_0 + s\,x_1$ runs from noise at $s = 0$ to data at $s = 1$ along the trajectory $\dot{z}_s = v(z_s, s)$ with v the (marginal) velocity field of §C.6. For two times $r \le t$ define the average velocity

$$u(z_r, r, t) \triangleq \frac{1}{t-r}\int_r^t v(z_\tau, \tau)\,d\tau, \tag{C.8}$$

the displacement of the flow over $[r, t]$ per unit time. Two immediate properties follow. First, the limit as $t \to r$ is the instantaneous velocity, $\lim_{t\to r} u(z_r, r, t) = v(z_r, r)$, by the fundamental theorem of calculus. Second, multiplying by $(t-r)$ makes transport across the interval a single algebraic step,

$$z_t = z_r + (t-r)\,u(z_r, r, t), \tag{C.9}$$

doi:10.6342/NTU202601285

exactly and with no solver — one evaluation at $(r, t) = (0, 1)$ maps noise to data. The catch is that (C.8) is an intractable integral, so u cannot be regressed directly.

Differentiating away the integral. Clear the denominator in (C.8),

$$(t - r)\, u(z_r, r, t) \;=\; \int_r^t v(z_\tau, \tau)\, d\tau,$$

and differentiate both sides with respect to the state time r, holding the horizon time t fixed. Along the trajectory the state moves with $dz_r/dr = v(z_r, r)$ and $dt/dr = 0$. The left-hand side, by the product rule, is

$$\frac{d}{dr}\Big[(t - r)\, u\Big] = -\, u(z_r, r, t) + (t - r)\, \frac{d}{dr} u(z_r, r, t).$$

The right-hand side, by the fundamental theorem of calculus applied to the lower limit (hence the leading minus sign), is

$$\frac{d}{dr} \int_r^t v(z_\tau, \tau)\, d\tau = -\, v(z_r, r).$$

Equating the two and cancelling the $-u$ against $-v$ rearrangement gives the MeanFlow identity

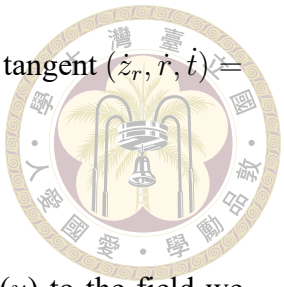
$$\boxed{u(z_r, r, t) \;=\; v(z_r, r) \;+\; (t - r)\, \frac{d}{dr} u(z_r, r, t),} \tag{C.10}$$

which is Equation (3.11) of the main text. The total derivative along the trajectory expands through the chain rule into partials,

$$\frac{d}{dr} u(z_r, r, t) \;=\; \partial_z u \cdot \dot{z}_r + \partial_r u \;=\; v(z_r, r)\, \partial_z u + \partial_r u, \tag{C.11}$$

doi:10.6342/NTU202601285

i.e. exactly the directional (Jacobian–vector) derivative of u along the tangent $(\dot{z}_r, \dot{r}, \dot{t}) = (v, 1, 0)$.



Why this is trainable. Equation (C.10) relates the field we want (u) to the field we can sample without bias ($v = x_1 - x_0$ along each conditional path, by §C.5) using only quantities available at a single point of a single trajectory — the path integral is gone. A network $u_\theta(z_r, r, t, c)$ is regressed onto the right-hand side of (C.10), with the derivative term (C.11) computed by one forward-mode Jacobian–vector product through u_θ along the tangent $(v, 1, 0)$ and the whole target held under stop-gradient. No higher-order gradients, no teacher, no integration, and no discretisation curriculum are required. A network achieving zero loss satisfies (C.10) everywhere, and therefore reproduces the average velocity of (C.8).

Boundary condition and the FlowCast special case. At $t = r$ the interval factor $(t-r)$ vanishes and (C.10) collapses to $u(z_r, r, r) = v(z_r, r)$: average velocity over a zero-length interval is instantaneous velocity. This is exactly the I-CFM/FlowCast regression of §C.6. MeanFlow therefore generalises the FlowCast baseline — the collapsed ($t = r$) subset of the batch trains the boundary that the $r < t$ subset propagates outward into the interval — which is what makes the comparison of Chapter 4 an A/B test on the objective alone.

Convention note. Geng et al. [5] place data at $s = 0$ and differentiate with respect to the horizon time t, obtaining $u = v - (t-r)\frac{d}{dt}u$. Here data sits at $s = 1$, so the differentiation runs through the lower limit of the integral and the sign of the interval term flips to a plus, as in (C.10). The two are the same identity in mirrored time; the loss and sampler of Section 3.6 are internally consistent with the convention used here.

doi:10.6342/NTU202601285